

INFERENTIALISM, BETWEEN RELATIVISM AND DOGMATISM

Kou Okada

Université de Lille, France

Universidade de Lisboa, Portugal

August, 2024

Supervisors: Prof. Shahid Rahman (Lille) and Prof. David Yates (Lisbon)

A thesis submitted to the Université de Lille and Universidade de Lisboa
in partial fulfillment of the requirements of the degree of

Master of Arts in Contemporary Philosophy

© Kou Okada, 2024

Abstract

Robert Brandom's normative inferentialism promises a theory of meaning that avoids the pitfalls of the now-dominant representationalist approach to meaning (viz. dogmatism) while steering clear of relativism of conceptual norms embraced by his pragmatist predecessors. In this thesis, I criticize Brandom's approach and offer an alternative. In particular, I argue that Brandom's attempt at securing the objectivity of conceptual norms as a way out of the dilemma between dogmatism and relativism fails (Chapter 1). Then, I generalize the insights of the debate on the principle of harmony within logical inferentialism (Chapter 2) to natural language, relying on insights from the dialogical tradition (Chapter 3). Finally, I offer a Kantian account of how we might ground semantic normativity—and in particular, the principle of harmony—on discursive agency (Chapter 4). The upshot is that normativity of conceptual norms has its source not in objective, mind-independent reality but in our nature as discursive agents.

Résumé

L'inférentialisme normatif de Robert Brandom promet une théorie de la signification qui évite les pièges de l'approche représentative de la signification, aujourd'hui dominante (à savoir le dogmatisme), tout en évitant le relativisme des normes conceptuelles adopté par ses prédécesseurs pragmatistes. Dans cette thèse, je critique l'approche de Brandom et propose une alternative. En particulier, je soutiens que la tentative de Brandom de garantir l'objectivité des normes conceptuelles pour sortir du dilemme entre dogmatisme et relativisme échoue (chapitre 1). Ensuite, je généralise les idées du débat sur le principe d'harmonie au sein de l'inférentialisme logique (chapitre 2) au langage naturel, en m'appuyant sur les idées de la tradition dialogique (chapitre 3). Enfin, je propose un compte rendu kantien de la manière dont nous pourrions fonder la normativité sémantique - et en particulier le principe d'harmonie - sur l'agence discursive (chapitre 4). La normativité des normes conceptuelles trouve sa source non pas dans une réalité objective et indépendante de l'esprit, mais dans notre nature d'agents discursifs.

Resumo

O inferencialismo normativo de Robert Brandom promete uma teoria do significado que evita as armadilhas da abordagem representacionista do significado, atualmente dominante (nomeadamente, o dogmatismo), ao mesmo tempo que evita o relativismo das normas conceptuais adotado pelos seus antecessores pragmatistas. Nesta tese, critico a abordagem de Brandom e proponho uma alternativa. Em particular, defendo que a tentativa de Brandom de garantir a objetividade das normas conceptuais como forma de sair do dilema entre dogmatismo e relativismo falha (Capítulo 1). Depois, generalizo as ideias do debate sobre o princípio da harmonia no âmbito do inferencialismo lógico (Capítulo 2) à linguagem natural, apoiando-me nas ideias da tradição dialógica (Capítulo 3). Por fim, apresento uma descrição kantiana do modo como podemos fundamentar a normatividade semântica - e, em particular, o princípio da harmonia - na agência discursiva (Capítulo 4). O resultado é que a normatividade das normas conceptuais tem a sua fonte não na realidade objetiva e independente da mente, mas na nossa natureza de agentes discursivos.

Table of Contents

Abstract	i
Résumé	ii
Resumo	iii
1 Brandom's Inferentialism and Objectivity	1
1.1 Representationalism and Inferentialism	2
1.1.1 Varieties of Inferentialism	3
1.1.2 Brandom's Inferentialism	6
1.1.3 Some Usual Objections and Replies	10
1.2 Brandom on Objectivity	12
1.2.1 The Need for Objectivity?	13
1.2.2 Brandom's Account	18
1.2.3 Three Kinds of Semantic Objectivity	22
1.3 The Problem with Brandom's Account	24
1.3.1 Objection 1: Objectivity proofs yield AT, but not PO	24
1.3.2 Objection 2: Scepticism about Incompatibility Semantics	27
1.4 Conclusion	33
2 Harmony: The Traditional Debate Revisited	34
2.1 Harmony as a Local Property	36
2.1.1 Inversion Principle and Intrinsic Harmony	37
2.1.2 Stability	40
2.2 Harmony as a Global Property	44
2.2.1 Separability and Atomism	45

2.2.2	Is Deducibility Enough?	48
2.2.3	Lorenzen on Admissibility	50
2.3	What is the Point of Harmony?	53
2.3.1	Principle of Innocence	54
2.3.2	“Syntax is insufficient for Semantics”	58
2.3.3	Alternatives to the Metalogical view	61
2.3.4	Semantic Significance of Harmony	63
2.4	Conclusion	66
3	Harmony for Natural Language	67
3.1	Brandom Against Harmony	68
3.1.1	Brandom on Intrinsic Harmony	68
3.1.2	Brandom on Conservative Extendibility	70
3.2	Dialogical logic	73
3.2.1	Basic Overview of Dialogical Logic	74
3.2.2	Dialogical Explanations for Defective Rules	81
3.3	Conclusion	85
4	Constitutivism about Semantic Normativity	87
4.1	Constitutivism	88
4.1.1	The General Idea	88
4.1.2	Korsgaard’s Kantian Constitutivism	90
4.2	From Metaethical to Semantic Constitutivism	95
4.2.1	Why Semantic Constitutivism?	95
4.2.2	Dialogical Constitutivism	97
4.2.3	Semantic and Practical Normativity are One	100
4.3	The <i>Shmagency</i> Objection	101
4.4	Conclusion	104

Chapter 1

Brandom's Inferentialism and Objectivity

What distinguishes a word or a sentence from mere sounds or scribbles on a piece of paper? A natural response would be that a word or a sentence (i.e. a linguistic expression) has *meaning*. There are two popular strategies to cash out the notion of meaning: the *representationalist* approach and *inferentialist* approach. In this chapter, I will do three things. In section 1, after defining the rough contours of the representationalism-inferentialism debate, I will present Brandom's inferentialism, which is the most worked-out account available of global inferentialism. Then, in section 2, I introduce one of the most contentious issue for Brandom's inferentialism: namely, how to account for objectivity of conceptual content, and why this is important. In section 3, I argue that Brandom's account fails for two reasons. First, Brandom's so-called 'objectivity proofs' are insufficient to account for the objectivity of conceptual content understood most straightforwardly. Second, Brandom's incompatibility semantics, which he relies on to defend his account of objectivity, undermines the very deontic (and dialogical) nature of meaning that his account presupposes.

1.1 Representationalism and Inferentialism

Meaning theories, or philosophical accounts of mental content and linguistic meaning, can be broadly classified under two camps: the representationalist approach and the use-theoretic approach. On the representationalist approach, conceptual contentfulness of linguistic expressions and mental states are essentially explained by appealing to the representational relation they bear to things 'out there': for example, a representationalist account for linguistic meaning might take sentences to represent state-of-affairs, singular terms to represent objects, predicates to represent properties, etc. So, mental states are understood as contentful insofar as they stand in correct representational relations with what it purports to represent; to understand the meaning of a linguistic expression is to know what it is that it represents.

In contrast, the use-theoretic approach maintains that the conceptual contentfulness should primarily be explained in terms of the rules according to which, or regularities that underlie, our use of linguistic expressions and concepts. The main usage of language/concepts that use-theorists have in mind, the one relevant for explaining conceptual contentfulness, is drawing inferences. Thus, a concept is taken to be contentful insofar as they stand in inferential relations to other concepts; to understand a linguistic expression is to be able to draw inferences containing them.

Crucially, representationalists need not reject that we perform inferences, and inferentialists need not reject that linguistic expressions and mental states have representational properties. In fact, they usually don't; intuitively, both representing the world and drawing inference are two important features of human cognition and so they constitute the *explananda* of any semantic theory. Rather, what the theorist is committed to is their respective order of explanation, i.e, to provide an account of one feature in terms of the other, more primitive feature. Relatedly, a semantic theorist will also usually assume an explanatory priority relation between language and mind, that is, whether mental content should be explained in terms of contents of linguistic expression or vice-versa. For example, Harman (1982) defines his influential account, Conceptual Role Semantics (CRS), explicitly as the commitment to a mentalist and inferentialist order of explanation:

“A theory of conceptual role semantics involves the following two claims: 1. The meanings of linguistic expressions are determined by the contents of the concepts and thoughts they can be used to express. 2. The contents of concepts and thoughts are determined by their functional role in a person’s psychology” (ibid., p.242).

Indeed, as with representationalism, inferentialism is a broad church. Thus, let me now provide a brief typology of inferentialisms in order to situate Brandom’s inferentialism within the debate.

1.1.1 Varieties of Inferentialism

There are six principal ways with respect to which inferentialist theories can differ:

1. *Causal vs. Normative*. On the causal inferentialist view, meaning of expressions are determined by their actual causal role when a particular agent engages in inferential activities (namely, reasoning or asserting). On the normative inferentialist view, the meaning of an expression is determined by rules that determine the correctness of inferential activities. This distinction is due to Peregrin (2014). Writing elsewhere, Peregrin says,

“The drawing of inferences is something that happens within the causal world (in the mind, and hence in the brain); whereas rules, though underlain by normative attitudes, which are events within the causal world, are not themselves states or events within the causal order. Unlike normative inferentialism, causal inferentialism says that the meaning consists in, or is caused by, certain events, namely individual drawings of inferences by individual speakers.”¹

Causal inferentialism is thus understood as a species of functionalism about the mind; theorists like Harman and Boghossian fall under this category. Inferentialists of the normative sort include Brandom, Peregrin, and Dummett.

¹From an unpublished manuscript, “What is Inferentialism?”, available from PhilPapers.

2. *Scope.* Inferentialist meaning-theory (i.e. those that take inferential role to be *sufficient* for determining meaning) can be divided into those that attempt to account for empirical expressions and those that aim only to account for logical—and perhaps some mathematical—expressions. The former, unlike the latter, must take into account quasi-inferential relations between perception and language (or, as Sellars’ puts it, ‘language-entry moves’) on one hand and between language and action (‘language-exit moves’) on the other. Brandom (2007) labels the former camp (to which he belongs) *strong inferentialism* and the latter camp *hyperinferentialism*.
3. *Which inferences?* Following Sellars (1953), we may divide inferences in two categories: *formal inferences* and *material inferences*. Formal inferences are inferences whose correctness is determined by rules that specify the meaning of logical vocabulary involved; this is another way of saying valid in virtue of logical form. Material inferences, on the other hand, are inferences whose correctness is determined by rules that specify the meaning of *non-logical* vocabulary involved. It includes, for example, the inference from ‘1 is a natural number’ to ‘2 is a natural number’ is a valid inference in virtue of the rules governing the meaning of the concept ‘natural number’; it also includes empirical (and thus defeasible) inference from ‘It is raining’ to ‘The streets are wet,’ which is valid in virtue of the rules governing the meaning of the concepts ‘rain’, ‘street’, and ‘wet’.² Logical inferentialists mostly focus on the former kind³; those working on the semantics of natural language must also account for the latter. There is also much to say about the relation between the two kinds of inference. Brandom, for instance, thinks that inferences whose validity depends on material content is explanatorily prior to those whose validity depends on the meaning of logical constants. In other words, formal inference can be defined

²An empirically-minded observer might object that the inference from ‘It is raining’ to ‘The streets are wet’ is actually an *enthymeme*, wherein the following premise is left implicit: ‘If it rains, the streets will be wet’. An addition of this premise would, indeed, turn the material inference into a banal formal inference, namely *modus ponens*. But this is precisely the point that Sellars (1953) argues against. The correctness of the original inference is explained by appealing to the meaning of the non-logical concepts involved, which is determined by network of material inferences an expression is involved in (and especially the counterfactuals it validates).

³Mostly but not entirely, because a number of recent work in proof-theoretic semantics attempt to provide a semantics for certain mathematical concepts.

in terms of material inference, but not vice versa (Brandom, 2000, p. 55).⁴

4. *Minimal semantic unit.* In one sense, the basic semantic unit of inferentialism is uniform: it is at the level of inferential relations, or sequents, as opposed to individual term/concept or propositions which is typical for representationalist approaches. But, there is another way in which the minimal unit of semantic significance may differ. Some inferentialists think that the inferential role of a given expression can be specified by rules in relative isolation from other rules. Call this view *atomism*. As discussed in the previous chapter, this view is common in logical inferentialists. In contrast, theorists like Brandom are holists. *Holism* states that the meaning of an expression is determined by the entire network of inferential connections that make up the language. Finally, others allow for ‘semantic clusters,’ a set of expressions whose meanings are determined by their inferential links to all the others in the set, but maintain that these clusters can be isolated. Dummett’s *molecularism* is one such view.⁵ On Dummett’s conception, the meaning (i.e. the inferential role) of a logical expression, while dependent on that of other logical expressions, bear no relation to that of non-logical expressions.
5. *Naturalism.* An inferentialist can claim that meaning-determining rules are either irreducibly normative or that it can be reduced, typically by appealing to evolutionary biology, to causal phenomena. Some inferentialists do not concern themselves with this question at all (e.g. Dummett and Schröder-Heister) while other explicitly endorse either the anti-naturalistic position (e.g. Brandom) or the naturalistic position (e.g. Peregrin (2024) and Tennant (1987)).
6. *Relation to logic.* Some inferentialists think that inferentialist semantics justify their reformist views of logic (in Dummett and Prawitz’s case, intuitionistic logic and in Tennant’s case, relevance intuitionistic logic) while others opt to remain agnostic about their choice of logic. Brandom, for example, has quietist views about questions concerning the correct logic, preferring to think of logic as a useful instrument

⁴Brandom is not the only one to think this. Cf. Lorenzen & Lorenz (1978).

⁵This typology is due to Steinberger & Murzi (2017).

to make our discursive commitments explicit.⁶

1.1.2 Brandom's Inferentialism

Brandom's inferentialism is the most worked-out global inferentialist meaning-theory. It is characterized by the following six features: 1) normative, 2) strong, 3) inclusive of formal and material inferences, 4) holistic, 5) anti-naturalist, and 6) logically quietist. More needs to be said about Brandom's inferentialism to appreciate the interconnectedness of these theses. Though I cannot give an exhaustive treatment of Brandom's several-treatises-length project here, I will nonetheless try to capture its overall shape.

Brandom's project aims to construct a meaning-theory (or, in his words, a 'theory of conceptual contentfulness') on the basis of observing the characteristics of what linguistically endowed creatures (namely, us) do in practice. A theory of what we *do*, according to Brandom, is explanatory prior to—and thus *grounds*—a theory of what we *say*; or, as Brandom puts it, “semantics is answerable to pragmatics” (1994, p.83). In this sense, Brandom picks up where Wittgenstein (1953) left off—meaning of a linguistic expression is how we use them in our social practices (viz. in a 'language game'). Unlike Wittgenstein, who held a pluralistic view about language games, according to which all the different kinds of use of an expression contribute equally to its meaning,⁷ Brandom thinks that the practice of asserting and drawing inference (“the game of giving and asking for reasons”) is the uniquely privileged locus of meaning. All other use of language are derivative in relation to the game of giving and asking for reasons. Brandom writes,

“What makes something a specifically *linguistic* [...] practice is that it accords some performances the force or significance of *claimings*, of *propositionally* contentful commitments, which can both serve as and stand in need of reasons. Practices that do not involve reasoning are not linguistic or (therefore) discursive practices. Thus the 'Slab' *Sprachspiel* that Wittgenstein introduces in the opening sections of the *Philosophical Investigations* should not, by these stan-

⁶Cf. R. Brandom (2018) for details about his 'expressivist' view on logic. See also footnote 12.

⁷See, for example, Wittgenstein, 1953, §18. Various uses of language (i.e. not limited to assertion) has been studied extensively, for example, by speech-act theory. Cf. Austin (1962).

dards of demarcation, count as a genuine *Sprachspiel*. It is a vocal but not yet a verbal practice. By contrast to Wittgenstein, the inferential identification of the conceptual claims that language (discursive practice) has a *center*; it is not a motley. Inferential practices of producing and consuming reasons are *downtown* in the region of linguistic practice. Suburban linguistic practices utilize and depend on the conceptual contents forged in the game of giving and asking for reasons, are parasitic on it. Claiming, being able to justify one's claims, and using one's claims to justify other claims and actions are not just one among other sets of things one can do with language. They are not on a par with other 'games' one can play. They are what in the first place make possible talking, and therefore thinking: sapience in general." (R. Brandom, 2000, p.14, original italics).

For Brandom, then, the very possibility of human cognition comes down to the fact that we engage in the 'game of giving and asking for reasons'. This 'game' amounts to the social practice in which speakers *endorse* claims (usually in assertion, but could also be in other forms of action) and *challenge* claims due to other speakers by providing *reasons*. Drawing inference is a necessary condition for being able to play this game; for, if some claim *p* were to serve as a reason for another claim *q*, we need to be able to infer *q* from *p*. In order to perform such an inference, language itself must be structured such that one sentence can *entail* another.

So, on the one hand, language must be structured by entailment relations as to enable the practice of giving reasons. On the other hand, asking for reasons involves challenging a claim, namely as something that needs justification. Brandom thinks that challenging a claim paradigmatically involves asserting a claim that is *incompatible* with it.⁸ So, in addition to entailment relations, language must be structured by incompatibility relations. To understand a language, then, is to have a practical mastery of entailment and incompatibility relations which is a precondition for being able to play the game of giving and

⁸Brandom's articulation of incompatibility relations relies on the law of excluded middle, which is only valid in classical logic. Most inferentialists think that if inferentialism is correct, precisely due to the requirement of harmony, we ought to endorse intuitionistic logic. This creates some friction between Brandom and other inferentialist, but I will not get into this thorny issue here.

asking for reasons. Accordingly, one way to think about the meaning of a given sentence is as the union of 1) the set of sentences that it entails and is entailed by and 2) the set of sentences that are incompatible with it. More precisely, and in a slightly different, epistemological formulation, we may put it this way: to understand the meaning of a sentence is to know the claims it follows from and the claims that follow from it, as well as the claims that cannot be endorsed simultaneously with it.

Crucially, this social practice—of giving and asking for reasons—is structured by *rules* (or, ‘norms’ in Brandom’s terminology) that specify which inferential moves are acceptable and which are not. This is the content of Brandom’s *normativism*. This simple move parries the vicious circularity objection often raised against inferentialism: the meaning of an expression is its inferential role, but inference involves transition from one meaningful expression to another (and not on uninterpreted string of symbols) and so appears to presuppose the meaningfulness of expressions prior to their use in inference. So, the thought goes, inferentialists are guilty of explaining meaning in terms of inferential role on one hand and inference in terms of meaning on the other.⁹

The rebuttal goes like this. The meaning of an expression is not its role in *actual* inferences that we draw or are disposed to draw, but the *correct* inferential role specified by socially-shared rules. So, while it is true that the inferential role of expression when we actually perform inferences is explained by our understanding of the meaning of the expression, the meaning of the expression itself is determined by role-specifying social rules that exists beyond any single instantiation of reasoning.¹⁰

Now, the origin and ontological status of these rules seem to be mysterious, especially given the current philosophical climate dominated by naturalized epistemology and materialist ontology. Brandom has a complex story to tell about the emergence of rules from a more primitive social practice of taking someone (including oneself) to be *committed* to and *entitled* to certain claims on the basis of other commitments and entitlements of that speaker (an activity he calls ‘deontic scorekeeping’). For instance, a collateral commitment that comes with asserting a claim (which is in itself a commitment) would be to

⁹Cf. P. Boghossian (2014) for how a (causal) inferential role semantics might overcome this issue.

¹⁰For a more careful inferentialist treatment of the circularity objection, see Peregrin (2018).

provide a reason when challenged; moreover, when a speaker asserts a claim, they entitle anyone else to assert the same claim. The main idea is that we attribute to others (and oneself) 'normative statuses' of commitments and entitlements; this results in a complex network of normative social relationships, which gives rise to certain rule-governed patterns of behaviors, from which meaning-constitutive rules are abstracted.

So, inferential role is defined in terms of preservation of commitments and entitlements. Commitments, in this context, can be understood in the Dummettian fashion as the *consequence* of applying a concept; *entitlements* are the *condition* for application of a concept. So, for example, given that one is committed to 'I am a human', they ought to also be committed to 'I am a mammal'; that one is entitled to 'It is raining' entitles them to 'The streets are wet.' Incompatibility can be derivatively defined the interplay of commitments and entitlements: given that one is committed to certain claims, they are excluded from being entitled to other claims. More precisely, p and q are incompatible iff commitment to p precludes entitlement to q and *vice versa* (Brandom, 1994, p.160). In sum, rules governing (both material and formal) inference, which determine the meaning of a given expression, is cashed out as the patterns of behavior that arise within the game of giving and asking for reasons. This is what Brandom means by "grounding a inferential semantics on a normative pragmatics" (1994, p.132) or "know-that in terms of know-how" (ibid., p.135).

All this talk of 'patterns of behavior' may be reminiscent of Daniel Dennett's naturalistically acceptable and deflationary account of intentionality, in which intentionality is understood simply as an heuristic device to explain patterns of behavior exhibited by something ('an intentional system').¹¹ But Brandom thinks that linguistic practice is "normative all the way down" (ibid., p.625) since what gives rise to rules is not simply a pattern of behavior, but a pattern of behavior *instituted* by the the speakers' attitudes ('normative attitudes) of taking something to have normative status. Indeed, Brandom, doubling down on his anti-naturalism, thinks that Dennett himself ought to be committed to the irreducibility of the normative for sake of coherence (cf. ibid., p.55-62).

¹¹Cf. for example, Dennett (1991).

1.1.3 Some Usual Objections and Replies

I think that many of the usual objections leveled against it are either off-target or are largely methodological and thus does not directly affect the cogency of Brandom's argument. Objections that I have in mind are usually raised by empirically-minded philosophers that are not sympathetic to the inferentialist project in the first place.¹² Objections of this sort include the circularity problem mentioned in the section 1, as well as the idea that Brandom's theory is 'too abstract' and has little theoretical use. For instance, Williamson notes that the traditional truth-conditional approach to semantics, pioneered by Tarski (for formal language) and subsequently Davidson (for natural language), "has been basic to the massive development of mainstream formal semantics over recent decades, in both linguistics and philosophy of language, for natural and artificial languages" (Williamson (2010)). The empirical success of truth-conditional semantics means that the revisionary inferentialist semantics ought to provide a very good reason to abandon truth-conditional semantics in favor of it, or at least, to be considered as an alternative. On Williamson's view, inferentialism fails to do so: "If you want an explicit theory of how some particular linguistic construction contributes to the meanings of sentences in which it occurs, the inferentialist is unlikely to have one. Better try the referentialist" (ibid.).¹³

It is worth dwelling on Williamson's line of thinking, as it can be either read as off-target or methodological. It is off-target because, as Steinberger & Murzi (2017) note, inferentialism is a metasemantic theory (i.e. what I have, following Dummett (1991), referred to as 'meaning-theory') rather than a semantic theory. The difference is that the former tries to identify the conditions under which meaning is possible (or, in the epistemic formulation, what the speaker needs to know to understand any sentence) while the latter tries to answer 'what is the meaning of p ?' where p is an arbitrary sentence, by providing a suitable interpretation. David Lewis makes a particularly instructive remark in this context:

"I distinguish two topics: first, the description of possible languages or gram-

¹²For example, the title of Fodor & Lepore (2007), "Brandom Beleaguered," attests to this claim.

¹³These quotes were taken from Williamson's review of R. Brandom (2009), so the arguments are brief and underdeveloped there. For a more fleshed out critique of inferentialism along these lines, see chapter 4 of Williamson (2007).

marks as abstract semantic systems whereby symbols are associated with aspects of the world; and, second, the description of the psychological and sociological facts whereby a particular one of these abstract semantic systems is the one used by a person or population. Only confusion comes of mixing these two topics” (Lewis, 1970, p.19).

Williamson’s objection is clearly guilty of this confusion. Once the confusion is cleared up, we can see that it is, in principle, coherent to model language truth-conditionally (for instance, for the sake of formal semantics) while subscribing to inferentialism for the metasemantic questions.¹⁴

Yet Williamson’s complaint can also be read as a methodological one: namely, that inferentialism does not cohere well with the best understanding of reality as described by the sciences, including linguistics. Furthermore, since Brandom thinks that normativity is irreducible to causal phenomena (namely events), his project is anti-naturalistic. If naturalistic acceptability is one of the theoretical desiderata of any semantic theory, then this objection would be powerful. But, in Brandom’s case, it is not. Brandom rejects naturalized epistemology, where science and philosophy are methodologically continuous.¹⁵ On the contrary, according to Brandom, philosophers ought to

“produce new vocabularies in which we can understand ourselves and each other, and they do that by thinking about the kinds of being we are, and about the role of such vocabularies in instituting and constituting the conceptual normativity that is the medium in which beings like us live our lives” (R. Brandom, 2009, p.149-150).

By subscribing to an anti-naturalistic methodological stance about philosophy, Brandom has no real reason to oppose anti-materialism, namely, to an ontology that accepts normative objects (namely, rules). Given this, naturalists would deem Brandom’s project unacceptable, both methodologically and substantially.

¹⁴Indeed, MacFarlane (2010) suggests such an approach.

¹⁵cf. Quine (1969).

I think that the naturalistic objection is warranted, even fair. Even if we accept Brandom's view of philosophy and its relation to the natural science, we still need answers as to how normative phenomena and causal phenomena relate, or why normativity is irreducible. In my view, Brandom has thus far been unable to provide any convincing response to these questions. Nonetheless, it should be noted that the critique is based on a naturalistic position that is *external* to Brandom's theory; such an external presupposition does not affect the coherence of Brandom's arguments themselves. So, I will not consider these important and interesting questions in any more depth.

Yet, there is a related objection, which, in my view, threaten the very enterprise of global inferentialism from criteria *internal* to Brandom's theory: namely, that Brandom's inferentialism cannot account for objectivity of conceptual content.

1.2 Brandom on Objectivity

Brandom thinks that being able to account for objectivity is a desideratum of any meaningful theory, and acknowledges the difficulty inferentialists face with respect to providing such an account. However, Brandom claims that his brand of inferentialism has the resources to account for objectivity. As we will see, many commentators disagree. It is far from obvious what this notion, *objectivity*, even is. In order to get the discussion started, let me suggest a working definition:

Objectivity A sentence is *objectively correct* iff it correctly represents mind-independent reality.

Objectivity, at least in the context of our discussion, is best understood as a property, or normative status, of some claim. Objectivity in this sense is closely related to the property of truth, understood classically as correspondence with reality. So, another way to put the above thesis is that a claim is objectively correct iff it is *true*. In this section, first I will identify the reason behind the idea that accounting for objectivity is a desideratum of any meaningful theory; then, I will summarize Brandom's own account of objectivity. Finally, I will present some objections to Brandom's account.

1.2.1 The Need for Objectivity?

As an inferentialist about semantics and a pragmatist about philosophical method, both in the sense of classical American pragmatism and in the sense of later Wittgenstein, not many theoretical resources are available to Brandom to account for objectivity. Most inferentialists are (in)famous for being metaphysical anti-realists—roughly, that reality can coherently be said to exist prior to being perceived and/or articulated into sentences by rational beings—which is a commitment that typically goes hand-in-hand with deflationism about truth.¹⁶ Classical pragmatists (like Dewey and James) also evade committing to realism and a robust conception of truth, opting for quietist views about such big metaphysical questions. But, contrary to most inferentialists and pragmatists, Brandom wants to recuperate realism and objectivity within his inferentialism. Speaking about Brandom (and other pragmatists of this generation), Levine (2010) writes,

“In recent years, a renascent form of pragmatism has developed which argues that a satisfactory pragmatic position must integrate into itself the concepts of *truth* and *objectivity*. This New Pragmatism, as Cheryl Misak calls it, is directed primarily against Rorty’s neo-pragmatic dismissal of these concepts. For Rorty, the goal of our epistemic practices should not be to achieve an objective view, one that tries to represent things as they are ‘in themselves,’ but rather to attain a view of things that can gain as much inter-subjective agreement as possible. In Rorty’s language, we need to replace the aim of objectivity with that of *solidarity*. While the New Pragmatists agree with Rorty’s ‘humanist’ and ‘anti-authoritarian’ notion that the world by itself cannot dictate us what we should think about it, they demur from his suggestion that this required us to give up the notions of truth and objectivity” (p.567-568, my italics).

Now, why should Brandom demur from giving up the notions of truth and objectivity? I take it that the main reason for this is Brandom’s dissatisfaction with the pragma-

¹⁶Indeed, this is the main thesis of Dummett (1991). The point is to ‘dethrone’ *truth* as the master concept of semantics, replacing it with *assertibility* and/or *validity*.

tists' (especially Rorty's) positive philosophical contribution. In order to better understand this, let us review the interrelation of Brandom's and Rorty's thoughts.

Brandom's philosophy is heavily influenced by that of Rorty, who was one of his doctoral advisor (the other being David Lewis).¹⁷ In particular, Brandom's philosophy takes heed of Rorty's *anti-representationalism*, which is one of the most radical critiques of philosophy as such. In his book, *Philosophy and the Mirror of Nature* (Rorty (1979)), Rorty sets out on a pragmatist crusade against representationalism. At the heart of Rorty's pragmatist and anti-representationalist thesis is a rather simple epistemological consideration. Rorty thought that epistemology is condemned to Agrippa's trilemma, whereby justification of knowledge is either circular, results in an infinite regress, or supposes an unjustified foundation on which all cognition rests. The first two options, according to Rorty, lead to skepticism. The first option appears to lead to solipsism: if justification indeed moves in a circle, rationality begins to appear disconnected from reality, or as John McDowell provocatively put it, like "a frictionless spinning in a void." (1994, p.11) The second option is equally troubling. We need only to make two plausible assumptions about the nature of justification: that justification is a binary relation between the justifier (e.g. "It is raining outside") and the justified (e.g. "The streets are wet"), and that justifiers can only transmit antecedently possessed rational warrant. Since it is, in principle, always possible to question the justificatory status of the justifier, the second option generates an infinite regress and thus leads to skepticism. In order to put a stop to the regress, one must posit a foundation. This remaining option, embraced by philosophers hitherto, is the main target of Rorty's attacks.

Rorty identifies and attacks two kinds of foundationalism: one that takes sense-experience to be a given that needs no further justification and another that admits a kind of cognition that is immediate and nonrepresentational, which serves as the foundation for all representational cognition. Empiricist philosophers, falling in the former camp, commits what Sellars dubbed the 'Myth of the Given,' which is, crudely, to falsely suppose that entities that cannot be reasons (i.e. propositionally contentless items like pure sense-datum) can stand in inferential or justificatory relation with those that can be (i.e. propositionally

¹⁷The clearest articulation of Brandom's intellectual debts to Rorty is found in R. B. Brandom (2022).

contentful items). Rationalist philosophers fall in the latter camp, assuming that subjects have privileged and infallible access to nonrepresentational states, paradigmatically Descartes' *cogito* and Plato's cognition of the Forms. In both cases, the foundations are deemed dogmatic by Rorty as they are irreducibly subjective—or based on first-person experience—and cannot be subject to further rational inquiry. This is what Rorty (1970) means when he says, “incorrigibility [is] the mark of the mental.”

But what exactly is the connection between representationalism and epistemological foundationalism? Rorty thinks that philosophers are forced into accepting foundationalism because they are cornered into the intolerable choices offered in Agrippa's trilemma, and what brings the trilemma into view is their shared commitment to representationalism, in particular, the idea that reality can be truthfully represented in language and in thought. This is the crux of Rorty's critique: representationalism necessarily commits us to traditional epistemology, which in turn leads us to foundationalism and thus dogmatism; thus, we must abandon the idea that language and thought essentially aims to represent reality. Rorty, in a pithy slogan, summarizes this point: “cognition is for coping, not for copying.” (1979, p. 11) This is, for Rorty, the essential pragmatist critique of epistemology and of representationalism that he attributes to later Wittgenstein, as seen in, for example, the following aphorism: “If I have exhausted the justifications, I have reached bedrock and my spade is turned. Then I am inclined to say: ‘This is simply what I do.’”

Given that Western philosophy from Plato onwards—and especially analytic philosophy—essentially aims at truthfully representing reality in language, according to Rorty, philosophy needs a radical reorientation. The validity of his argument leading up to such a radical conclusion notwithstanding, Brandom acknowledges the force of Rorty's epistemological criticism of representationalism. Rorty's critique can be seen as a fertile ground from which the philosophy of Brandom sprouts. Indeed, Brandom's philosophy can be understood as an attempt to flesh out a constructive theory of meaning, contra ‘therapeutic’ quietism. In doing so, Brandom rejects that constructive theorizing is futile, while staying true to the lessons from Rorty's critique of representationalism, namely its two central theses of the incorrigibility of the mental and social pragmatism about normativ-

ity.¹⁸

One of the upshots of the forgoing discussion is that reality (i.e. what we purportedly represent) cannot on its own normatively constrain us. In other words, the world and things in it are not the sort of things that can justify our beliefs and assertions, since to suppose that it can is to fall victim to the Myth of the Given. This ‘anti-authoritarian’ move leaves an explanatory vacuum for beliefs and assertions formed on the basis of perceptual experience—if we cannot appeal to ‘how things are’ as the ultimately source of justification for them, how are perceptually-formed beliefs and observational reports possible?

In fact, Brandom is acutely aware of the obvious pitfall of his *doktorvater*’s views: if all norms that determine the correctness of a claim is conventional, then we are faced with the threat of relativism (i.e. the correctness of a claim depends solely on the conventions of a linguistic community). From there, it is a short step to anti-rationalism—along the lines of the oft-cited Nietzschean dictum, “there are no facts, only interpretations.” Since Rorty embraces relativism of this kind, critiques leveled against Rorty along these lines are legion—including critiques of the so-called ‘postmodernism’, a movement wherein Rorty is sometimes placed. I will refrain from developing these critiques further, since I think that it is rather easy to see why untamed relativism about epistemic normativity is troubling.

Since linguistic rules, according to Brandom, arise within some a given practice, this seems to imply that rules are *social conventions*. Conventionalism may not be a problem for some specific and isolated semantic clusters (e.g. for rules governing the meaning of logical or mathematical vocabulary¹⁹), it is not, in general, attractive for the holist. Conventionalism implies relativism, in the sense that rules that are conventional are also only rules in the context where such conventions hold, namely in the context of a particular linguistic community. Problems arise in particular for rules governing empirical concepts as conventionalism seems to imply that the conditions and consequences of applying them

¹⁸A worked-out interpretation of Brandom’s philosophical project as a response to Rorty’s can be found in Bouché (2020).

¹⁹Since claims brought forward in these domains are typically understood as *analytic*, the meaning of concepts belonging to these categories can be conventional and yet lead to substantive truths.

(for example, 'electron') is a matter of social convention. But shouldn't the meaning of such concepts be, at least in part, determined by the 'things out there'?

This brings us to the fore one of the principal criticism of global inferentialism: in Williamson's (2010) words, "since inferential relations do not fix truth and reference, meaning has not been adequately tied to the language-independent world." This is a problem that Brandom inherits from Dummett's assertibilist account of meaning. Dummett's assertibilist account seeks to replace the central role occupied by truth-conditions in traditional semantic theorizing with assertibility conditions. But such a project runs into a familiar problem: norms that governs assertibility (the norm of rational warrant) appears to be distinct from the norm of truth, which determines the correctness of an assertion in a way that is fundamentally different from the correctness conceived of as rational warrant. This problem gives rise to what Wright and Hale (2010) call the *truth question*:

"I may with full warrant – hence commendably – assert what is actually not so, and I may unwarrantedly – hence reprehensibly – assert what, luckily for me, is so. But untruth is still a form of failure, a way for even the best-warranted assertion to be in bad standing; and truth is still a form of success, a point of commendation for even the most capricious claim. how, using just the kind of materials which the Assertibilist allows himself, can a conception of the content of moves be constructed which will allow them – as they must be, if they are assertions – to be subject to both these apparently contrasting norms?" (p.276-277).

In addition to an account of truth, inferentialism needs an account of a related but nonetheless distinct phenomena of objectivity. Assertibility conditions are, in general, relative to the situation of the epistemic agent (including what information she has access to); however, the content—understood either in inferential and representational terms—is not in general relative.²⁰ This observation gives rise to the objectivity question: "can [Brandom] provide for the emergence of assertoric moves whose contents are objective

²⁰Special cases may include the use of the first-person pronoun.

in the specific sense that they do not describe the situation of the agent – what is or is not permitted, or prohibited, for her in that particular context – nor other features of her attitudes?” (ibid., p.277).

If all rules are understood as conventional, the inferentialist view amounts to giving up on both truth (as something distinct from norms that govern assertibility conditions and consequences) and objectivity understood as perspective-independence of content. Such a move would bring inferentialism dangerously close to Rorty’s view, and charges of anti-rationalism and skepticism about the very possibility of *genuine* normativity (which is to be contrasted with arbitrary rules, like that of etiquette) start to creep in. Thus, the chief motivation behind Brandom’s attempted rehabilitation of objectivity can be seen as a way out of the unpalatable consequences relativism brings.

1.2.2 Brandom’s Account

Acknowledging the threat of conventionalism, Brandom writes,

“The identification of objectivity with intersubjectivity [...] is defective in that it cannot find room for the possibility of error regarding that privileged perspective. [...] The alternative is to reconstrue objectivity as consisting in a kind of perspectival form, rather than in a nonperspectival or cross-perspectival content. What is shared by all discursive perspectives is that there is a difference between what is objectively correct in the way of concept application and what is merely taken to be so, not what it is – the structure, not the content” (1994, p. 599-600).

So, Brandom is clear in rejecting Rorty’s proposal of replacing objectivity with *solidarity* (or ‘intersubjectivity’), conceived of as the collective agreement by epistemic subjects within a linguistic community. The perspective of the ideal linguistic community is still, on Brandom’s view, just another perspective; it can be wrong, and thus, be criticized. In doing so, he rejects what he calls the ‘I-we’ account of sociality, which, as Kiryushchenko (2021) notes, coincides, in essence, with Peirce’s model of inquiry under which the ultimate epistemic subject is the entire epistemic community, which constitutes a regulative

ideal for rational inquiry. On Peirce's view, the ideal epistemic community provides a public criteria of truth, in the sense that what would be held true by the ideal community would coincide with truth. So, objective truth on this view is defined as a privileged perspective of the epistemic community at-large that meets certain conditions.

Brandom's alternative construal of objectivity is built on top of his pragmatic model of deontic scorekeeping, which upholds an 'I-thou' account of sociality. According to the 'I-thou' account, the perspective of each epistemic subject is "at most *locally* privileged in that it incorporates a structural distinction between objectively correct applications of concepts and applications that are merely subjectively taken to be correct" but, in contrast to the 'I-we' account, "none of these perspectives is privileged in advance over any other" (Brandom, 1994, p.600). Here, Brandom echoes Davidson's explanation of the pragmatic origins (i.e. within the context of primitive forms of linguistic communication) of objective representational content. Davidson (1984), expounding on a model of communication which he calls 'radical interpretation', writes,

"[...] the concepts of objective truth and error necessarily emerge in the context of interpretation. The distinction between a sentence being held true and being in fact true is essential to the existence of an interpersonal system of communication. [...] Someone cannot have a belief unless he understands the possibility of being mistaken, and this requires grasping the contrast between truth and error [...] [but this contrast] can emerge only in the context of interpretation, which alone forces us to the idea of an objective, public truth" (p.170).

Like Davidson, Brandom thinks that the ability to appreciate the distinction between 'what is taken to be true' (subjectively) and 'what is true' (objectively) emerges only in his model of deontic scorekeeping—which is "the social practices of attributing and acknowledging commitments and entitlements, which implicitly institute [normative] statuses" (Brandom, 1994, xvii) In fact, 'deontic scorekeeping practices' can be seen as the systematized generalization of Davidsonian radical interpretation.²¹

²¹Seen in this light, Brandom's model of pragmatics (viz. deontic scorekeeping) can, in principle, be cut

The exact structure of ‘deontic scorekeeping,’ which is essential to fully understanding Brandom’s account of objective representational content, is (perhaps overly) complex. For now, noting a few things about it should suffice. The fundamental move in deontic scorekeeping is the *attribution* of commitments. By attributing a commitment p to someone, the scorekeeper (or the discursive agent) notes three things. First, the scorekeeper attributes to the subject the assertion/belief that p ; second, that the subject commits to provide a reason when the truth of p is challenged; and third, that the subject commits to all of the claims that the scorekeeper believes are (materially) entailed by p . Whether the subject acknowledges them or not, claims that stand in material entailment relation to p are thus understood as the subject’s *undertakings*.

Using the resources of the deontic scorekeeping model (as well as his semantic model based on entailment and incompatibility relations), Brandom produces a pair of what he calls “objectivity proofs.”²² In particular, he shows that the following two inferences are, in general, impermissible:

No First-Person Ignorance (p) [$p \rightarrow$ I claim that p]

No First-Person Error (p) [I claim that $p \rightarrow p$]

The main idea of the proofs are that in both cases (of the above inferences), the antecedent and the consequent have differing incompatibility relations, due to the fact that attribution of a commitment to an epistemic subject does not automatically entitle her to that commitment. To illustrate this point, consider the following claims:

1. The swatch is red.
2. The claim that the swatch is red is properly assertible by me now.
3. Rational beings never existed.

off from his inferentialism about meaning. In fact, Macfarlane (2010) argues that the Davidsonian truth-conditional semantics is compatible with Brandom’s pragmatics. This threatens Brandom’s project at-large due to Brandom’s claim that his semantic theory grounded on his model of pragmatics; if his pragmatics is compatible with truth-conditional semantics, it is not clear, by Brandom’s own lights, why we should prefer an inferentialist semantics.

²²I will not treat these proofs in depth for the sake of concision. Cf. *ibid.*, p.605 for details.

It is clear that 1. and 2. have the same assertibility conditions. But, their propositional contents are different, due to the fact that their consequences differ. In particular, 1. is compatible with 3. but 2. and 3. are incompatible. The asymmetry between 1. and 2., shows that objectivity can be explained entirely on inferential grounds: some claims, namely empirical ones, are objective in the sense that their truth does not depend on any attitude (of endorsement or rejection). Thus, Brandom writes, “The fact is that the distinction between sentences sharing assertibility conditions and sharing truth-conditions ... can be made out in terms of commitments and entitlements, without the need to invoke the notion of truth” (1994, p. 604). The idea is that semantic correctness—i.e. proprieties of an inference—does not collapse into any given scorekeeper’s perspective. Furthermore, if we were to give up this simple idea, every scorekeeper would take herself to be omniscient and incorrigible (ibid., p.605), bankrupting the entire deontic scorekeeping model. Therefore, in line with the Davidsonian take on radical interpretation, Brandom thinks that this recognition must be shared by all speakers, constituting a necessary structural feature of deontic scorekeeping practices.

Now, recall my provisional characterization of objectivity: a claim is objectively correct iff it correctly represents mind-independent reality. Compare this with Brandom’s own take:

“The objectivity of conceptual norms requires that any attitude of taking, treating, or assessing as correct an application of a concept in forming a belief or making a claim be coherently conceivable as mistaken, because of how things are with the objects the belief or claim is about” (1994, p.63).

Given that the primary way in which we apply conceptual norms is in asserting a claim, Brandom would, I think, assent to my characterization of objectivity, albeit in a qualified manner. The qualification would come in at least three ways. The first and most obvious qualification would be that the above objectivity thesis is not to be interpreted as a *metasemantic* claim: namely, as a metalinguistic statement specifying the semantic representational content of the sentence in the object language, akin to Tarski’s celebrated T-schema and its appropriation by Davidson in his account of meaning (viz.

truth-conditional semantics). The truth condition of a claim is never settled in the sense that on one is, nor would they ever be, in an epistemic position to assert, for arbitrary sentence *S*, the claim “*S* is objectively true” without being subject to challenges or being exonerated from the possibility of error. In other words, objectivity does not consist of a theoretical rendering of representational content from a perspective-free ‘view from nowhere.’

This brings us to the second qualification: the representational content of an expression²³ in general, and the meaning of the predicate ‘correctly represents mind-independent reality’ (i.e. the classical interpretation of the truth predicate) in particular, is to be explained in terms of its *expressive roles*, namely, as locutions that allow speakers to make explicit exactly what someone’s inferential commitment is, in virtue of which deontic scorekeeping is done (as in the example given above). As such, an expression’s representational content is ultimately explained in terms of the expression’s role in deontic scorekeeping practices.²⁴

1.2.3 Three Kinds of Semantic Objectivity

Before moving onto objections to Brandom’s account of objectivity, let us now summarize the discussion up until now. Reinikainen (2021) offers a useful typology of the notions of objectivity discussed by Brandom:

- (AI) A norm *n* is attitude-immanent for community *C* iff it is not possible for everyone in *C* to be mistaken about the correct applications of *n*.
- (AT) A norm *n* is attitude-transcendent for community *C* iff it is possible for everyone in *C* to be mistaken about the correct applications of *n*.
- (PO) A norm *n* is properly objective for community *C* iff the world of objects partially determines the correct applications of *n*.

²³This includes including that of subsentential expressions and, in Brandom’s terminology, ‘*De re* ascription’ of propositional content.

²⁴Brandom promises us an inferentialist explanation of representational content and of a realist notion of objective truth, but strictly speaking, this is inaccurate; he explains representation by appealing to his pragmatic theory, which can, in principle, be severed from his inferentialism. See footnote 24 for more.

The weakest form of objectivity is AI. Mostly this form of objectivity applies to non-discursive norms such as norms of etiquette. For such norms, “it makes no sense to suppose that [the community] could be wrong about this sort of thing” (Brandom, 1994, p.53). This form of objectivity is clearly insufficient for Brandom’s aim of avoiding relativistic consequences about norms and thus rational practices.

Next, AT stands in contrast to AI in that the conditions for applying norms that satisfy AT are not necessarily and sufficiently determined as correct by the linguistic community (or a subgroup of experts therein). In other words, the consensus of the linguistic community can be wrong about circumstances under which norms are applied. Importantly, Brandom says that norms governing the inferential role of expressions fall under this category: they are distinct from AI norms precisely because only conceptual norms are rightly called attitude-transcendent (1994, 53-54). In this sense, AT is a more demanding criteria for norms than AI.

Finally, PO states that at least part of determinant of correct concept application is the mind-independent world. It is by far the strongest form of objectivity considered here. To satisfy PO, the negative view of objectivity, as consisting in acknowledgement of the possibility of error even from the point of view of the linguistic community, is not sufficient; the norms themselves must be subject to correctness of the representational content determined by the world of objects.

As Reinikainen (2021, p.63-66) points out, according to Brandom, the three kinds of semantic objectivity are layered and continuous, in the sense that AT is grounded on AI and PO on AT. The original source of semantic normativity, according to Brandom, is the normative attitudes (of ‘taking-true’) of the speakers. This is a view he calls *phenomenalism about norms* (1994, 25). Once norms are sufficiently entrenched (or ‘instituted’) in practice, then they satisfy AI. Then, once the primitive discursive practice develops sufficiently, the thought goes, some norms—namely conceptual norms—begin to satisfy AT. As mentioned earlier, the explanation for the shift from AI to AT is given essentially along Davidsonian lines. This is when discursive practice—or deontic scorekeeping practice—can be said to be fully up-and-running. It is also at this point, according to Brandom, that norms begin to satisfy PO: in recognizing the possibility of error, speakers also rec-

ognize that the source of semantic normativity is not just their normative attitudes. The source is extended to the world of objects, such that the objects come to be ‘incorporated’ in the practice and as such begin to dictate what our conceptual norms should be. So, while semantic normativity is, at base, grounded in normative attitudes, the recognition that the world in part ought to shape conceptual norms allows the world to exert normative authority over our discursive practices. This is the view Brandom calls *normative phenomenalism* (ibid., p.627).

1.3 The Problem with Brandom’s Account

Brandom’s account of objectivity is persistently attacked by critics from all kinds of different considerations. Here, I will develop two such objections: i) that Brandom wants to endorse PO, but in fact his explanation, mainly consisting in his objectivity proofs, only support AT and ii) that if Brandom wants to (forcibly) explain PO in terms of incompatibility, he needs not only an *deontic* modal, but an *alethic* modal notion of material incompatibility relations. In either case, Brandom cannot vindicate objective representational content, it comes without heavy metaphysical commitments.

1.3.1 Objection 1: Objectivity proofs yield AT, but not PO

Brandom promises us accounts of AT and PO which, according to him, naturally emerge in deontic scorekeeping practices. In particular, Brandom writes that his objectivity proofs show that “the facts as construed in this work are independent of what anyone or everyone is committed to” (i.e., AT), which as a result make possible “genuine fact-stating discourse, for they incorporate practices of asserting claims and inferences according to their objective correctness—a kind of correctness that answers to how things actually are, rather than how they are taken to be, by anyone (including oneself) or everyone” (i.e., PO) (1994, p.606-607).

But, as various commentators have noted under different guises, Brandom’s account of objectivity really amounts to AT and fails to deliver on PO.²⁵ In particular, even if we

²⁵See, for example, Levine, 2010; Gibbard, 2010; Reinikainen, 2021.

assume that Brandom's objectivity proofs go through and that the threats of the *No First-Person Ignorance* condition and the *No First-Person Error* condition have been dealt with, it still does not show that PO holds. Let us assume that AT holds, namely, in virtue of the fact that any one perspective can in principle be wrong.

But how do we get from AT to PO? Brandom suggests that acknowledging the possibility of error of any given perspective implies an acknowledgement of the normative force (or 'authority' in Brandom's terminology) of something other than the normative attitudes that institute social norms: namely, objective reality. To illustrate the general idea, consider Brandom's favorite example of how PO might emerge out of AT (1994, p.332-333):

1. The subject has a normative attitude such that she is disposed to draw the inference "If some substance tastes sour, it will turn litmus paper red."
2. The subject perceptually experiences a substance that tastes sour.
3. The subject then sees that in fact the substance turns litmus paper blue.
4. The subject acknowledges her practical failure, nixing her commitment to infer "If some substance tastes sour, it will turn litmus paper red."

In this example, the subject's normative attitudes undergo a change owing to the subject's experience of a chain of causal events. So, normative phenomena are intertwined with causal phenomena. The example undoubtedly shows that the world enforces causal constraints on our normative attitudes, in the sense that certain physical changes in the world caused changes in the subject's disposition to draw certain inferences or accept certain claims. But, this causal constraint is describable in a purely naturalistic vocabulary; it is far from obvious how the world can *normatively* constrain our normative attitudes, that is, how facts can *justify* the subject's moves in the deontic scorekeeping model. The subject *ought* to alter her commitments given her acknowledgement of her own error; the fact that her disposition to draw certain inferences changed due to causal events in the world does is just that: a fact. A fact could only exert normative force (the force of

'ought') on the subject only on her acknowledgement that inferences (and thus concepts) she endorses ought to be in accordance with facts.

But PO requires the claim that objects themselves constrain norms: if we want to say that, for instance, norms governing the concept 'litmus paper' is objective in the sense specified by PO, we need an explanation of why/how litmus paper (the physical object) can partly set the standard of *correctness* of applying such norms. Since correctness must always be cashed out in normative terms, it is at best unclear, on Brandom's explanation, how objects (or perceptual experience) themselves, rather than our acknowledgement of the importance of them, can exert normative force on us.

Brandom goes on to tell us, rather programmatically, that in order to understand how the world can normatively constrain us, we must give up the opposition between reality on one side and thought/talk on the other. Brandom appeals to a Tractatrian picture of the world as composed of states-of-affairs ("The world is everything that is the case, a constellation of facts" (1994, p.333)) and quotes with approval the following aphorism from the *Investigations*: "When we say, and mean, that such-and-such is the case, we-and our meaning-do not stop anywhere short of the fact; but we mean: this-is-so." The discussion remains extremely opaque here, but Brandom's stance seems to suggest a kind of metaphysical idealism, wherein thought/talk is identified as such with the world. "What is lost" by assuming this metaphysical thesis, according to Brandom, "is only the bifurcation that makes knowledge seem to require the bridging of a gap that opens up between sayable and thinkable contents-thought of as existing self-contained on their side of the epistemic crevasse-and the worldly facts, existing on their side" (ibid.).

Brandom elaborates on these programmatic remarks in subsequent works, most notably on his recent works grappling with Hegel's idealism.²⁶ But this is not the place to evaluate Brandom's metaphysical views. We will touch on this topic when we get to Brandom's construal of the principle of harmony in Chapter 3. For now, it suffices to notice that such a radical metaphysical stance requires more substantiation and justification than a mere few lines of allusion to Wittgenstein can deliver. In fact, if indeed Brandom's account of objectivity relies on his quasi-idealistic metaphysics, he fails to deliver on the

²⁶Cf. R. Brandom (2019)

promise of accounting for objectivity from the theoretical resources afforded by his deontic scorekeeping model of discursive practice. That is, his main project of the second half of *Making It Explicit*, namely to derive an account of objectivity and objective representational content by taking the pragmatic model of discursive practice as the sole theoretical primitive, falls short by his own lights.

1.3.2 Objection 2: Scepticism about Incompatibility Semantics

The trouble for Brandom's account of objectivity doesn't end there. Recall the two inferences (viz. No first-person ignorance condition and no first-person error condition) that threaten the very possibility of objectivity (conceived of as AT) within the deontic scorekeeping model. The content of Brandom's objectivity proofs is to show that, in each of the two cases, the antecedent and the consequent do not share the same extension for incompatibility relations. Given Brandom's account of objectivity relies heavily on the notion of material incompatibility, any criticism raised against incompatibility semantics threatens to take down Brandom's account of objectivity with it.

Once again, incompatibility is explained pragmatically in terms of the interplay of commitments and entitlements: p are q incompatible iff a commitment to p precludes entitlement to q and vice versa. Now, the immediate question is what this 'preclusion' amounts to. On the most obvious reading, the nature of this preclusion should be understood in *deontic* modal flavor: a commitment to p makes it *impermissible* to be entitled to q , and vice-versa. In other words, given an antecedent commitment to p , one ought to be not committed to q and vice-versa. But, this simple analysis collapses the two kinds of deontic status—i.e. commitment and entitlement—into one, namely as commitments. So, this brings us to an interpretative problem: why do we need to appeal to the interplay between 'commitment' and 'entitlement' to cash out the notion of incompatibility?

The answer is that in order to fully appreciate the difference between the deontic statuses, one must understand that, for Brandom, assertions are "implicit knowledge claims" (1994, p.200); and, to cash out the notion of knowledge, one must appreciate the distinction of social perspective between *attributing* a commitment (to someone else)

and *acknowledging* or undertaking it (oneself). On one hand, a subject's (S) attribution of a commitment (p) to another subject (A) amounts to the claim, 'S is committed to 'A is committed to p '; on the other hand, (S)'s acknowledgment of A's commitment (p) amounts to the claim, 'S is committed to p .' Leveraging the distinction between attribution and acknowledgment, Brandom can refashion the classic tripartite analysis of knowledge as justified true belief in his deontic scorekeeping model in the following way: attribution of knowledge to someone as a conjunction of attribution of a commitment (corresponding to the belief condition), attribution of an entitlement (corresponding to the justification condition), and acknowledgement of a commitment oneself (corresponding to the truth condition). Therefore, as Brandom notes, "the distinction between the norm of truth and the norm of warrant only becomes practically visible when one is keeping score on someone else's deontic statuses. For it is only from that point of view that belief, truth, and warrant can come apart" (2010, p.361-362).

So, the correct pragmatic analysis of incompatibility relation is necessarily multi-perspectival, or, more specifically, dialogical.²⁷ So, it is rather misleading to define incompatibility in a monological (or uni-perspectival) way as 'a commitment to p precludes entitlement to q and vice versa.' But while the deontic modal rendering of incompatibility (i.e. in terms of rights and obligations) secures us the distinction needed for AT, it is once again insufficient for PO. The deontic reading secures us AT because it allows us to draw a distinction between the propositional content of objective statements (like 'The swatch is red') and non-objective statements (like 'That the swatch is red is now assertible by me'). Showing that assertibility conditions on their own do not determine content (i.e. consequence of application is just as important) is enough to show that at least some conceptual norms are not dependent on any particular discursive perspective. However, to vindicate the claim that content is partly determined by objects themselves (PO), you need more than that. As Hale & Wright (2010) note,

²⁷Brandom is at times very clear on this point: "The conceptual contents employed in monological reasoning, in which all the premises and conclusions are potential commitments of one individual, are parasitic on and intelligible only in terms of the conceptual contents conferred by dialogical reasoning, in which the issue of what follows from what essentially involves assessments from the different social perspectives of interlocutors with different background commitments" (1994, p.497).

“In order merely to distinguish the members of the relevant kind of objective/non-objective pairings, Brandom has only to show that certain of their inferential connections are somehow different. However merely somehow distinguishing objective statements from (putatively) co-assertible non-objective ones is not enough. In addition, they have to be distinguished fully and correctly: the Assertibilist has to show specifically how he can make a contrast between the members of such pairs which is a correct reflection of the conceptual relations which the propositions in question actually respectively bear to propositions at large. The challenge is not merely to get some kind of distinction going between objective/non-objective pairs but to track the detail of the conceptual compatibility/incompatibility relations which they respectively actually sustain across contents in general” (p.282).

So: while the deontic modal rendering of ‘preclusion’ (i.e. in terms of rights and obligations) secures us the distinction needed for AT, it is once again insufficient for PO. For, what is needed in order to vindicate PO is a alethic modal rendering (i.e. in terms of possibility and necessity), which allows us to define the semantic notion of *inconsistency* in terms of the pragmatic notion of incompatibility. If Brandom’s explanation for objectivity can be read so as to vindicate PO, the incompatibility between, for example, ‘That the swatch is red is now assertible by me’ and ‘Rational beings never existed’ must hold in virtue of the world, or, given that Brandom echoes the Wittgensteinian claim that the world is composed of facts, in virtue of objective facts that make up the world. In other words, the incompatibility between the above claims must be involving an impossibility, understood alethic modally: if p and q are incompatible, then it is impossible for $p \wedge q$ to be true.

Brandom seems to be aware of this issue. Despite his insistence that incompatibility can only arise in a dialogical setting (i.e. in the context of ‘I-thou’ sociality), Brandom is open to the idea that a semantics, understood monologically without explicit reference to the underlying pragmatic context, can be abstracted and coherently articulated as a formal semantics in the form of incompatibility semantics (although the pragmatic layer would necessarily be part of this theory, only implicitly). Already in *Making it Explicit*,

Brandom claims *en passant* that p and q are *inconsistent* given that they stand in material incompatibility relation, according to which inconsistency is understood as a formal semantic notion grounded on a more primitive material notion of incompatibility, analogue to how formal inference is explained in terms of material inference (1994, p.160, p.672). But the idea that incompatibility relations can yield a formal semantics comes to fruition in *Between Saying and Doing* (2008). The suggestion there is that we could represent the propositional content expressed by a sentence by the set of sentences that express propositions incompatible with it (or, more precisely, represent propositional content as a function that takes as inputs set of sentences and output the set of sets of sentences that are incompatible with it). Incompatibility is cashed out, alethic modally,²⁸ along the following lines: p and q are incompatible iff it is impossible for p and q to hold simultaneously. Furthermore, for Brandom, incompatibility semantics is ‘directly modal’ by which he means, “it does not proceed by defining (non-modal) truth at an index, and then generalizing over indices.” Here again, lurking in the background is Brandom’s idealism.²⁹

Rosenkranz (2001) provides an interesting critique of Brandom’s account of objectivity based on this alethic modal reading of ‘preclusion’, where the tension between the two claims are cashed out as inconsistency. According to Rosenkranz, “it is part of our ordinary conception of truth as objective that there are some claims which it is not inconsistent to think true even while there are no warrants on whose basis they may be asserted” and that “this much even the anti-realist is willing to concede” (p.236).³⁰ For example, consider Goldbach’s conjecture: it is possible to imagine that it is true (or that it is false), even if we have no way of justifying it (or its negation) and therefore asserting it (or its negation). Accordingly, Rosenkranz thinks Brandom is minimally committed to the following claim:

If p is apt for objective truth, then ‘ p & $\neg Wp$ ’ is not inconsistent

²⁸This might be a controversial claim, but I think nonetheless it must be attributed to Brandom; if incompatibility is still understood as a deontic notion even at the semantic level, it is subject to an obvious objection along the lines of Rosenkranz (2001). Rosenkranz’s argument will be evaluated in what follows.

²⁹There are many obvious objections to this project. In particular, Brandom rejects any kind of possible-worlds semantics for modal terms, and it is unclear whether his account adequately captures the meaning of modal terms—he says it can; others (e.g. Turbanti (2011)) say it cannot. But here I will merely flag this issue.

³⁰Here, he cites Crispin Wright, Dummett, and Brandom himself to substantiate the claim.

where Wp is read as ‘There are no warrants for p .’ Then, Rosenkranz proves that according to Brandom’s characterization of incompatibility in terms of entitlements and commitments, there are no claims that are apt for truth. In particular, the proof shows that Brandom’s definition of commitment, entitlement, and incompatibility entails the following two conditionals:

$$1. (Cp \& E \neg Wp) \rightarrow \#$$

$$2. (C \neg Wp \& Ep) \rightarrow \#$$

Here, Cp is read as ‘a commitment to p is undertaken,’ Ep as ‘There is an entitlement to p ,’ and $\#$ as a symbol for inconsistency, “which is akin to that for *Falsum* but whose exact interpretation I leave open” (Rosenkranz, 2001, p.236). To give a concrete example, the first conditional might be the something like: the commitment to ‘The swatch is red’ precludes entitlement to ‘It is not the case that the swatch is red is properly assertible by me now’. The second conditional would then be: the commitment to ‘It is not the case that the swatch is red is properly assertible by me now’ precludes entitlement to ‘The swatch is red.’ So, ‘The swatch is red’ and ‘It is not the case that the swatch is red is properly assertible by me now’ is inconsistent. Given the definition of incompatibility, this entails that for any arbitrary p , ‘ $p \& \neg Wp$ ’ is inconsistent. Even informally, this much is intuitive.

I will not go through the proof in detail, contenting myself with two observations. The first is that Rosenkranz’s proof arguably shows not only that PO cannot be sustained within Brandom’s theory, but also AT. If indeed Brandom’s characterization of the deontic scorekeeping practices entails the claim that ‘ $p \& \neg Wp$ ’ is inconsistent in the sense that it is impossible for a claim to be true and none of us have justification for it, it seems to imply that we could never be wrong about anything we assert with warrant. (It should be noted that Rosenkranz’s argument is reminiscent of the infamous *knowability paradox* due to F. Fitch (1963) which plagues most anti-realist views, which says that “all truths are knowable” entails “all truths are known.”) And of course, these are not consequences that Brandom would accept. By now it should be quite obvious that the source of these troubles is the alethic modal construal of preclusion (i.e. in terms of inconsistency) which

completely drops the inherently deontic and dialogical dimension of material incompatibility. Understood deontically, the incompatibility of ' $p \ \& \ \neg Wp$ ' is straightforwardly analyzed as a typical example of structural irrationality: one ought not to endorse p given her antecedent commitment to $\neg Wp$, and vice-versa. This is completely uncontroversial, as it says nothing about objective facts.

However, given that Brandom wants to account for PO and doing so requires an alethic modal reading of preclusion, it is admittedly not clear where exactly the deontic dimension should fit in. Yet again, Brandom's arguments seem to trade on his metaphysical view developed most fully in R. Brandom (2018), under which there is a certain kind of identity between deontic and alethic modalities. Here it suffices to note that such a story incurs heavy metaphysical commitments, and that no such story is available within *Making it Explicit*, nor is it possible to tell it within the theoretical aims of the book.

Second, it should be noted that Rosenkrantz's proof is valid in classical logic, but not in intuitionistic logic: his proof makes use of *reductio* (but, of course, it is possible to create a *reductio*-free proof, only we would need to introduce the law of excluded middle or other intuitionistically invalid theorems). As noted earlier, Brandom, unlike most other inferentialists, is logically quietist: he endorses an expressivist view of logic, according to which logical vocabulary is distinguished by its 'expressive role' in *Making It Explicit* the inferential relations that articulate the semantic contents of the concepts expressed by the use of material concepts. If one offers this logically expressivist, semantically inferentialist answer to [the question, "how to distinguish between logical and non-logical vocabulary?"], the [question "which logic is correct?"] lapses" (Brandom, 2018, p.70). So, Brandom thinks that classical and intuitionistic logic are compatible, in the sense that they both help us make our commitments explicit, albeit in different contexts. This is a point of tension between Brandom and most other inferentialists, who think that intuitionistic logic is the correct (or, at least, more coherent) logic in virtue of the fact that operational rules for some classical connectives are inharmonious. This point will be developed further in the following two chapters.

1.4 Conclusion

The purpose of this chapter was three-fold: i) introduce readers unfamiliar with Brandom's inferentialism to its basic tenets, ii) provide a detailed exegesis of Brandom's take on objectivity, and iii) identify some issues with Brandom's account of objectivity. With regards to iii), the upshot is that PO cannot be coherently defended, at least, by relying on resources on Brandom's main (i.e. non-ad hoc) theoretical assumptions. To generalize slightly, Brandom's realist ambitions seem to stand in tension with his commitment to inferentialism; trying to account for PO even risks giving up AT, which is an aspect of objectivity that, in my view, Brandom is able to account for.

But why does Brandom want to account for PO? As far as I can tell, there is little to no such need.³¹ If I am correct in identifying Brandom's main concern as preventing our discursive practices to go off the rails (i.e. something that Rorty-style relativism might imply), an account of objectivity is, while sufficient, not strictly necessary. All we need to avoid the charge of relativism is an adequately articulated normative constraint on meaning-conferring linguistic rules in order to discriminate between defective rules (and thus incoherent concepts) and legitimate rules (and thus coherent concepts). As we will see in the next chapter, logical inferentialists try to articulate such a normative constraint for rules in the form of the principle of harmony. The full relevance of the principle of harmony for natural language will not be clear until the third chapter.

³¹Besides, in my view, it is probably a lot less metaphysically taxing to argue for PO by incorporating insights from the empiricist tradition, i.e. the argument that perceptual experience can normatively constrain our beliefs and assertions. I will touch upon this in my last chapter, but a full treatment of this topic is beyond the scope of this project.

Chapter 2

Harmony: The Traditional Debate Revisited

Logical inferentialism (sometimes called *proof-theoretic semantics*¹) attempts to explicate the meaning of logical constants in terms of their use in deductive practice. Inferentialists typically distinguish between two types of use of a sentence that determine its meaning: the conditions for application of the concept (paradigmatically in assertion), and the consequences that follow from the application of such a concept. According to such a *two-aspect model* of language use, specifying the conditions for and consequences of the application of a given concept is sufficient to determine the meaning of such a concept. As Dummett (1973) puts it,

“crudely expressed, there are always two aspects of the use of a given form of sentence: the conditions under which an utterance of that sentence is appropriate, which include, in the case of an assertoric sentence, what counts as an acceptable ground for asserting it; and the consequences of an utterance of it, which comprise both what the speaker commits himself to by the utterance and the appropriate response on the part of the hearer, including, in the case of assertion, what he is entitled to infer from it if he accepts it.”(p. 396)

¹This term was coined by Schroeder-Heister (1991). Strictly speaking, proof-theoretic semantics is broader than logical inferentialism due to its ambition to give a semantics for non-logical concepts (paradigmatically mathematical ones). Nonetheless, they operate on the same meaning-theoretic assumptions and goals.

Interpreting the two-aspect model in the natural deduction setting, the information specifying the conditions for application of a given logical constant is encoded in its introduction rule (I-rule). The I-rules specify the circumstances under which it is correct to introduce into the discourse (or proof) the complex formula whose principal operator is the logical constant in question. The consequences of application, on the other hand, include everything that is derivable from a given formula. In particular, the elimination rule(s) (E-rule) of a logical constant specify what we can correctly derive from a formula where such a constant figures in as the principal operator. Thus, introduction rules and elimination rules together completely determine the meaning of a logical constant. For example, the meaning of the constant $\rightarrow$ is understood to be captured by the following rules:

$$\frac{[A] \quad B}{A \rightarrow B} \rightarrow\text{-I} \quad \frac{A \quad A \rightarrow B}{B} \rightarrow\text{-E}$$

One of the first major objection to the logical inferentialist project is due to Prior (1960), where he (implicitly) gave a *reductio* argument against logical inferentialism by inventing the logical connective *tonk* governed by the following pair of operational rules:

$$\frac{A}{A \text{ tonk } B} \text{tonk-I} \quad \frac{A \text{ tonk } B}{B} \text{tonk-E}$$

Assuming that transitivity holds—as it should—in the proof system to which it is added, *tonk* allows us to derive any arbitrary *B* for any arbitrary *A*. In other words, the addition of *tonk* to any proof system renders it deductively trivial. Of course, friends and foes of inferentialism alike agree that *tonk* is not a legitimate logical operator. But, Prior seems to argue, if meaning is wholly determined by its operational rules, why shouldn't *tonk*-like connectives count as legitimate and meaningful? From this, Prior's concludes that the inferentialist meaning-theoretic enterprise is untenable.

The objection amounts to the following: inferentialists are necessarily committed to conventionalism about logical laws and thus relativism about logical normativity. The usual inferentialist response to 'tonkitis' is to insist that the pair of operational rules that govern *tonk* are defective. Such introduction and elimination rules, according to inferen-

tialists, are unjustified because they are not in *harmony* (a term coined by Dummett (1991)). Now, how exactly to spell out the idea that some rules are ‘defective’ or ‘inharmonious’ is still widely discussed. Because inferentialists are divided on the philosophical motivation for the requirement for harmony, we find in the literature various ways to formally cash out this idea.

This chapter contains three sections. In the first, I review the original articulation of the requirement of harmony as a local property of the pair of inference rules due to Gentzen (1969 [1935]) and developed in the proof-theoretic tradition by Prawitz (1965) and others. In the second section, I consider harmony as a global constraint on the deductive system as a whole; this view is originally due to Belnap (1962), but has enjoyed influential support from Dummett (1991). Finally, in section 3, philosophical motivations for the principle of harmony are considered. Despite the efforts of logical inferentialists, I argue that there is not a clear positive reason to adopt it.

2.1 Harmony as a Local Property

Prior’s challenge primarily concerns inferentialists that subscribe to the two-aspect model of use, as the *tonk* connective exploits the possibility of the ‘mismatch’ between the introduction and elimination rules to render the proof system deductively trivial. The original inferentialist position, however, is immune to Prior’s objection as it subscribes to a *one-sided model*, under which the two kinds of use (viz. conditions for concept application and the consequences for doing so) do not contribute equally to the meaning: the introduction rules are understood to fix the meaning and thus to determine the corresponding elimination rules. In the paper that launched the logical inferentialist tradition, Gentzen (1935; English translation in 1969) proposed that the meaning of logical constants is captured by rules that introduce them into a proof. According to him,

“To every logical symbol $\&, \vee, \forall, \exists, \rightarrow, \neg$, belongs precisely one inference figure which ‘introduces’ the symbol—as the terminal symbol of a formula—and which ‘eliminates’ it. [...] The introductions represent, as it were, the “definitions” of the symbols concerned, and the eliminations are no more, in the

final analysis, than the consequences of these definitions. This fact may be expressed as follows: In eliminating a symbol, the formula, whose terminal symbol we are dealing with, may be used only “in the sense afforded it by the introduction of that symbol.” [...] By making these ideas more precise it should be possible to display the E-inferences as single valued functions of their corresponding I-inferences, on the basis of certain requirements.” (Gentzen, 1969, p. 294-295)

An example might help illustrate what Gentzen has in mind. Take the pair of operational rules for the conditional described in the previous section. According to $\rightarrow$ -I, we are permitted to endorse (in judgment or assertion) $A \rightarrow B$ only if we can derive B from assuming A. Thus, what the conditional makes explicit in the judgment is this inference and its hypothetical nature. What $\rightarrow$ -E (also known as *modus ponens*) prescribes is that once there is a deduction of A, together with the fact that we can infer B from assuming A, we can directly endorse B. As such, elimination rules’ sole function is to make explicit what we are entitled to in the first place when we introduced $\rightarrow$ in accordance with its meaning-determining introduction rule.

According to Gentzen’s one-aspect model, then, i) introduction rules are definitional and thus self-justifying and ii) elimination rules are justified only if its application yield judgments whose grounds are already contained within grounds for applying the introduction rules.

2.1.1 Inversion Principle and Intrinsic Harmony

Prawitz (1965) develops Gentzen’s programmatic remarks in his landmark study on natural deduction. He proposes that the relationship between the I- and E-rules can be captured in what he, following Lorenzen (1955), calls the *inversion principle*:

“Let α be an application of an elimination rule that has B as consequence. Then, deductions that satisfy the sufficient conditions [...] for deriving the major premises of α (if any), already “contain” a deduction of B; the deduction

of B is thus obtainable directly from the given deductions without the addition of α ." (Prawitz, 1965, p.33)

This principle is closely related to Gentzen's suggestion that the sole function of elimination rules is to 'undo' the inference made in accordance with the corresponding introduction rule. Importantly, the inversion principle guarantees that no novel information is produced when applying the elimination rules. For instance, let's take modus ponens ($\rightarrow$ -E). The inversion principle tells us that the conditions for inferring $A \rightarrow B$ (i.e. the conditions for applying $\rightarrow$ -I, which is a deduction of B from $[A]$) and a deduction of minor premise A simply constitutes the deduction of B . That is, once a deduction of B from $[A]$ and a deduction of A is available, we can directly infer B without applying $\rightarrow$ -I and subsequently $\rightarrow$ -E. This is because when the application of an introduction rule serves as a major premise of an elimination rule (which is the case for modus ponens), the deduction takes a detour which can be removed. Once all such detours are removed from the deduction, it is said to be *normalized* (or in the normal, or canonical, form).

The formal specification of this principle takes the form of the following *inversion theorem*:

"If $\Gamma \vdash A$ then there is a deduction of A from Γ in which no formula occurrence is both the consequence of an application of an I-rule and major premise of an application of an E-rule." (Prawitz, 1965, p.34)

Let us call any formulae that occurs as the consequence of an application of an I-rule immediately followed by the application of an E-rule the *maximal formulae*, as it is the most logically complex formula compared to others in the surrounding deductive path. Prawitz then goes on to prove through induction that this inversion theorem holds for major deductive systems, including classical, intuitionistic, second-order, and modal logics.² In particular, Prawitz shows that for any (non-defective) pair of rules for a connective, there exists a procedure through which the deduction containing a maximal formulae

²The inversion theorem needs a slight addition to deal with the complication that arises with the $\perp$ rule for classical logic. While this fact may have bearing on the debate between classical and intuitionistic logicians, it is inconsequential for our purposes. Cf. Chapter 11 and 12 in Dummett (1991) for an attack on classical logic based on similar considerations.

can be reduced to one that does not contain one. Dummett refers to these reduction procedures as “levelling local peaks” (1991, p. 250) and operational rules that allow us to level local peaks as being in *intrinsic* harmony. Intrinsic harmony is a property belonging to the pair of rules governing the connective, and thus is *local*. Later, this local notion of harmony will be contrasted with a *global* notion of harmony which will be explained as the property of the deductive system as a whole. But first, let’s take the case of the conditional to see how the leveling procedure works:

$$\begin{array}{c}
 \Pi_1 \quad \frac{A \quad \frac{\frac{[A] \quad \Pi_2}{B} \rightarrow I}{A \rightarrow B} \rightarrow E}{B} \rightarrow E \quad \Pi_3 \\
 \Pi_1 \quad \frac{[A] \quad \Pi_2}{[A] \vdash B}
 \end{array}$$

The deduction on the left side contains a maximal formulae, namely $A \rightarrow B$, but this peak can be leveled by arranging the demonstration of A (i.e. Π_1) and of $[A] \vdash B$ (i.e. Π_2) in a more linear fashion, as shown on the right side. Since there is a leveling procedure for the pair of rules governing $\rightarrow$, we have shown that they exhibit local harmony.

In like manner, Prawitz repeats this process with the four other connectives of first-order predicate logic. This inductive step of the proof of the inversion theorem, which utilizes leveling procedures, is integral to the main result of Prawitz’s study, known as the *normalization theorem*: that every derivation in a system of natural deduction can be transformed into a normal form.³ This result is, in essence, the natural deduction equivalent to Gentzen’s *Hauptstaz* (also known as the *cut-elimination theorem*) for systems of sequent calculus. Indeed, as Prawitz and Gentzen (1938) himself notes, the *Hauptstaz* is also based on the insights articulated in the inversion principle.

But where does the requirement for harmony, conceived of as an existence of a reduction procedure, fit within the inferentialist meaning-theory? According to Martin-Löf (1996), “the place where [reduction procedure] belongs in the meaning theory is precisely in the semantical explanation, or justification, of the elimination rule” (p.33). Since, ac-

³It should be noted that leveling procedures, while integral to the proof of the normalization theorem, do not on their own establish it since the reduction of non-normal derivation to its normal form sometimes involve rearranging the deductive path to create a maximal formula in case where the succession of introductions and eliminations is not immediate. This procedure is known as *permutation*.

According to the Gentzen-style one-sided model, introduction rules are self-justifying, the existence of a reduction procedure completes the justification of the pair of rules: expressions governed by intrinsically pair of rules can thus be considered fully justified.

To conclude this section, let us see how intrinsic harmony can help us with Prior's challenge. It should be clear that the `tonk` connective does not have a leveling procedure. In particular, Prior's runabout inference (i.e. the 'deduction' of an arbitrary B from an arbitrary A via the application of `tonk-I` and `tonk-E`) has a maximal formulae, namely $A\text{tonk}B$, that occurs both as the consequence of application of an I-rule and as the major premise of an application of an E-rule; but, this deduction cannot be reduced to a normal form that avoids the detour since no such deduction exists. The problem with `tonk` can be thus understood as the following: its elimination rule outstrips (i.e. asserts more than what is granted by) the grounds for applying its introduction rule. As such, the rules that give meaning to the `tonk` constant do not exhibit local harmony. The explanation for this behavior is that the condition for applying `tonk-I` (namely, the proof of A) does not contain the grounds to judge B . So, the proof-theoretic diagnosis of the problem with `tonk` comes down to the following: the elimination rule for `tonk` is, given its introduction rule, unjustified.

2.1.2 Stability

Having shown tentatively that intrinsic harmony blocks `tonk`, we may now ask broader questions about Gentzen-style one-sided semantic model. Firstly, one might wonder why introduction rules are taken to be semantically primitive, as opposed to the elimination rule. In other words, why shouldn't elimination rules fix the meaning? Indeed, some have suggested that in the case of the conditional (Rumfitt, 2000, p.790), the universal quantifier (Dummett, 1991, p.275) and the disjunction operator (Tennant, 1987, p.90), it would be in some sense more natural to grant elimination rules definitional priority. Those working to vindicate proof-theoretic semantics often follow Gentzen in adopting the priority of the introduction rules over elimination rules, but seem to take this idea for granted. The choice to grant the introduction rules a kind of semantic priority requires justifica-

tion, which is surprisingly missing from the vast literature on proof-theoretic semantics. We will provide a natural explanation for the priority of the introduction rules in the following chapter, drawing on the fact that we (i.e. rational agents) are embodied creatures that draw inference in time and are thus subject to temporal constraints.

Furthermore, granting that intrinsic harmony is a necessary property for any purportedly meaning-conferring pair of operational rules, we might still ask whether it on its own is constitute the sufficient condition. In other words, can intrinsic harmony rule out all *tonk*-like constants? In an influential argument, Dummett shows that intrinsic harmony is not sufficient in guaranteeing that the semantic properties of a constant is well-behaved. In particular, he concocts a counterexample by considering the case of the quantum logical disjunction.

The quantum-logical disjunction ($\ddot{\vee}$) differs from ordinary disjunction ($\vee$) in that its elimination rule does not allow for collateral hypotheses to be part of the minor premises. Since this is the sole difference, Prawitz’s demonstration that a reduction procedure exists for $\vee$ —and is therefore intrinsically harmonious—can be modified slightly to show that $\ddot{\vee}$ also is intrinsically harmonious. Dummett then imagines a system, S , containing $\ddot{\vee}$ and other intrinsically harmonious constants, with the notable exception $\vee$. Let’s say that we want to create an extension of this system, S' , by adding $\vee$ to it. All of the constants that of this system is intrinsically harmonious. Yet, as Dummett notes, something goes awry. In S' , by exploiting $\vee$ -I and $\ddot{\vee}$ -E, we can derive $A \vee B$ whenever we have $A \ddot{\vee} B$, thus collapsing quantum disjunction into ordinary disjunction:

$$\frac{A \ddot{\vee} B \quad \frac{[A]}{A \vee B} \vee\text{-I} \quad \frac{[B]}{A \vee B} \vee\text{-I}}{A \vee B} \ddot{\vee}\text{-E}$$

In other words, in S' , the conditions for asserting $A \ddot{\vee} B$ is sufficient to assert $A \vee B$. What exactly has gone wrong here? In short, the problem is that the elimination rule for $\ddot{\vee}$ is excessively restrictive in comparison to its introduction rule; the conditions for asserting $A \ddot{\vee} B$ is stronger than what the consequence of applying $\ddot{\vee}$ -E makes explicit. So, like *tonk*, there is a mismatch between the introduction and the elimination rule, only the mismatch is of opposite nature. Whereas *tonk*-E is exceedingly permissive with respect to *tonk*-I, $\ddot{\vee}$ -E is exceedingly restrictive with respect to $\ddot{\vee}$ -I.

So far, the diagnosis has lead to an identification of a local problem, i.e. for the pair of operational rules for $\ddot{U}$. But the addition of $\vee$ to S creates problems at the *global* level as well, i.e. for the deductive system as a whole. For one, S' validates formula that are invalid in S ; most notably, while the law of distributivity— $A \wedge (B \ddot{U} C) \iff (A \wedge B) \ddot{U} (A \wedge C)$ —is invalid in S , it becomes derivable for $\ddot{U}$ in S' .⁴ S' can thus be described as a *non-conservative extension* of S . Conservative extendibility is a key notion that serves as the basis for the requirement for harmony for Dummett and Belnap (1962); more on this will be said in the next section.

What's more, as Steinberger (2011) notes, a reduction procedure for a proof that contains both $\vee$ and $\ddot{U}$ does not exist in some cases. Let's say, for example, we want to extend the above proof of $A \ddot{U} B \vdash A \vee B$ by adding an application of $\vee$ -E at the end.⁵ In order to set up the reduction procedure by creating a maximal formulae, let us permute the proof in the following way:

$$\frac{A \ddot{U} B \quad \frac{\frac{[A]}{A \vee B} \vee\text{-I} \quad \frac{\Gamma_1, [A]}{C} \quad \frac{\Gamma_2, [B]}{C} \vee\text{-E} \quad \frac{\frac{[B]}{A \vee B} \vee\text{-I} \quad \frac{\Gamma_3, [A]}{C} \quad \frac{\Gamma_4, [B]}{C} \vee\text{-E}}{C} \ddot{U}\text{-E}}{C}$$

Notice that in such a configuration, the application of $\ddot{U}$ -E is not, in general, acceptable; it is only acceptable when the contexts (Γ_{1-4}) are all empty. What this shows is that such a proof cannot be reduced to a normal form, and consequently, that S' does not exhibit the global property of normalizability despite the fact that S does.

What the foregoing consideration—both at the local and global level—shows is that rules governing quantum-logical disjunction, despite them being intrinsically harmonious, do confer a *stable* meaning. For those who intend the requirement for harmony to be a cure for *tonkitis*, this should not be a worry as the requirement for intrinsic harmony takes care of Prior's counterexample. But for most logical inferentialists, the requirement for harmony is much more than that: they take it that a general account of harmony should specify the necessary and sufficient conditions for operational rules to genuinely confer meaning, thus assigning harmony a foundational role in the inferentialist meaning-theoretic project.

⁴The failure of distributivity is in fact one of the key features of systems of quantum logic.

⁵This example is due to Steinberger (2011).

Of course, Dummett belongs to the latter camp of theorists, and thus concludes that intrinsic harmony is insufficient to qualify as such an account of harmony as such. He suggests that the requirement for intrinsic harmony should be complemented by counterpart, *stability*, which would rule out excessively restrictive elimination rules (Dummett, 1991, p.287). Although Dummett himself does not develop this notion very much, it has recently attracted the attention of many working to vindicate proof-theoretic semantics.⁶ If intrinsic harmony is the requirement that a reduction procedure exists for any purportedly harmonious pair of rules, then stability may be formally cashed out as the requirement for an *expansion procedure*.⁷ Take the conditional for example:

$$\frac{\frac{\Pi \quad A \rightarrow B}{A \rightarrow B} \quad \frac{\frac{\Pi \quad A \rightarrow B \quad [A]}{B} \rightarrow\text{-E}}{A \rightarrow B} \rightarrow\text{-I}}$$

As shown, any proof Π of $A \rightarrow B$ (on the left side) can be expanded into a derivation that has an application of $\rightarrow\text{-E}$ followed subsequently by an application of $\rightarrow\text{-I}$ (on the right side). Since the pair of rules governing $\rightarrow$ has an expansion procedure, they can be said to exhibit stability.⁸

Interestingly, Dummett thinks that while the requirement for intrinsic harmony and stability⁹ together constitute a sufficient condition for a general requirement of harmony, he claims that they are not necessary conditions. In other words, if all the constants in a given deductive system exhibit harmony and stability, then it suggests that we are in good shape; however, it is not strictly the *point* of harmony, according to Dummett. Instead, Dummett endorses the global (and relational) property of conservative extendibility as the most adequate general account of harmony. Dummett writes,

“It is [harmony-as-conservative extendibility] that must prevail if the *point* of the requirement of harmony is to be attained, namely, that, for every logical constant, its addition to the fragment of the language containing only the other

⁶See, for example, Jacinto & Read (2017) and Tranchini (2018).

⁷Cf. Tranchini (2024) for details.

⁸The above expansion procedure in fact exists for the quantum logical disjunction; this

⁹These two aspects of harmony as a local property are aptly named by Schroeder-Heister (2014) as the ‘criterion of reduction’ and the ‘criterion of recovery’ respectively.

logical constants should produce a conservative extension of that fragment”
(Dummett, 1991, p.290, original italics).

Let us now turn to this idea.

2.2 Harmony as a Global Property

Belnap (1962) was the first to directly respond to ‘tonkitis’ on inferentialists’ behalf. The main problem with admitting tonk to a deductive system, according to Belnap, is that it perturbs our current deductive practice and its central principles to which we are committed. He writes,

“We are not defining our connectives *ab initio*, but rather in terms of an antecedently given context of deducibility, concerning which we have some definite notions. By that I mean that before arriving at the problem of characterizing connectives, we have already made some assumptions about the nature of deducibility.” (1962, p.131)

Now, how exactly to spell out the “antecedently given context of deducibility” is a delicate issue that requires substantial discussion. For now, we shall content ourselves with the identification of such a context with a particular deductive system. A deductive system is defined by a set of rules, both operational and structural, and axioms that permit us to draw inferences. Concretely, what the admission of tonk undermines is the commitment to transitivity, which holds in any proof system worthy of the name.¹⁰ Transitivity may remain implicit in the set of operational rules that constitute the deductive system (as is the case in natural deduction) or may be stated explicitly as a structural rule (as is the case in sequent calculus). Belnap adopts the latter approach, opting to explicitly codify transitivity in the form of a structural rule known as *cut*:

$$\frac{\Gamma \vdash A, \Delta \quad \Gamma', A \vdash \Delta'}{\Gamma, \Gamma' \vdash \Delta, \Delta'} \text{Cut}$$

¹⁰Some recent developments in substructural logics (for example, Weir (2013)) challenge the idea that transitivity is indispensable to a deductive system, but I won’t get into this issue here.

Given this antecedent commitment to transitivity, we can begin to formulate some criteria for admitting constants (and their corresponding rules) to the deductive system. In particular, in order to rule out constants governed by defective operational rules from ever entering our deductive system, Belnap proposes two criteria to determine the admissibility of logical constants to a given system. Of the two criteria for admissibility proposed, *conservativeness* is the key to ruling out *tonk*-like connectives.¹¹ Conservativeness amounts to the proviso that admitting a set of rules governing logical constants must not result in the derivability of new sequents (or, in Belnap’s terminology, ‘deducibility-statements’) involving only those constants drawn from the system to which it is added. In other words, newly derivable sequents must involve the novel constant. It should be noted that, unlike intrinsic harmony and stability, conservative extendibility is a global property, i.e. a property of the entire deductive system. Furthermore, it is a relational property that holds between the base system and its extension.

Now, the trouble with *tonk*, according to Belnap, is that adding its operational rules to a deductive system yields a non-conservative extension of the previous system. By translating the operational rules for *tonk* as sequents and by exploiting the cut rule, we can show that, for any arbitrary *A* and *B*, *B* is derivable from *A*:

$$\frac{\Gamma, A \vdash A \text{tonk} B, \Delta \quad \Gamma', A \text{tonk} B \vdash B, \Delta'}{\Gamma, A, \Gamma' \vdash \Delta, B, \Delta'} \text{Cut}$$

Given that the derived sequent—namely, $\Gamma, A \vdash \Delta, B$ —was not already valid in the system prior to the addition of *tonk*, introducing it to a system yields a non-conservative extension. So, based on the conservativeness criterion, Belnap suggests that *tonkish* rules should not be admitted to our deductive system.

2.2.1 Separability and Atomism

Having shown that the requirement that the addition of a novel constant ought to yield a conservative extension of the base system safeguards against novel *tonkish* constants,

¹¹The other criterion, *uniqueness*, guarantees the economy of the language. It can also be understood in some sense as a dual of conservativeness (cf. Dosen & Schroeder-Heister (1985)). Indeed, uniqueness can be seen as a global counterpart of stability, as it blocks the addition of a pair of rules in which elimination rules are too restrictive in comparison to the introduction rules.

we may now ask whether conservative extendibility is adequate as a general account of harmony that can serve as the foundation for the inferentialist meaning-theoretic project.

The most obvious point of departure is Dummett's (1991) influential discussion, since he understands the requirement of harmony as "a precondition for the possibility of a compositional meaning-theory" (p.247), and, as noted earlier, thinks that such a requirement should be identified with conservative extendibility: "the point of the requirement of harmony is [...] that, for every logical constant, its addition to the fragment of the language containing only the other logical constants should produce a conservative extension of that fragment" (p.290). In other words, Dummett thinks that for a language (or a fragment thereof) to be good working order, the rules governing the inferential role of expressions must satisfy the conservative extendibility requirement in relation to expressions comprising the rest of the language.

Indeed, it is far from obvious why Dummett favors the identification of harmony with conservative extendibility, or more generally, harmony as a property of the entire deductive system, rather than as a local property of the pair of operational rules (viz. intrinsic harmony and/or stability). The general motivating thought seems to be the idea that the meaning of logical constants are determined completely by the pair of operational rules that specify its inferential role. In other words, the presence or absence of any other constant in the deductive system should not affect its inferential role in anyway. From this it follows that the addition of a novel constant should not validate formulas that are composed entirely of the logical vocabulary of the base system. As such, to uphold that the meaning of any logical constant is exclusively determined by its rules, Dummett, following Belnap, thinks that the conservativeness constraint ought to be in place.

The idea that the meaning of each logical constant should be exhaustively determined by its operational rules naturally leads to the requirement that deductive systems ought to be *separable*, i.e. that every derivable formula in the system has a proof that only involves either operational rules for the constant that figure in that formula, or structural rules. Furthermore, separability, combined with the usual inferentialist thesis that to understand the meaning of an expression is to know all of the roles it can play when drawing inferences (see, for instance, P. A. Boghossian (2003)), entails inferential atom-

ism: the idea that a speaker should be able to understand the meaning of a particular expression (say, $\wedge$), without understanding other expressions (say, $\vee$) of the same language.

In fact, many logical inferentialists subscribe to atomism. Tennant, for example, writes, “it should not matter in what order one learns [...] the logical operators” (Tennant, 1997, p.315). Dummett, too, shares this sentiment, at least when it comes to logical vocabulary. He writes, “to understand $A \vee B$, one need not understand $A \wedge B$ or $A \rightarrow B$ ” (Dummett, 1991, p.223). So, Dummett’s identification of harmony with conservative extendibility can be seen as a way to guarantee the separability of logical constants, and, as a result, vindicating the inferentialist claim that the meaning of a logical constant is determined solely by its operational rules. The requirement for conservative extendibility, then, could be understood as the foundation of an inferentialist meaning-theory.

But, separability and the atomic conception of inferentialism—which, as we saw, go hand in hand—are controversial assumptions. As Milne (2002) argues, there are no positive arguments for the principle of separability beyond the assertion of an initially plausible but vague inferentialist credo. Furthermore, when inferentialist semantics is generalized to beyond logic to natural language, the principle of separability begins to appear implausible: why should one expect that we can learn the meaning of ‘north,’ for example, independently of learning the meaning of ‘south’?¹² This latter consideration does not directly affect the cogency of the principle of separability when applied to deductive systems; however, logical inferentialists then owe us an explanation of i) why the semantics of logical constants should be treated in a privileged manner with respect to that of the rest of language and ii) what the criterion of logicality is, if we are to isolate logical constants and their meaning explanations from its natural language counterpart.

Dummett seems to be aware of these issues. This is why, as Steinberger (2011) notes, Dummett appears conflicted on whether or not to fully embrace the principle of separability: on the one hand, Dummett singles out logical constants “as a uniquely simple case” (Dummett, 1991, p.223) of expressions that can be learned in isolation from other expressions, on the other hand Dummett rejects that all introduction rules must be be

¹²Cf. Gareth Evans’ discussion of the *generality constraint* for an argument supporting this idea.

‘pure’ like the Gentzen-Prawitz model we saw in the previous section. Commenting on the Gentzen-Prawitz requirement that each operational rule must mention only one logical constant (so as to guarantee separability), Dummett writes,

“Reflection shows that this demand is exorbitant. An impure c-introduction rule [where c is some logical operator] will make the understanding of c depend on the prior understanding of the other logical constants figuring in the rule. Certainly we do not want such a relation of dependence to be cyclic; but there would be nothing in principle objectionable if we could so order the logical constants that the understanding of each depended only on the understanding of those preceding it in the ordering” (Dummett, 1991, p.257).

If Dummett’s claim that harmony should be understood as conservative extendibility rests on the principle of separability, then it appears to rest on shaky grounds, even for Dummett’s own standards. But the problem for harmony-as-conservative extendibility does not end there.

2.2.2 Is Deducibility Enough?

Earlier we bracketed the question of how exactly to spell out Belnap’s ‘the antecedently given context of deducibility’, and this is a good time to revisit it. Deducibility is by definition relative to each deductive system, so harmony as conservative extendibility, if it were to serve as the justification for operational rules, trades on the earlier question of what deductive system we are committed to in the first place. As Belnap notes,

“It is good to keep in mind that the question of the existence of a connective having such and such properties is relative to our characterization of deducibility. If we had initially allowed $A \vdash B$ (!), there would have been no objection to tonk, since the extension would then have been conservative. Also, there would have been no inconsistency had we omitted from our characterization of deducibility the rule of transitivity.” (1962, p.133)

In a similar vein, as Ripley (2015) notes, once we give up the unrestricted use of the cut rule entirely, the addition of tonk to a system of sequent calculus yields a conservative extension, and thus, by Belnap's criteria, counts as a legitimate and meaningful connective. So, the question is, what determines—or better yet, *justifies*—the choice of a deductive system? Belnap seems to be relatively unbothered by this question, as he simply assumes that the context he has given (which includes the axiom $A \vdash A$ and four structural rules, viz. weakening, permutation, contraction, and transitivity) would be unassailed:

“In accordance with the opinions of experts (or even perhaps on more substantial grounds) we may take this little system as expressing all and only the universally valid statements and rules expressible in the given notation: it completely determines the context.” (1962, p. 132)

Belnap suggests that perhaps the justification of his deductive system would either consist of justification by the relevant authority or ‘perhaps on more substantial grounds’. The first option is unacceptable for two reasons. First, in general, appeal to authority is a weak form of justification because it simply pushes the burden of justification, i.e. by giving a valid *epistemic reason* pertinent for a given claim, onto the authority in question. In the absence of a valid epistemic reason, justification granted by the authority amounts to nothing more than a disgruntled parent's declaration ‘Because I said so!’, accompanied by the illocutionary force of a command. Such an act lands us in the territory of dogmatism.¹³ Secondly, justification by appeal to authority is problematic in this particular case because there is no consensus among the experts as to which deductive system expresses ‘all and only the universally valid statements and rules’, as the lively debate on logical pluralism and substructural logic attest to.¹⁴ The problem for Belnapians is reformulated as the following: Can we still salvage harmony as conservative extendibility without the presupposition that there is a uniquely privileged deductive system?

The short answer is no. To be sure, there is no problem with the idea that a deductive system (call it S') containing a new operator is a conservative extension of the base system

¹³It might happen to be justified by a valid moral or pragmatic reason—e.g. ‘clean your room, it's filthy!’—but is nonetheless epistemically unjustified.

¹⁴For example, I have mentioned in a previous footnote some logicians who work on intransitive logic, i.e. those that deny the structural rule of transitivity.

S given that $\Gamma \vdash_{S'} A$ only if $\Gamma \vdash_S A$, and to that extent S' displays harmony *relative* to S . But the problem with this construal of harmony is its focus on deducibility, which makes it an inherently relative notion. In the absence of the antecedent commitment to a particular deductive system, harmony-as-conservative extendibility could only serve as *conditional grounds* for the inferentialist meaning-theoretic project, i.e. conditional on the endorsement of a deductive system or another as the ‘one true logic’. In other words, the justification for harmony-as-conservative extendibility rests on a prior justification for and endorsement of a particular deductive system. If the requirement for harmony is to serve as the foundation for a meaning-theory, such an approach seems too far-fetched if possible in principle.¹⁵

Now, contrast this conclusion with that of the previous section: “Since, according to the Gentzen-style one-sided model, introduction rules are self-justifying, the existence of a reduction procedure completes the justification of the pair of rules: expressions governed by intrinsically pair of rules can thus be considered fully justified.” Therefore, intrinsic harmony (and stability) appear to be more promising than conservative extendibility as far as formulations of the requirement of harmony go.

2.2.3 Lorenzen on Admissibility

Before moving on from the topic of harmony-as-conservative extendibility, we should note that Belnap’s criterion of conservativeness is historically preceded by Lorenzen (1955) who proposed similar restrictions to the *admissibility* of rules to a formal system (§2, 1955):

“A rule R is called admissible in a calculus K , if its addition to the primitive rules of K —resulting in an extended calculus $K + R$ —does not enlarge the set of derivable sentences. If $\vdash A$ denotes the derivability of A in K , then R is admissible if $\vdash_{K+R} A$ implies $\vdash_K A$ for every sentence A .” (Schroeder-Heister, 2008, p.218)¹⁶

¹⁵In fact, I think that any theory of meaning that tacitly relies on the knowledge that our current inferential practice is correct is doomed from the get-go. I will comment more on this when we arrive at what the requirement for harmony amounts to when applied to the semantics of natural language.

¹⁶Much of the subsequent exegesis of Lorenzen (1955) is also due to Schroeder-Heister (2008).

Some clarification about Lorenzen's project and terminology is in order. The main goal of Lorenzen's *Operative Logic* (1955) is to justify intuitionistic logic based on principles of admissibility. The basic idea is not dissimilar from Belnap's: to begin with admissibility criteria, which are then applied to determine which rules of inference (in particular, operational rules for logical constants) are justified. However, Lorenzen's approach is more systematic and more general, since, unlike Belnap, Lorenzen does not start out by assuming an 'up-and-running' deductive context but instead begins with a general theory of formal systems (or what Lorenzen calls *Protologic*) in order to flesh out principles of admissibility; what Lorenzen refers to as 'calculus' in the above quote is equivalent to Curry (1951)'s notion of formal system, which include basic expressions and rules for producing complex expressions out of basic ones. Indeed, it would not be a stretch to say that Lorenzen was already trying to articulate, in the form of principles of admissibility, a theory of harmony understood in the Dummettian sense as the "precondition for a compositional meaning-theory," some 40 years before the term was even coined.

In Lorenzen's (1955) conception of logic, the notion of admissibility plays a fundamental role. In the above quote, implication is explained in terms of admissibility. Now, it is tempting to explain admissibility in terms of implication,¹⁷ but obviously, such a strategy would yield viciously circularity. So, Lorenzen explains admissibility by referring to the notion of an *elimination procedure* (Lorenzen, 1955, §3). As Schröder-Heister sums up,

"R is admissible in K, if every application of R can be eliminated from every derivation in K + R. The implicational relation between existential statements expressed in [the statement $\vdash_{K+R} A$ implies $\vdash_K A$] is reduced to the insight that a certain procedure reduces any given derivation in K + R in such a way that the resulting derivation does no longer use R" (Schröder-Heister, 2008, p.218).

So, admissibility is analyzed as the existence of a kind of a procedure—a kind of 'doing' (i.e. a concrete task) rather than a kind of 'saying' (i.e. an assertion), to quote Brandom's oft-cited turn of phrase.¹⁸ The specification of this procedure is the content of,

¹⁷ Along the following lines: $\vdash A$ implies $\vdash_R A$ only if R is admissible.

¹⁸ Cf. R. Brandom (2008).

in Lorenzen's terminology, the 'operative meaning' of admissibility. In this sense, Lorenzen's notion of admissibility shares the same pragmatist (i.e., use-theoretic) underpinning as intrinsic harmony, which was analyzed earlier in the chapter as the existence of a procedure (namely, to reduce any derivation into its normal or canonical form).¹⁹ By analyzing admissibility in terms of the elimination procedure, Lorenzen understands implication to signify more than derivability, namely, as an expression of the *understanding* gained through the elimination procedure which underpins the notion of admissibility.

Let us now compare Lorenzen's notion of admissibility with Belnap's conservativeness constraint. *Prima facie*, admissibility is just the conservativeness constraint without the proviso Belnap adds in order to allow for complex statements involving novel constants. But this is too quick. While Lorenzen insists on a distinction between the notion of admissibility and derivability,²⁰ Belnap's sole focus is on derivability. Admissibility, as we saw, has an epistemic component: the operative meaning of admissibility is based on the ability to perform the elimination procedures, which yields further understanding through such performances and at the same time demands the appropriate kind of know-how to do so. On the other hand, derivability is a formal notion that simply describes the syntactic relationship between assumptions and theorems. Belnap's failure to appreciate the epistemic significance of the activity of reasoning, in my view, is part of the reason why his criteria of conservative extendibility, which describes the relationship between systems with differing 'deducibility-statements', is irreducibly relativistic and thus fails to *justify* the logical inferentialist project in any normatively significant way.²¹

To repeat my diagnosis of the problem with harmony-as-conservative extendibility, conservative extendibility has no all-things-considered normative significance when taken as a purely syntactic constraint. Lorenzen's approach to admissibility is illuminating in

¹⁹As was noted earlier, Lorenzen was also the first to articulate the 'inversion principle'—from which intrinsic harmony springs—as one of the principles of admissibility.

²⁰As Schröder-Heister (2008) notes, "the standard proof-theoretic example of a rule admissible but not derivable is that of the cut rule in the first-order sequent calculus" (p.220).

²¹Indeed, it is the conceptual distinction between admissibility and derivability is what prompts Schröder-Heister to write, "it is the admission of this sort of evidence which makes Lorenzen an intuitionist rather than a formalist" (ibid.). Here, 'intuitionist' is understood roughly as taking 'construction,' an epistemic achievement of the reasoner, to be fundamental to logic (and mathematics) while 'formalist' is understood roughly to take structures describable in purely syntactic terms to be fundamental to logic.

that it allows us to conceive of the constraint we set for meaning-determining rules in a explicitly normative terms since justification is built into the very notion of admissibility. Having said that, Lorenzen's (1955) project of *Operative Logic* failed with respect to capturing the meaning of implication (namely, the fact that the assumption can be discharged when introducing the connective without possessing the proof of it). This led Lorenzen to the what is known as his 'dialogical turn,' wherein features related to meaning (of constants) was treated on one level (the 'play level') and features related to validity was treated on another level (the 'strategy level'). For now, I will merely flag this issue only returning to it in the final chapter.

2.3 What is the Point of Harmony?

So far, I have introduced the requirement for harmony as a inferentialist response *contra* Prior's *tonk* challenge. I have shown that, despite their differences, both the local account of harmony (i.e. the relationship between each introduction and elimination rules must exhibit intrinsic harmony and stability) and the global account (i.e. the deductive system as a whole must exhibit properties like conservative extendibility) are formally adequate in safeguarding against tonkish constants. Tonkers' reply would naturally be to question the legitimacy of harmony. Imposing such a requirement on every set of rules governing connectives, the thought goes, seems to run contrary to the oft-cited inferentialist slogan of *meaning as use*. By all means, it is true that the *tonk* connective is not intrinsically harmonious (locally) and its addition to first-order predicate calculus creates a host of problems at the global level: it creates a non-conservative extension of the system, alters the inferential roles (and thus the meaning) of other constants, and renders the system non-normalizable. But if the meaning of logical constants are solely determined by the rules governing them, why shouldn't *tonk* count as perfectly meaningful if undesirable or unjustified? That is, the claim that *tonk* is defective because its addition to standard deductive systems render them useless (a practical consideration), the claim that rules governing *tonk* are unjustified (an epistemic consideration), and the claim that rules governing *tonk* do not really confer meaning (a semantic consideration) appear to be

three distinct claims.

Here, the inferentialist has two realistic options.²² They might concede that tonkish constants are perfectly meaningful if its pair of rules unjustified; or, they might press on with the stronger claim, namely, that tonkish constants are not meaningful at all. In either case, we need an explanation of why the requirement of harmony could serve as the justification of logical laws or as the criterion for meaningfulness of logical constants. Since the epistemic and semantic significance of the requirement of harmony is not often kept apart, we find much confusion in the literature. So, it would be prudent to take extra care to keep them apart:

- **Epistemic interpretation of harmony:** Rules that fail to satisfy the requirement of harmony are unjustified.
- **Semantic interpretation of harmony:** Expressions governed by rules that fail to satisfy the requirement of harmony are incoherent or meaningless.

To be sure, both the epistemic interpretation and the semantic interpretation of the requirement of harmony is individually sufficient to refute Prior's initial argument. If rules governing *tonk* are defective, i.e. unjustified or incoherent, then Prior's initial objection to logical inferentialism loses its bite. But, in order to truly vindicate the inferentialist approach to meaning, we need to interpret harmony first and foremost as a semantic thesis rather than an epistemic one.

2.3.1 Principle of Innocence

While the literature on harmony is vast, much of the attention has been paid to technical details; those that explicitly deal with the motivation of the requirement harmony is scant. In a rare treatment of this issue, Steinberger (2009) makes the following plausible suggestion regarding the philosophical motivation for the requirement for harmony:

The principle of innocence: logic alone should not be a source of new information. That is, it should not be possible, solely by engaging in deductive

²²Here, I rule out the practical reply for the same reasons that I gave when criticizing Belnap's argument in section 2.3.

reasoning, to discover hitherto unknown (atomic) truths about the world that we would have been incapable of discovering (at least in principle) independently of logic. (Steinberger, 2009, p.60)

The principle of innocence follows from an intuitive and commonly accepted understanding of what a deduction is. Unlike other forms of reasoning—for example, induction or abduction—deductive reasoning does not aim at uncovering novel information about the (empirical) world. Since the principle of innocence only motivates intrinsic harmony and conservative extendibility, given the preceding discussions, we may also add its counterpart to the effect that there is no information loss, in order to motivate the local property of stability (and the global property of uniqueness-preservation): it should not be possible, solely by engaging in deductive reasoning, to say less about the world than when we started off. Given this proviso, the principle of innocence seems to motivate all forms of harmony discussed so far. Thus, according to Steinberger, “The role of harmony [...] is precisely to protect the innocence of logic” (ibid.). Indeed, Dummett at times also seems to make this very point.²³ Dummett’s identification of harmony with conservative extendibility could be thus be understood as resulting from his commitment to the principle of innocence—that the addition of logical laws should not validate hereto invalid formula and thus change the meanings of expressions already contained in the language.

Now, the principle of innocence implies that the logical fragment of language should be clearly demarcated from the rest of the language. For, if inferential relations that constitute the meaning of logical expressions and those that constitute the meaning of non-logical expressions were connected, of what should logic be innocent? This observation gives rise to the following thesis:

The principle of autonomy: the logical fragment is self-contained. That is, the meaning of a logical constant cannot depend on the meanings of non-logical expressions. (Ibid.)

The principle of innocence and the principle of autonomy swim or sink together. Not only is the principle of autonomy implied by the principle of innocence, the converse also

²³cf. for example Dummett, 1991, p.219

holds: without a clear demarcation, we begin to lose the boundary between expressions that are ‘innocent’ (i.e. whose meanings do not affect and are not affected by conceptual changes that alter the meaning of ordinary expressions) and those that aren’t.

If, as Steinberger suggests, the principle of innocence motivates the requirement of harmony, then the requirement of harmony is best understood as a justification for laws of logic (i.e. inference rules for logical constants); in turn, the justification of the requirement of harmony rests on the justification of the principle of innocence and on the tenability of the principle of separability and of autonomy (which are collateral commitments of the principle of innocence). In this way, logical laws are justified by appealing to our understanding of what logic is, or ought to be.

I will not deal here with this larger question about the nature of logic, in particular, whether or not the principle of innocence correctly captures an important feature of logic. For our purposes, it suffices to observe that whatever the ‘correct’ conception of logic is, appealing to the principle of innocence as the basis of the requirement of harmony has several important consequences for the larger inferentialist meaning-theoretic project. Firstly, it implies that the semantics of natural language can in principle proceed in isolation from logical inferentialism. So long as the principle of autonomy holds, there is no positive reason to explain the semantics of natural language in inferential terms. For example, a theorist could plausibly subscribe to truth-conditional semantics for natural language while explaining the meaning of logical constants in inferential terms.²⁴

Relatedly, given the principle of innocence, the requirement of harmony now is most straightforwardly seen as a *criterion for logicality*. This amounts roughly to the claim that, in order for an expression to count as a logical constant, it must satisfy the requirement of harmony (however we formally cash it out). So, quasi-connectives like tonk are ruled out as *logical* constants. They are ruled out on the basis that its introduction and elimination rules, since they violate the requirement of harmony, are unfit to count as a logical constant; that is, given a prior understanding of what logic ought to be, tonk ’s meaning-conferring rules are unjustified.

²⁴That said, it should be noted that most likely no logical inferentialist would adopt this position. This mere fact already indicates that there is probably a reason more fundamental than the principle of innocence that motivates the requirement for harmony.

Crucially, the choice of motivating the requirement of harmony through the principle of innocence seems to commit us to the epistemic interpretation of harmony. Again, the foregoing discussion shows that the principle of innocence *justifies* the requirement of harmony, which in turn justifies the rules for (legitimate) logical constants. Indeed, there is no obvious way in which the principle of innocence might sustain the semantic interpretation (unless logicality is paraphrased as ‘meaningfulness qua logical constant’). On this view, tonkish expressions seem to be perfectly meaningful (as the meaning is adequately specified by introduction and elimination rules) if the rules that give meaning are unjustified.

So far we have been exploring the explanation and some consequences of justifying requirement of harmony by appealing to the thought that logic should be innocent. This argument is intuitive enough, as it rests on the commonly accepted conception of what logic—in particular, deductive reasoning—ought to be. Unfortunately, this strategy is problematic, at least for two reasons. Firstly, as I have hinted at earlier, the fact that the motivation so conceived for the requirement of harmony is logic-specific (i.e. does not apply to the semantics of the rest of language) is a problem for the larger inferentialist project. Beyond the fact that the principle of innocence fits in an ad-hoc manner within the deeper pragmatist (viz. ‘meaning as use’) theoretical picture that underpins inferentialism, objections along Prior’s lines are legion regarding inferentialism about natural language. Furthermore, if we think, as inferentialists typically do,²⁵ that it is instructive to “look at the contents of logical concepts as providing the key to conceptual content generally” (R. Brandom, 2007, p.161), the appeal to the principle of innocence as the basis for the requirement harmony—which holds the key to the vindication of logical inferentialism—dooms the project from the get-go.

The applicability of the requirement of harmony to natural language semantics is a substantial topic that requires careful attention; thus, I will dedicate the next chapter to this issue. For now, it is enough to note that even theorists like Dummett, who subscribes to the principle of innocence, think that rules governing any purportedly meaningful

²⁵For example, R. Brandom (2000), Dummett (1991), and Peregrin (2014), amongst others.

expression, logical or otherwise, should be harmonious.²⁶ This suggests *prima facie* that there is a deeper motivation behind the requirement harmony beyond the principle of innocence.

The claim that the point of harmony is to secure the innocence of logic creates a rift between semantics of logical constants and of non-logical expressions, but of course, this rift on its own does not directly affect the cogency of the argument when presented in isolation from global inferentialist commitments. Nonetheless, the rift should be understood as a symptom of a underlying problem, namely that considerations from the principle of innocence fails to support the semantic interpretation of harmony. As it will be shown in the following discussion, the interpretation of harmony as an epistemic constraint is, while necessary, not sufficient for inferentialists; the interpretation of harmony as a semantic constraint—namely, that expressions governed by inharmonious rules are incoherent or meaningless—is crucial in order to vindicate the inferentialist meaning-theoretic project at-large against usual doubts put forth by its opponents.

2.3.2 “Syntax is insufficient for Semantics”

The philosophical intuition that underlies Prior’s doubt runs deeper than what the *tonk* argument makes explicit. Following Searle (1980)’s (in)famous Chinese room argument, we might put it as a slogan: *Syntax is neither sufficient for nor constitutive of semantics*. Indeed, it is the very intuition shared by most contemporary theorists of semantics for both formal and natural languages that are skeptical about the inferentialist meaning-theoretic project. My contention is that the correct understanding of harmony—namely, the semantic interpretation—will help dispel this worry and vindicate the inferentialist approach.

According to Stevenson (1960), the moral that should be drawn from Prior’s argument is that operational rules cannot define the meaning of a connective—which leaves room for *tonk*—since meaning, properly speaking, should be stated in terms of truth-functional statements in the metalanguage. According to this line of thinking, there is an

²⁶We will discuss Dummett’s view—that any meaningful expressions should exhibit harmony—at length in the following chapter.

antecedent notion of what, for example, “or” (or more precisely, statements of the form “A or B”) means independently of the role it plays in inference; an effective way to capture this antecedent meaning is by specifying the truth conditions, either in the metalanguage or via truth tables. What the invention of tonk shows, then, is that in the absence of a truth-functional—and more generally, value-functional—explanation of meaning of connectives in the metalanguage, operational rules lose its semantic (and thus normative) significance since the semantic significance of rules come down to how well it behaves in accordance with the semantics we specify antecedently and independently of setting the rules.

There is a sense in which Stevenson’s conclusion is trivially true. In modern logic after Tarski (1935), one tends to conflate the term ‘semantics’ with model theoretic semantics, according to which formal systems are first assumed to be uninterpreted symbols which needs to then be assigned a meaning by specifying the suitable function that maps uninterpreted symbols (in the object language) to its ‘extension’ (in the metalanguage). One of the assumptions of this project is that syntax—in the narrow sense as concerning rules determine the well-formedness of a given string of symbols—is largely independent of semantics so conceived.²⁷ This line of thought originates in the metamathematics tradition inaugurated by Hilbert, whose goal was to show that mathematical reasoning can be explicitly stated in a precise formal language. As Sundholm (1997) notes,

“[...] so called metamathematical expressions, that is, inductively generated strings, or trees, of atomic metamathematical “signs”, are not made to express anything, but on the contrary are themselves, like any other mathematical objects expressed by means of ordinary linguistic expressions. Metamathematical expressions are not tools for speaking with, but objects to be spoken about” (p.191-192).

²⁷Largely but not entirely, since it is not difficult to see that the meaning of a complex formula is composed *syntactically* from atomic formulae, even if atomic formulae are themselves given a model-theoretic semantics. So, the very commitment to compositionality (which is usually a desiderata of any meaning-theory) suggests that syntactic features are at least part of the meaning of some expressions. Taking this idea further, Read (2008), drawing on examples from modal logic, observes that “[The demand for a truth-functional meaning explanation] is clearly unreasonable, as there are many non-truth-functional connectives, forming complex propositions whose truth-value is not determined by any finite set of “truth”-values of their constituents” (p.286).

Even after Gödel conclusively demonstrated that Hilbert's program (viz. formalizing all mathematical statements) cannot be sustained, its founding assumptions have lived on. Indeed, its effects can be felt throughout the history of modern logic, of which Stevenson's elaboration of Prior's argument is one. Sundholm writes, "It is fair to say, however, that without the Hilbert program, and the ensuing metamathematical codification by, say, Gödel, Tarski, and Bernays one would not have thought of studying formal languages consisting of uninterpreted strings of objects" (ibid., p.191). This metalogical view is, at this point, deeply entrenched in the philosophical imagination. Indeed, metalogical assumptions are usually embedded in critiques of logical inferentialism, old and new.²⁸

Now, inferentialists characteristically resist the bifurcation of the analysis of language, both formal and natural, into syntax and semantics. If we abandon the idea that formal language—i.e. at the object language level—consists purely of meaningless symbols, then Stevenson's objection loses the bite. The entire inferentialist meaning-theoretic project rejects that an 'antecedent meaning,' namely the extensional specification of the referents, exists for logical connectives and atomic formulae alike. So, Stevenson's objection appears to be misguided. Nonetheless, Stevenson's exegesis of Prior's *tonk* argument is illuminating because it makes explicit the disagreement between *tonkers* and *anti-tonkers* which runs at this deeper level: the rub is created by what each camp understands to be the correct relationship between syntax and semantics. In order to dispel doubts about inferentialism without begging the question, we need to deal with this conceptually prior issue.

In the following sections, I aim to show that harmony—conceived as a criterion for meaningfulness—is fundamental to the inferentialist defense of syntax and semantic as irreducibly connected features which can be made explicit at the same level (viz. the object language).

²⁸Compare, for example, Rumfitt (2017): "there is a far better explanation of why '*tonk*' is meaningless than that it violates Harmony. It is not meaningful, because a formula whose main connective it is does not say anything; such a formula does not say anything because it does not have truth-conditions."

2.3.3 Alternatives to the Metalogical view

Inferentialists are united in rejecting the metalogical assumption, namely, that formal language consists of uninterpreted string of symbols whose meanings are specified in the metalanguage.²⁹ But what exactly is wrong with it? Consider the following critique of Tarski's semantic project due to F. B. Fitch (1944):

"The metalanguage in which a general semantics would be formalized would not be of much use unless it could take itself as one of its object languages, for otherwise a general semantics could never deal in detail with a sufficiently large group of formal languages. It would always have to treat its own formalization as outside the scope of its pronouncements" (p.68).

Fitch's concern is that, from a practical, computational point of view, the metalanguage needs to be given a semantics by embedding itself to the object language. But the same point could be argued from the pragmatist philosophical point of view. The driving idea is that meaning cannot be settled from 'the view from nowhere,' i.e. outside (the object) language, whether natural or formal. And, if meaning is essentially use, then the project of semantics qua specifying meaning using language would never be complete, as practices change and evolve; this is another meaning of Brandom's turn of phrase, 'Semantics must answer to pragmatics.' Given this philosophical backbone, if Sundholm is correct to characterize metamathematical expressions not "as tools for speaking with, but objects to be spoken about," the metalogical assumption must give. (In fact, the reasons behind logical inferentialists' rejection of the metalogical view is similar to that of pragmatists' rejection of representationalism as presented in the previous chapter.)

The alternative to the metalogical view is to understand language as *always* interpreted. The most influential source of this alternate view can be found in Frege's early writings. In the *Begriffsschrift* ('Concept-script'), Frege tacitly assumes that any well-formed formula (of the object-language) are meaningful, i.e. interpreted. Objects that

²⁹It should be noted that there are other criticisms to the Tarskian definition of truth (and of logical consequence as necessary truth-preservation given this notion of truth) from within the model-theoretic tradition, most notably Etchemendy (1990).

are derived in Frege's system are judgments (or assertions), which are contentful; judgments stand in contrast with well-formed formula of modern logic, which are syntactic objects that do not express content unless interpreted.³⁰ Because assertions are expressed in the object language (thanks to the use of the turnstile, which serves as the 'force indicator'), Frege's formalism allows one to make explicit inferential relations involving assertions (with propositional content) in addition to derivability relations involving (uninterpreted) well-formed formula.

Since assertion is an implicit knowledge claim (as explained in the previous chapter), the epistemological slant was part and parcel of Frege's early inferentialism. In §3 of the *Begriffsschrift*, he gives an inferential definition of propositional content. Frege analyzes the propositional content of a judgment A as what B has in common with any judgment that can be substituted in its place as a premise in valid arguments. In other words, A and B have the same propositional content just in case in any arbitrary context Δ and for every formula C ,

$$\Delta, A : C \text{ is valid if and only if } \Delta, B : C \text{ is valid.}$$

Since Frege's work precedes the metamathematical turn in the early 20th century, he does not explicitly criticize a metalogical approach to meaning. The (rather underdeveloped) meaning explanation that Frege gives in §29, however, is clearly incompatible with the metalogical approach. For Frege, as Sundholm notes, "meaning explanation proper for the mathematical language is not itself mathematical in nature: one does not give a semantics for the language of mathematics by doing further mathematics, for example, by proving theorems" (1997, p.193).

Frege's early epistemological approach to logic and foundations of mathematics was then developed in the *constructivist* (or *intuitionistic*; I shall use the terms interchangeably) tradition of Brouwer, Heyting, and Kolmogorov. Constructivists reversed the metalogical order of explanation: rather than beginning with a formal axiomatization then specifying a suitable semantics, which then is used to prove the soundness of that language, con-

³⁰Sometimes it is suggested that Frege abandoned his earlier inferentialism and embraced a more metalogical approach to semantics after his *Sinn und Bedeutung*. This need not bog us down for the purposes of our discussion.

constructivists took as explanatory primitives the notion of a *proposition*. Then, they defined how we can use them in our deductive practices, namely what it is for a proposition to be true, when someone is entitled to assert the proposition, what acts of inference are truth-preserving, etc.³¹

Most important for our discussion is that, in the constructivist tradition, proposition is analyzed as its proof-conditions (i.e. what one needs to do in order to produce an adequate demonstration), or in an alternative but essentially similar veil, as a problem to be solved.³² Thus, it is fair to say that two theoretical orientations form the foundation of constructivist semantic theorizing: first, the metaphysically-tinted question, ‘what is the meaning of x?’ is replaced by the epistemologically-tinted question, ‘what is it for a speaker to understand the meaning of x?’; second, propositional knowledge (“know-that”) is explained in terms of ability (“know-how”). Inferentialists, including Brandom and especially Dummett, adopt both of these insights—i.e. the epistemological and pragmatist orientation to theorizing—from constructivism.

2.3.4 Semantic Significance of Harmony

In light of this diagnosis of the disagreement between *tonkers* and inferentialists, what can we say about the requirement of harmony? Let us return to Dummett’s claim that harmony is “a precondition for the possibility of a compositional meaning-theory,” for now we have the resources to appreciate this claim fully.

First, let’s clarify why compositionality is a desideratum of a meaning theory. There is the empirical fact that all natural languages are *productive*, in the sense that no language has an upper bound for the number of well-formed, contentful sentences. The syntactic explanation for the productivity of language is Chomsky’s celebrated notion of *recursion*; nonetheless, the fact that those sentence are meaningful (and thus interpreted) can only be explained by invoking the semantic notion of *compositionality*: roughly, that principle that the meaning of a complex expression is determined by the meanings of its constituents and the rules used to combine them. Recall that, to the question, ‘What is it for a speaker

³¹See Sundholm (1983) for a clear and thorough presentation of this idea.

³²The former equivalence is due to Heyting (1930) and the latter to Kolmogorov (1932).

to understand the meaning of an expression?', inferentialists would answer, 'To be able to use it in accordance with the correct rules of inference involving that expression.' In other words, it is the theoretical desideratum of a general meaning theory to articulate the what kind of epistemic conditions must be in place for speakers to engage in inferential practice using specific expressions. So, if we want to articulate a meaning theory of how speakers actually use language, compositionality must be accounted for.

But why is harmony strictly necessary for this phenomenon? Elaborating on Brouwer's criticism of classical mathematics, Dummett writes,

"[Language] must stand up to the critical scrutiny of its functioning [and such scrutiny] aims at a systematic means of ascribing content to the expressions and sentences of the language, in terms of which accepted modes of operating with it (including the rules of inference observed) can be justified, or, better, are evidently justified. The mere existence of a common practice in using it, a practice that can be learned and that incorporates a means of distinguishing conventionally correct uses from incorrect ones, did not, for Brouwer, guarantee that that practice was coherent. *It is incoherent when there is disharmony between its component conventions, or equivalently, when it cannot be rendered comprehensible by any systematic ascription of content*—and this means, eventually, *when it would be impossible to construct a meaning-theory that accorded with all features of the practice*. We have seen that a practice may exist, and be learnable, but still open to criticism as failing to achieve the purpose that a linguistic practice should serve. That was precisely what Brouwer believed to be true of the practice of classical mathematics" (1991, p.240-241, my italics).

Why does Dummett think that disharmony between rules can create a situation where it is "impossible to construct a meaning-theory that accorded with all features of the practice?" This is left opaque in *The Logical Basis of Metaphysics*, but a brief thought experiment will illustrate the intuition. Imagine a (formal or natural) language composed only of *tonkish* expressions—i.e., expressions whose condition for application and consequences of application are intrinsically inharmonious and unstable, and/or expressions whose

addition to the base language creates a non-conservative extension. Even if the external observer (say, a radical interpreter in Davidson's sense) observes some regularities in patterns of behaviors of the 'speakers' of this language, there could not possibly be any property to properly call *meaning*, as the 'inferential' role of any expression could not possibly be learned, at least by any human, and probably by any rational being. Indeed, the relations between expressions can only be understood quasi-inferentially, in that some expression literally follows another. So, one can only concoct the *tonk* example against a background of a meaningful language, i.e. a sufficiently coherent set of rules and expressions that (mostly) respect the principle of compositionality.

So, on Dummett's count, harmony is first and foremost a semantic constraint. Expressions (or a group of them) governed by rules that fail to satisfy the requirement of harmony are incoherent or meaningless. Indeed, the fact that such rules give rise to incoherence is the basis on which Dummett advances the epistemic interpretation: Rules that fail to satisfy the requirement of harmony are unjustified.

Does Dummett's argument succeed? There are reasons to doubt it. Dummett rightly thinks that compositionality is a desideratum of any meaning theory. He also thinks, for reasons unrelated to the discussion on harmony, that the inferentialist approach is the correct way to theorize about meaning. Given these two assumptions, the principle of harmony must hold. But the inferentialists' opponent (like Prior) attack inferentialism on the basis that it leads to conventionalism and thus relativism about conceptual norms. In order to dispel the opponent's worries without begging the question, Dummett must provide a *positive* reason for why the principle of harmony should hold. Indeed, Dummett (1991) doesn't seem to provide any positive reason to regard harmony as something that should hold independently of our antecedent commitment to the inferentialist approach to meaning. That is, since inferentialist approach to semantics is correct and since compositionality must be accounted for, Dummett thinks harmony must hold if we are to 'harmonize' these two commitments. But even if Dummett is right to think broadly that inferentialism is correct, we can always put into doubt the cogency for the requirement for harmony (as, we will see in Chapter 3, Brandom does) within the larger inferentialist enterprise.

2.4 Conclusion

This chapter took a historical tour of the literature on the principle of harmony, which was formulated by a number of theorists in the wake of Prior's (1960) classic objection against logical inferentialism. In the discussion, two themes emerged: first, that harmony, if such a requirement could be vindicated, should take the form of a *local property* of the pair of rules governing a constant (*qua* intrinsic harmony and stability) over Belnap's and Dummett's *global* formulation (*qua* conservative extendibility). Secondly, we examined the philosophical motivations for the principle of harmony: in particular, we looked at the Principle of Innocence (as a justification for the epistemic interpretation of harmony) and Dummett's argument for the semantic interpretation (based on considerations from compositionality). In either case, neither of these strategies were found to be convincing.

The discussion of this chapter, in conjunction with Chapter 1 (on Brandom's inferentialism and its account of objectivity), sets up the second half of this work. If the principle of harmony can be adequately grounded, then we secure one form of objectivity, understood as the *attitude-transcendence* of norms. Recall that a norm n is attitude-transcendent (AT) for community C iff it is possible for everyone in C to be mistaken about the correct applications of n . Even if a linguistic community has unanimous agreement about the correct applications of n , if n results in disharmony, then it is wrong to endorse n .³³ Given that the principle of harmony secure us AT, two questions arise: Could the principle of harmony apply to rules governing concepts of natural language, and what *grounds* such such a constraint? The first question is the topic of Chapter 3; the second question is the dealt with in the Chapter 4.

³³Of course, the principle of harmony itself is a norm; thus, it cannot be precluded that, even if we endorse harmony, we are nonetheless collectively wrong to endorse this norm. In this sense, harmony does not secure us attitude-transcendence of norms. But this is a more general problem about philosophy (or theorizing more generally) and the possibility of objectivity. This will, hopefully, be a topic I will explore on another occasion.

Chapter 3

Harmony for Natural Language

Taking the practice of asserting and drawing inference to be the ‘downtown’ of language naturally leads to the idea that the basic unit of semantic significance is a proposition, paradigmatically expressed by an assertible sentence; thus, all other units of meaning that is presupposed in any compositional meaning-theory, including term and (unsaturated) predicate, are explained in terms of propositional content (Brandom, 1994, p.79). Propositional content of an assertion is of course explained inferentially, namely by the *conditions* under which the assertion may be correctly inferred and the inferential *consequences* that follow from it. This is a point that Brandom inherits from Dummett’s inferentialism based on assertibility.¹

Notice that in assuming that the two-aspects of inferential role—viz. condition and consequence—contribute equally to the meaning of an assertion, Brandom departs from Gentzen-style one-sided approach to meaning prevalent in contemporary logical inferentialism.² For, Gentzen’s assumption was that introduction rules, or rules codifying when a statement is assertible, *fixes the meaning*; elimination rules, or rules codifying what can be correctly asserted from it, only makes explicit the consequences of the application of

¹Here, I am simplifying the discussion for the sake of brevity. As explained in section 1, on Brandom’s count, propositional content is explained by incompatibility and entailment (semantic notions) which are then explained by attributions and acknowledgements of commitments and entitlements (pragmatic notions). Since commitments and entitlements are, at base, conditions and consequences of making a claim, I think that making the equivalence between Dummett’s and Brandom’s theories, at least in this case, is safe.

²Brandom also depart from the one-sided approach to meaning that takes the converse side (namely, the consequences) to fix the meaning. Interestingly, Brandom attributes this approach to classical pragmatists like Peirce. Cf. Brandom, 1994, p.123.

the introduction rule. The animating thought is that consequences of any assertion are already implicitly ‘contained’ in the conditions for assertion. In this sense, the normativity of elimination rules are derived from that of the corresponding introduction rule. As we saw in chapter one, this consideration motivates the principle of harmony, which renders expressions governed by intrinsically inharmonious rules meaningless. This observation gives rise to questions regarding the role of harmony within Brandom’s inferentialism. In particular, must rules governing the use of non-logical (or materially contentful) expressions also exhibit harmony?

Brandom comments, rather briefly, on the requirement for harmony, in both its local and global variant, and its relation to his brand of inferentialism. In section 1, I will summarize Brandom’s claim that neither intrinsic harmony nor conservative extendibility is plausible as a criterion for meaningfulness when it comes to materially contentful expressions. (In its stead, Brandom suggests that the requirement for harmony should be understood as a *process* of “harmonizing commitments.” A full treatment of Brandom’s notion of harmony is beyond the scope of this project.)

The plausibility of Brandom’s views notwithstanding, I argue in section 2 that Brandom is wrong to abandon intrinsic harmony altogether. I draw on the Dialogical tradition in logic to show that a proper appreciation of the irreducibly dialogical constitution of meaning allows us to see that harmony is actually an internal feature of any meaningful dialogical interaction between discursive subjects. Not only is harmony a precondition for a compositional inferentialist meaning theory, it is a precondition for a theory of meaningful communication between subjects as such.

3.1 Brandom Against Harmony

3.1.1 Brandom on Intrinsic Harmony

In criticizing Gentzen-style one-sided model of language use, Brandom (1994, p.121) asks us to consider the following pair of assertions:

1. I foresee that I will write a book on Hegel.

2. I will write a book on Hegel.

According to Brandom, the sentences share the same conditions under which asserting it is correct, but differs with respect to what we can infer from it. He writes,

“Any circumstances under which one is entitled to one of these claims (or to acquire the belief it expresses) are circumstances under which one is entitled to the other. [...] Yet what follows from the claim is quite different. If I will write a book about Hegel, then I will write a book about Hegel. Yet that I will write a book about Hegel does not follow from my foreseeing or predicting that I will” (Brandom, 1994, p.121).

Brandom’s reasoning is *prima facie* plausible. Whenever we assert something about the future, we rely on knowledge about what we think will happen in the future; since human beings are not omniscient, to require more, to the effect that future-tensed expressions are assertible only in case the speaker knows that something will happen, appears exorbitant. Conversely, when a speaker makes a prediction about the future, they must have a valid reason (or evidence) to support this claim. This implies that they are also entitled to assert the future-tensed claim without the epistemic qualification (viz. ‘I foresee [...]’).

As for the claim that the consequences of 1. and 2. differ seems self-explanatory. Besides, the fact that at least one of the two aspects of use (i.e. either the conditions or the consequences) comes apart for the two expressions should not come as a surprise; if both the conditions and consequences were identical, then, according to any inferentialist account, the two expressions would express the same propositional content. But it is clear that 1. and 2. express different propositions. So, given that 1. and 2. share the condition for assertion, their consequences must be different.

Brandom’s simple argument is controversial for many reasons. One reason is that it implicitly rejects the requirement for intrinsic harmony, which serves as the cornerstone of logical inferentialism. Recall that logical inferentialism is built on the very idea that condition for asserting *fixes the meaning*; the consequences of asserting should not outstrip the grounds for making that assertion. Expressions governed by inference rules that fail to comply with this requirement, on the logical inferentialist view, is incoherent.

Not only does Brandom think inferential practices display this kind of disharmony, Brandom builds his account of objectivity by appealing to the contrast between sentences like 1. and 2. So, Brandom thinks that disharmony, insofar as it allows us to distinguish between norms instituted by perspectives within discursive practice and norms that 'report' to objective reality, is a precondition for objectivity and thus genuine normativity (i.e. the possibility of norms that can issue rational 'oughts'). In this sense, then, there appears to be a fundamental rift between Brandom's inferentialism and logical inferentialism.

3.1.2 Brandom on Conservative Extendibility

Brandom thinks that logical vocabulary must satisfy the requirement of harmony-as-conservative extendibility, but that it is wrong to require non-logical (or material) concepts, to satisfy it. Brandom begins his rejection of the conservative extendibility requirement for non-logical concepts by considering the following influential passage from Dummett (1981):

"[Conceptual change in ordinary discourse can be] motivated by the desire to attain or preserve harmony between two aspects of an expression's meaning. A simple case would be that of a pejorative term, e.g. 'Boche'. The condition for applying the term to someone is that he is of German nationality; the consequences of its application are that he is barbarous and more prone to cruelty than other Europeans. We should envisage the connections in both directions as sufficiently tight as to be involved in the very meaning of the word: neither could be severed without altering its meaning. Someone who rejects the word does so because he does not want to permit a transition from the grounds for applying the term to the consequences of doing so. The addition of the term 'Boche' to a language which did not previously contain it would produce a non-conservative extension, i.e. one in which other statements which did not contain the term were inferable from other statements not containing it which were previously not inferable" (Dummett, 1981, p.454).

In a language that does not contain the concept 'Boche', (presumably) the claim 'Germans are prone to cruelty' is not valid; but, the addition of 'Boche' validates this claim, creating a non-conservative extension of the base linguistic system. So, the main take-away from this example, according to Dummett, is that linguistic practices governed by rules that result in disharmony (understood as non-conservative extension) is, even if it is currently in use and thus learnable, defective. Again, it is not immediately clear whether the 'defectiveness' involving 'Boche' should be understood epistemically or morally (i.e. as unjustified), or semantically (i.e. as incoherent). The immediate impulse is to understand the rules governing 'Boche' as unjustified (either morally or epistemically). But, as we saw in the previous chapter, Dummett understands the defect ultimately as a semantic one. Rules governing the use of 'Boche' are unjustified *precisely because* they give rise to incoherence. Recall that, for Dummett, expressions are incoherent if they "cannot be rendered comprehensible by any systematic ascription of content—and this means it would be impossible to construct a meaning-theory that accorded with all features of its practice" (Dummett, 1991, p.241). And, according Dummett's analysis, if any expression of a given language does not harmony-as-conservative extendibility, we cannot give a coherent inferentialist meaning theory.

Brandom agrees with Dummett that rules governing 'Boche' is defective. However, he explicitly rejects Dummett's explanation for their defectiveness, namely as creating a non-conservative extension of the base language. The problem with 'Boche', Brandom writes, is "not that once we explicitly confront the material inferential commitment that gives them their content, it turns out to be *novel*, but that it can then be seen to be indefensible and inappropriate" (R. Brandom, 1994, p.127). So, according to Brandom, rules governing 'Boche' are unjustified, not because it is semantically incoherent; it is unjustified because it clashes with deeper commitments that we already endorse (perhaps due to moral or epistemic considerations), namely that it is not the case that all Germans are prone to cruelty.

Brandom's explanation is wanting. The most obvious objection would be that, in a holistic system like Brandom's where all meaning-determining rules are in one basket (i.e. intertwined in a complex network), it is practically impossible to determine which

rules are defective and which rules aren't. For instance, imagine someone that has a German friend, Hans, who she knows to be kind; she also is an active member of the French Résistance, regularly employing the concept 'Boche.' The two commitments—i.e. to endorse that Hans is German and kind on one hand and to endorse statements that contain the expression 'Boche'—are clearly incompatible, but it is unclear which commitment should override the other. In particular, within a holistic inferential network, the material inferential role of 'German,' 'kind,' and 'Boche' are interdependent; the perspective from which one can judge norms as “indefensible and inappropriate” are always shifting. Indeed, this is the kind of objection raised by Prior (1960) in the previous chapter: how can we determine that some concepts are defective if its meaning is simply inferential role? It was precisely to combat this type of skepticism that the principle of harmony was articulated.

Brandom also gives another reason for rejecting that conservative extendibility is a theoretical desideratum of a meaning theory, at least for those concerning languages that involve material content: namely, that introducing concepts with substantive material content (i.e. one which is not already implicitly contained in the base linguistic system) is not, in principle, undesirable. Brandom substantiates this claim by pointing to conceptual progress in the sciences, where novel concepts are introduced all the time, yielding a non-conservative extension of the base system. Consider the following example:

The candle burned $\vdash$ The candle released phlogiston

In a discursive context that endorses the phlogiston theory, according to the conceptual norms governing the concept *phlogiston* and predicate *x burns*, this inference can be seen as valid. But of course, this inference is *invalid*, given that the phlogiston theory is debunked in the current discursive context. So, the issue at hand is the following: since any theory can be wrong in principle, the conceptual norms that determine inferential roles can themselves be wrong. Indeed, a cursory glance at the history of science attests that certain inferences, one understood at the time to be valid in virtue of the meaning of concepts involved, turned out to be invalid because a better theory—i.e. a better set of conceptual norms—came along to replace the existing one. Brandom takes it that concep-

tual progress of this kind shows that it is wrong to uphold the conservative extendibility requirement for conceptual norms governing non-formal (or material) concepts. After all, introducing the concept *oxygen* to our discursive practice created a non-conservative (and non-uniqueness preserving) extension of the base system of inferential relations; but this, Brandom wants to say, surely is a welcome change.

I am sympathetic to Brandom's line of thinking. In particular, I agree that the problem with expressions like 'Boche' is not that it creates a non-conservative extension of the base system. But I disagree that rules governing 'Boche' are coherent if unjustified. The real problem with rules governing expressions like 'Boche' are that they are intrinsically inharmonious: by introduction and elimination of the concept 'Boche', we can conclude more (i.e. an increase in information) than what is granted by the conditions for applying the concept. That is, 'He is cruel' does not follow from 'He is German,' so there is no way to level the local maxima created by the introduction and subsequent elimination of the concept 'Boche.' This is why 'Boche' is not only morally and epistemically unjustified, but also incoherent.

Notice that this explanation is not available for Brandom as he rejects that intrinsic harmony should hold for rules governing natural language expressions. In the next sections, we will see how we can account for Brandom-style counterexamples to intrinsic harmony by drawing on the Dialogical tradition in logic.

3.2 Dialogical logic

Dialogical logic is not a 'logic' in the sense of deductive systems (like classical or intuitionistic logic) but a "general framework having a rule-based approach to meaning (instead of a truth-functional or a model-theoretical approach) which allows different logics to be developed, combined and compared within it" whose foundational idea is that "meaning and rationality are constituted by argumentative interaction between epistemic subjects" (Rahman et al., 2018, p.57). As such, it can be seen, to some extent³, as an implementation of Brandom's game of giving and asking for reasons. As Marion (2010) notes,

³The reason for this qualification will be soon clear.

“My suggestion is simply that dialogical logic is perfectly suited for a precisification of these ‘assertion games’. This opens the way to a ‘game-semantical’ treatment of the ‘game of giving and asking for reasons’: ‘asking for reasons’ corresponds to ‘attacks’ in dialogical logic, while ‘giving reasons’ corresponds to ‘defences’. In the Erlangen School, attacks were indeed described as ‘rights’ and defences as ‘duties’, so we have the following equivalences:

Right to attack $\iff$ asking for reasons

Duty to defend $\iff$ giving reasons

The point of winning ‘assertion games’, i.e., successfully defending one’s assertion against an opponent, is that one has thus provided a justification or reason for one’s assertion” (p.490).

Indeed, there are good reasons to think that dialogical logic can make explicit the process by which meaning comes into view of discursive agents via the game of giving and asking for reason, which is ironically left rather implicit by Brandom. Since, according to R. Brandom (1994) himself, “The conceptual contents employed in *monological reasoning*, in which all the premises and conclusions are potential commitments of one individual, are parasitic on and intelligible only in terms of the conceptual contents conferred by *dialogical reasoning*, in which the issue of what follows from what essentially involves assessments from the different social perspectives of interlocutors with different background commitments” (p.497), it is rather puzzling that his formal semantic projects (viz. incompatibility semantics) abstracts away from this essential dialogical dimension of reasoning.

With that said, let me give an overview of how Dialogical Logic works.

3.2.1 Basic Overview of Dialogical Logic

Note: This subsection relies heavily on Clerbout & McConaughy (2022), since it is the clearest exposition of the framework available in English. Many parts of it—namely the explanation of how to build a play—are direct quotations from this source with minor changes.

Dialogical logic studies dialogues, and takes the form of dialogues, thereby giving semantics a game-theoretical treatment.⁴ In a dialogue, two parties (players) argue on a thesis (a certain statement that is the subject of the whole argument) and follow certain fixed rules in their argument. The player who states the thesis is the Proponent, called **P**, and his rival, the player who challenges the thesis, is the Opponent, called **O**. In challenging the Proponent's thesis, the Opponent is requiring of the Proponent that he defends his statement. Actions in a dialogue are called moves; they are often understood as speech-acts involving declarative utterances (statements) and interrogative utterances (requests). The rules for dialogues thus never deal with expressions isolated from the act of uttering them.

The rules in the dialogical framework are divided into two kinds of rules: particle rules, and structural rules.

Particle Rules

Particle rules (*Partikelregeln*), or rules for logical constants, determine the legal moves in a play and regulate interaction by establishing the relevant moves constituting challenges and defences, depending on the main logical constant in the expression at stake. The appropriate challenges and defences (interaction) are the appropriate ways of asking for reasons and of giving them, and are specific to each kind of statement. (These rules are analogous to how the pieces move in chess.)

The below table shows particle rules for constants that make up first-order logic. Two things to note immediately about the notation: i) **X** and **Y** represent anonymous players so that the meaning of the logical constants stays player-independent; ii) the exclamation mark after the player indicates that the player asserts the following proposition, the question mark indicates that the player is making a request (i.e. asking for reasons).

⁴Here, I will discuss the framework as it pertains to the Lorenzen and Lorenz tradition (e.g. Lorenzen & Lorenz (1978)), though there are many other traditions working broadly on dialogue-based logical/argumentation frameworks. Cf. Clerbout & McConaughey (2022).

Table 1: Particle Rules for First-order Logic			
	Statement	Challenge	Defense
Conjunction	$\mathbf{X!}A \wedge B$	$\mathbf{Y?} L^\wedge$ or $\mathbf{Y?} R^\wedge$	$\mathbf{X!}A$ respectively $\mathbf{X!}B$
Disjunction	$\mathbf{X!}A \vee B$	$\mathbf{Y?}_\vee$	$\mathbf{X!}A$ or $\mathbf{X!}B$
Implication	$\mathbf{X!}A \rightarrow B$	$\mathbf{Y?}A$	$\mathbf{X!}B$
Negation	$\mathbf{X!}\neg A$	$\mathbf{Y?}A$	–
Universal quantification	$\mathbf{X!}\forall x P(x)$	$\mathbf{Y?}[x/a_i]$	$\mathbf{X!}P[x/a_i]$
Existential quantification	$\mathbf{X!}\exists x P(x)$	$\mathbf{Y?}_\exists$	$\mathbf{X!}P[x/a_i]$

Structural Rules

Structural rules (*Rahmenregeln*) determine the general course of a dialogue game, such as how it is initiated, how to play it, how it ends, and so on. In contrast to the particle rules, the point of these rules is not to spell out the meaning of the logical constants by specifying how to act in an appropriate way; it is rather to specify the structure according to which interactions take place.⁵ It is one thing to determine the meaning of the logical constants as a set of appropriate challenges and defences (as above), it is another to define whose turn it is to play and when a player is allowed to play a move. (These rules are analogous to the starting position, what constitutes a checkmate, etc. in chess.) There are four basic structural rules in the standard dialogical framework, defining which conditions must be respected for a given sequence of moves to constitute a legitimate play.

SR0: Starting Rule A play starts with the proponent stating a proposition called the thesis. After the thesis has been stated, the opponent chooses a positive integer which will be her repetition rank during the play. Then the proponent chooses his own repetition rank.

SR1: Development Rule Players move alternately. Once the repetition ranks have been chosen, each move is either attacking or defending a move made by the other player, in accordance with the particle rules.

⁵In order to avoid any confusion, it should be noted that these rules are quite different in kind from what is known as structural rules in sequent calculus (i.e. cut, weakening and contraction, permutation).

SR2: Copy-cat Rule (also known as *Formal* or *Socratic* Rule) **P** can play an atomic formula (e.g., Pa) only if **O** has stated it previously in the play.

SR3: Winning Rule The play ends when it is a player's turn to make a move but he has no available move left. That player loses and the other player wins.

Dialogical Meaning Explanation: Play vs. Strategy Level

So far, we have been focusing on how actual plays are constructed. But the dialogical framework also allows for one to study games in terms of collections of plays. The former layer—in which actual games are carried out—is known as the *play level*; the latter layer—in which a collection of possible plays for a given thesis is studied—is known as the *strategy level*.

A strategy for player **X** in a dialogical game is defined as a *complete conditional plan* of action. It is conditional in the sense that the strategy informs how **X** plays depending on the moves of the adversary. It is complete in the sense that it must inform how **X** plays for every possible choice of move by the adversary. Given a formula A , a strategy for **X** in the game for A can be described as a function that assigns an **X**-move M to every non-terminal play Δ ending with a **Y**-move such that extending Δ with M results in a play for A . Such a strategy is *winning* if all terminal plays resulting from it are **X**-terminal (i.e., won by **X**). In other words, a winning strategy for a player is a complete conditional plan of action leading to the player's victory no matter how the adversary plays.

The notion of winning strategy is important because it corresponds to the proof-theoretic notion of *validity*. Indeed, various algorithms have been devised to turn the procedure for construction of winning strategy within the Dialogical framework into proofs in other formal frameworks (and vice-versa).⁶ As mentioned in passing in Chapter 2, the clear separation of the play-level and the strategy-level is one of the founding insights of Dialogical Logic in the Lorenzen and Lorenz's tradition. What the distinction allows for, in essence, is to designate the play-level as the layer where meaning is constituted, while associating the with validity features. This distinction becomes obvious once we reflect on the fact

⁶For example, in Chapter 9 of Rahman et al. (2018), the correspondence between winning strategies and demonstrations in Martin-Löf's Constructive Type Theory is demonstrated.

that it is possible for proponent (or opponent) to lose an individual play even if the thesis is valid (or invalid), that is to say, has an associated winning strategy. What the player needs to know in order to count as having the grasp of the meaning of the expression is explained purely pragmatically (i.e. as a know-how): knowing how to (legally) take part in play in accordance with the particle rules and strategy rules.

On the Dialogical view, the meaning of an expression is constituted in the actual interaction of the players during a play. Within the play level, there is a fine-grained distinction *local* meaning and *global* meaning. The *local meaning*, which pertains to logical constants, is set by the particle rules that determine the proprieties of interaction. Notice that some particle rules specify which player gets to choose which elementary proposition to defend. For example, the difference between a conjunction and a disjunction, is that in the case of a conjunction, the challenger chooses the conjunct that will be defended, whereas in the case of a disjunction, it is the defender who chooses the disjunct (and similarly for universal and existential quantification). The meaning of logical constants is then not captured by which proposition it figures into (i.e. in an assertion or request) but also by the *choices* governing which player can do what with it. So, it is the actual plays ground meaning. Crucially, particle rules set local meaning, which is *player-independent*: the rules do not change depending on which player (i.e. **P** or **O**) defends or challenges.

In contrast to the local meaning of constants which are specified by player-independent rules for interaction, the global meaning is governed by *player-dependent* structural rules, which specify the structure according to which interactions must take place. The contrast between these two levels is made obvious by the fact that one can achieve the validity features of different deductive systems simply by modifying the structural rules (i.e. maintaining the same particle rules). The difference between local and global meanings are made salient when we compare how validity features of different logics can be expressed in the dialogical framework. The particle rules and structural rules defined in this subsection is a dialogical rendering of classical logic. We can, for example, yield validity features (or the consequence relation) of intuitionistic logic by simply replacing **SR1**, by its intuitionistic counterpart:

SR1i: Intuitionistic Development Rule After the repetition ranks have been chosen, each

move is a challenge or a defence in reaction to a previous move, in accordance with the particle rules shown previously. Each player can challenge the same previous move at most n times—where n is the player's repetition rank—or defend against the adversary's last unanswered challenge.

The point is that we can keep fixed all of the particle rules, thus maintaining the local meaning, and yet by changing the structural rules, we thereby change the global meaning and thus validity features associated with the strategy-level.⁷

Thus, meaning, both local and global, is a feature of the play-level (structured by particle and structural rules). If meaning is to be constituted in a dialogue (i.e. interactions of the players following rules), then dialogue games must be characterized something humans can actually perform. Therefore, the games must be finite: *Dialogue-definiteness* is thus introduced as a constraint of all dialogues. This notion is important for two reasons: first, it captures the human cognitive constraint; second, it allows us to define what a *proposition* is: proposition is a *dialogue-definite expression*, that is, as an expression A such that there is an individual play about A that can be said to be lost or won after a finite number of steps, following some given rules of dialogical interaction.⁸

Finally, an important philosophical point is in order. According to the dialogical perspective, it is the particle rules that ground the structural rules, and the play-level that grounds the strategy-level. Particle rules ground the structural rules in the sense that the structural rules ought to be formulated in accordance with the particle rules: the sole purpose of structural rules are to specify when and how we can implement that particle rules of interaction (i.e. it provides the context in which we play). In turn, the strategy-level is grounded on the play-level in the sense that strategies, and by extension *winning strategy* (i.e. validity), only make sense to talk about when a course of choices implemented in

⁷There is a similarity between this idea and the proof-theoretic idea that we can distinguish between classical and intuitionistic logic within sequent calculus without modifying the inference rules, instead specifying whether conclusions must be single (yielding classical logic) or multiple (yielding intuitionistic logic). In this sense, the dialogical framework vindicates the idea behind *Meaning Invariant Logical Pluralism* due to Beall & Restall (2000), though of course there are important differences due to the fact that the latter proof-theoretic approach focuses only on inferential relations (or, in dialogical terms, the strategy-level).

⁸Cf. Lorenz (2001, p. 258): for an expression to *count* as a proposition A there must exist an individual play about the statement $X!A$, in the course of which X is committed to bring forward a local reason to back that proposition, play which must result in either win or loss after a finite number of moves according to definite particle and structural rules.

the play-level by each player is abstracted for the purposes of analysis. (Consider: what would a strategy for playing chess consist in if there were no such game as chess?) As Rahman & Keiff (2005) put it,

“Indeed, the distinctive feature of the pragmatic semantics of dialogic is its connection to proof, both in its elementary and its complex level. In its elementary level it is associated to prime formulæ which express a proposition understood as encoding an informal proof, conceived as the predicate atomic language games. In this context, the particle rules show how to play a game fixing the semantics for complex formulæ based on the atomic (game) meaning assigned to prime formulæ – but (possibly) independently of any determinate assignation. Now, one very important point, and a quite often misunderstood tenet of the dialogic, is that this theory of meaning should set the basis for the formal concept of proof leading to the notion of validity. Once more, *the meaning in use of the notion of informal proof underlying the local semantics should furnish the way to formal proof, and not the other way round*” (p.19, my italics).

In fact, the fine-structure of dialogues takes care of the Dummettian claim that meaning theories should be compositional without an explicit appeal to the principle of harmony (though harmony, as we will soon see, implicitly holds in some sense within the dialogical framework). We should expect any dialogical framework, regardless of which logic it implements, to give rise to a compositional meaning theory: the meaning of an atomic formulae (if any) are constituted in the dialogical interaction that are pre- or informal; the meaning of complex formula are determined by the rules of interaction and the meaning of atomic ones.⁹

⁹The standard dialogical framework does not have the expressive power necessary within the object language to model what has been characterized here as the ‘informal’ proof, and thus cannot provide a meaning theory for atomic formula. The standard dialogical framework, enriched with the expressive resources of Martin-Löf (1984)’s Constructive Type Theory, can deal with the meaning of atomic formulae, cf. Rahman et al. (2018).

3.2.2 Dialogical Explanations for Defective Rules

I have promised that a proper appreciation of the dialogical constitution of meaning allows us to see why Brandom is too quick to reject the requirement for harmony. I will now show that Dialogical logic has the resources to account for linguistic phenomena that purportedly undermine the requirement for harmony.

Tonkitis

The dialogical framework gives us a straightforward explanation of why certain inference rules, namely those governing the meaning of logical constants, are defective. The case of Prior's *tonk* will once again be the test case. Recall that the inferential role of *tonk* is given by the following rules:

$$\frac{A}{A \text{ tonk } B} \text{ tonk-I} \quad \frac{A \text{ tonk } B}{B} \text{ tonk-E}$$

Now, the challenge is to reformulate these rules, which describe inferential validity, into particle rules, which describe proprieties of interaction, within the dialogical framework (given, lets say, the structural rules that yield classical logic). The short dialogical answer to why *tonk* is defective is that we simply cannot articulate coherent particle rules that capture the validity features given by inference rules governing it.

Informally, it's not difficult to see why. Suppose that we want to implement the run-about inference (i.e. from A to B via $A \text{ tonk } B$) in the dialogical framework. Accordingly, we set B as the thesis, and **O** concedes A to begin the play. Now, **P** asserts $A \text{ tonk}_1 B$. The opponent will now challenge this claim. Now, what would consist in an appropriate challenge? There are two pairs of possibility here. The first pair consist of asserting an atomic formulae (i.e., ' $\mathbf{O}!A$ ' or ' $\mathbf{O}!B$ ') and the second pair consists of a request, (' $\mathbf{O}?R^{\text{tonk}}$ ' or ' $\mathbf{O}?L^{\text{tonk}}$ ' or ' $\mathbf{O}?_{\text{tonk}}$ '). Now, **SR2**, or the Copy-cat Rule, states that **P** can only state an atomic formula only if **O** has stated it previously in the play. Therefore, the only option to secure a winning strategy for **P**, which is what is required to capture the inference rules for *tonk* given the equivalence between validity and winning strategy, is for **O** to challenge by stating the right-side formula, i.e. ' $\mathbf{O}!B$ '. (Note also that the concession made by

the opponent, i.e. A , was not even invoked in the dialogue at all.)

There is clearly something circular about this meaning explanation. In fact, since the challenge ' $X!B$ ' and the defense ' $Y!B$ ' are in fact identical assertions, tonk may not be seen as a dyadic connective at all: the meaning would reduce to that of B . The philosophical point here is that the Copy-cat rule secures a form of analyticity that emerges in the interaction of the proponent and the opponent. Given that, for example, only A has been conceded by the opponent (perhaps as a result of a previous material dialogue concerning the elementary statement), no thesis stronger than A can be associated with a winning strategy. Indeed, the Copy-cat rule can be seen formally as the dialogical rendering of the principle of (intrinsic) harmony (Rahman et al., 2018, p.80).¹⁰

Nonetheless, we might force an interpretation of tonk such that the particle rules make it clear that tonk is a dyadic connective and still model its validity features. In that case, we must sacrifice the player-independence of particle rules:

Table 2: (Proposed) Particle Rules for tonk			
	Statement	Challenge	Defense
tonk_1	$X!A\text{tonk}_1B$	$Y?\text{tonk}_2$	$X!A$ or $X!B$
tonk_2	$X!A\text{tonk}_2B$	$Y?L^{\text{tonk}_1}$ or $Y?R^{\text{tonk}_1}$	$X!A(\text{resp.})O!B$

The particle rules for tonk_1 are essentially modeled after disjunction: the defender has the choice in which atomic formula to request. The particle rules for tonk_2 , by contrast, is modeled after conjunction: the challenger has the choice in which atomic formula to request. We can only implement the inference rules for tonk (i.e. build a winning strategy for $\mathbf{P}$ given by Prior) in the dialogical framework iff tonk_1 is applied when $\mathbf{P}$ states and tonk_2 is applied when $\mathbf{O}$ states. But particle rules, in fixing the local meaning for both parties, should be player-independent. Since tonk must violate player-independence in order to achieve the desired validity features, it does not have a stable local meaning.

So, tonk is ruled out in the dialogical framework in two ways: by symmetry, based on

¹⁰It would be interesting to see if the Copy-cat rule also guarantees *stability*, which in the dialogical framework would be the constraint that the proponent should not be able to say less than what is granted by the opponent. It is *prima facie* implausible that it does; however, it would be worth investigating whether stability should be incorporated into the dialogical framework, and if so, what the adequate formulation of the structural rule should be.

considerations pertaining to local meaning, and by intrinsic harmony, based on considerations pertaining to global meaning. Indeed, symmetry and intrinsic harmony are crucial aspects of the dialogical approach to meaning. As Rahman et al. (2018) put it, these are the very “principles for *immanent reasoning*” (p.81, my italics). Reasoning in accordance with these principles is immanent in the sense that the reasoning requires no external assumption or perspective to those which are contained within a given dialogue.¹¹

Let me close this discussion with a quick methodological point. As explained, a constant like `tonk` could never arise from a dialogical understanding of meaning as constituted in the play-level, due to both constraints at the local and global level. Only when validity features given by inference rules at the strategy-level are imposed onto the play-level in a top-down manner do these problems arise.

Asymmetry between conditions and consequences

Dialogical logic also has a straightforward explanation of the seeming asymmetry in the (identical) assertability conditions and (differing) consequences of application for claims that Brandom invokes to substantiate his account of objectivity, like the following two pairs:

1. (a) I foresee that I will write a book on Hegel.
(b) I will write a book on Hegel.
2. (a) The swatch is red.
(b) The claim that the swatch is red is properly assertible by me now.

Brandom objects to Gentzen-style one-sided meaning theories and, with it, the harmony conceived of as a local property, on the basis that the first claim of each pair and the second of each pair share the same assertibility conditions but differ in their consequences. The dialogical view is to insist that Brandom—and inferentialists more generally—have the play-level (pragmatic/semantic layer) and the strategy-level (ontological layer) confused.

¹¹This notion of immanent reasoning has interesting resemblances and connections with the Hegelian tradition of *immanent critique* popular in Critical Theory, cf. Jaeggi (2018). My intuition is that immanent reasoning could complement the project of immanent critique by providing a semantic foundation for it.

Let us review why Brandom thinks that 1a and 1b share the same assertibility conditions. (I will use Brandom's first pair of examples, even though analogous points apply for the second pair.) The motivating idea is that whenever we have enough evidence (e.g., an intention to write the book, the fact that 'I' have been in the past capable of fulfilling my intentions, etc.) to serve as a reason for 1a, the same evidence should suffice as a reason for 1b. Given that humans are not omniscient beings, we cannot know for sure whether 1b will realize; so, assuming that we can coherently assert 1b, it must be the case that 1a and 1b must share assertibility conditions.

But from a dialogical perspective, the assertibility conditions of the 1a and 1b are different. Recall that, in technical terms, 'inference' belongs to the strategy-level, as the minimal semantic unit for which the notion of validity applies to. Thus, when we are engaged in giving and asking for reasons (i.e. the play-level), we are not drawing inferences in this technical sense. Since conditions for application (of conceptual norms) and consequences of application are *inferential relations*, they belong to the strategy-level. So, we may make the respective equivalence between the inferentialist questions, "What are the conditions for asserting A?" and "What are the consequences of asserting A?" and the dialogical questions, "Is there a winning strategy for A, and if so, how can we build it?" and "What are the claims for which winning strategies can be built, on the basis that there is a winning strategy for A?"

There are, in general, no winning strategies available for theses that involve empirical vocabulary, since empirical reasoning is defeasible.¹² Nonetheless, we might ask, "What sort of conditions must obtain for something to count as a winning strategy for a claim?" One can make explicit, by treating time modally (as in branching-time temporal logic¹³), that 1a has an associated winning strategy iff we can show that there is at least one history (i.e. course of events), or perhaps most histories, there is a point at which 'Brandom writes a book on Hegel' is true. On the other hand, 1b has an associated winning strategy iff we

¹²What about empirical analytic statements? A case could be made that winning strategies can be built for such statements, but this reader of Quine's "Two Dogmas of Empiricism" remain unconvinced. In dialogical terms, 'subplays' can be opened for any elementary statements. This involves playing *material dialogues*, which I do not have the space to discuss here. But the general idea is that rules governing any empirical statements are subject to change as new information are introduced. This is the same point Brandom makes by invoking the example of conceptual progress in the sciences.

¹³Cf. for example Prior (1967).

can show that in all possible histories, there is a point at which ‘Brandom writes a book on Hegel’ is true. So, the ‘winning-strategy conditions’ for 1a and 1b differ. To translate this back to Brandomian language: the conditions for and consequences of assertion for 1a and 1b differ.

If the conclusion that there are no winning strategies for empirical statements were to imply that we cannot meaningfully use empirical concepts (in assertions), this would be a blow for the dialogical framework. But, recall the dialogical insistence that meaning is constituted at the play-level, and that proposition is a dialogue-definite expression. The point is that **P** and **O** can play to win according to discursive rules, thereby constituting the meaning of the expressions. In addition, if the play results in **P**’s win, she is able to assert the propositionally contentful expression, provided that the normative validity for the assertion is understood locally within the context of the dialogue. For an assertion of a universally valid sort, **P**’s reason for asserting must account for all the possible plays that constitute a strategy. In particular, **P** must be in possession of the know-how to build a winning strategy.

In short, the dialogical approach allows us to account for the asymmetry in meaning between statements like 1a and 1b, while at the same time upholding the principle of intrinsic harmony at the level of inference, understood as a certain necessary coordination (as prescribed by the Socratic rule) between **P** and **O** for a meaningful dialogue.

3.3 Conclusion

I think that Brandom is too quick to dismiss the requirement of intrinsic harmony, since intrinsic harmony is just the requirement that wards off worries about conventionalism about the meaning of logical constants understood inferentially. Furthermore, appreciating the dialogical constitution of meaning dispels Brandom-style counterexamples to intrinsic harmony, wherein two expressions seem to have identical conditions for application but differing consequences.

An important consequence of this result is that Brandom’s explanation for objectivity (as attitude transcendence of norms) appears to be fundamentally misguided, as his

objectivity proofs presuppose the idea that expressions governed by inharmonious rules are not only coherent but necessary to make sense of objectivity. So, the question, “Is objectivity possible in Brandom’s system?”, is, in my opinion, moot. Brandom’s efforts to refashion objectivity is futile. What we need in order to ward off the charge of relativism and anti-rationalism is to identify the *source* of normativity: in virtue of what are rules genuinely normative? The answer to this question lies, as we will see in the next chapter, in intrinsic harmony.

Chapter 4

Constitutivism about Semantic Normativity

Now, let us bring together the various strands that have been developed throughout this thesis:

1. Brandom-style inferentialism must account for some form of objectivity (minimally construed as attitude-transcendence of norms) in order to avoid relativism about conceptual norms, but whether Brandom's own account succeeds is at least questionable, if not incoherent (Chapter 1).
2. For logical vocabulary, the requirement that intrinsic harmony must obtain between rules governing the meaning of an expression provides this normative standard viz. attitude-transcendence of norms, but the principle is not grounded on substantial fact/principle (Chapter 2).
3. A proper appreciation of the dialogical constitution of meaning shows that intrinsic harmony is part and parcel of meaningful communication between discursive subjects. (Chapter 3).

In this chapter, I aim to make explicit the grounds for semantic normativity. The principal question is the following: how do conceptual norms generate genuinely normative reasons for concept application, paradigmatically in assertions and judgments? That is,

in virtue of what do conceptual norms have their normative authority over our thoughts? My answer is that semantic normativity can be grounded on constitutive principles of discursive agency.

In section 1, I summarize the constitutivist strategy to ground practical normativity in metaethics, with particular focus on Korsgaard (2009)'s Kantian Constitutivism. Then, in section 2, I show that we can apply the same general strategy to conceptual norms. There, I present my approach to grounding semantic normativity, which I call *Dialogical Constitutivism*. In the final section, I deal with some possible objections to my account.

4.1 Constitutivism

A popular strategy in metaethics is to explain the normativity of ethical claims by appealing to facts about the nature of agency. *Constitutivism* is a family of views that adopts this strategy (e.g. Korsgaard (2009), Velleman (2000), Katsafanas (2013), Street (2010)). Its central idea is that certain features of action are constitutive features of action, such that in the absence of those features, an event would cease to be an action. In other words, some features demarcate *action* from mere *behavior*. The constitutive features then provide us with normative standards against which actions, and derivatively, ethical claims, may be evaluated.

4.1.1 The General Idea

The standard example that illustrates how constitutivism works is the game of chess. When a subject plays chess, she aims at winning (i.e. checkmating her opponent), or at the very least, obtaining a draw. If she simply moves around pieces on the board in accordance with what is permitted by the rules of chess, then she is not playing chess at all. So, all moves that the subject makes implicitly have the aim of checkmating. Thus, the aim of checkmating one's opponent is a *constitutive aim* of playing chess; making arbitrary moves on the chess board without this aim, therefore, does not count as playing chess.

Two immediate qualifications are in order. First, winning is rarely the sole aim of play-

ing chess. In fact, it would be rather sad and dull if it were. Trying out a new, and thus risky, opening and purposefully putting oneself in a bad (*qua* more likely to be checkmated) position to take on a challenge are things chess players regularly do. As these examples show, auxiliary aims often are in direct conflict with the primary aim of playing chess, which we analyzed as winning. But what makes these aims auxiliary or secondary and the aim of winning primary is precisely the fact that these kinds of deviation from the aim of winning can only go so far before one ceases to be engaged in the game of chess.

Relatedly, there is another worry: it seems possible to play chess with the active intention of losing, for instance, when a more experienced player (say a father) wants to encourage a less experienced player (his daughter) to keep playing and improving. The father's aim actively undermines the supposed aim of chess, but surely, the objector would point out, that he is nonetheless still playing chess. A typical constitutivist reply is the following is that if all chess players were to play with the intention of losing at all times, playing chess would cease to be an activity, or at least, it would radically change the nature of the activity such that we can no longer say that it is the same activity. Thus, playing to checkmate is a constitutive aim of chess playing as it is practiced today.

This reply points towards a general call for a more fine-grained analysis of what events count as what activity. This is the second qualification. For example, the constitutivist might say that the activity in question (i.e. chess playing the with intention of losing) is a more complex variant of playing chess. As Katsafanas (2018) writes,

"The important point is that describing the agent simply as playing chess in these cases elides important details. It's as if we describe the person interviewing for a job as *talking*—it's true, but it leaves out further aims that she has, aims that modify the activity in significant ways, and thereby require that someone who analyzes the features of the activity describe it more precisely. Just so with chess: if you find out that your opponent is playing with the intention of losing, then in ordinary circumstances it does not matter whether we describe him as *playing chess* or *playing chess with the intention of losing*. However, the latter, more specific description picks out important features that distinguish this action from an ordinary case of playing chess" (p.368, original

italics).

Now, constitutivists agree that actions in general have constitutive features that distinguishes them from other events. Highly contentious is what the constitutive features consist in. Velleman (2000), for example, suggests that action constitutively aims at *self-understanding*; contrarily, Katsafanas (2013) argues that action has two constitutive aims: “(1) agential activity, which [is] analyzed as requiring stable approval of one’s action given complete information about its etiology, and (2) encountering and overcoming resistance to one’s ends” (p.376). The general idea is that these constitutive aims issue normative reasons for action, in the sense that we have all-things-considered reasons to successfully achieve the aim because that is just what practical rationality consists in.

4.1.2 Korsgaard’s Kantian Constitutivism

Up to this point, we have given an overview of the dominant kind of constitutivism which extracts constitutive features based on a special kind of intention (viz. ‘aim’) of the practical agent. But this is not the only way to think about constitutive features of action. Another approach is to identify the constitutive feature of action with its *constitutive standard* or norm. Thus, we arrive at two kinds of constitutivist views: aim-based (or Humean) accounts and rule-based (or Kantian) accounts. The former takes aims to be the constitutive feature of action, while the latter takes norm-governedness to be the constitutive feature of action. The main proponent of the latter view is Christine Korsgaard. Since it is the latter view that is relevant for the purposes of this paper, I will spend this section outlining Korsgaard’s main ideas.

To get the discussion started, consider the Aristotelian conception of nature under which categories of objects are individuated by their respective *telos* or functions. What makes something a hammer, for example, is that we can use it to hammer in nails. The objects’ characteristic function, then, give rise to normative standards for evaluation: a hammer whose handle is broken and thus cannot be used to hammer a nail in place is a *defective* hammer. Thus, on the teleological view of nature, functions give rise to constitutive standards.

Furthermore, if the hammer turns out to be unusable *qua* hammer and is beyond the point of repair, then that object would cease to be a hammer. This is what Korsgaard (2009) calls the ‘constitution requirement’: conforming to the standards that apply to an object *qua* its kind is what constitutes the object as the kind of object that it is. That said, this does not preclude the possibility of the object falling short of a constitutive standard that apply to its kind, and yet still belong to it. But, as Korsgaard (2019) writes,

“There is a limit—if the object falls too far below the standard, the object ceases to exemplify the kind. A standard is constitutive in this specific sense not merely if it arises in some general way from the nature or function of the object, but if it arises from (something like) the very concept of the object. A house is, by definition, a kind of shelter. It may have a leaky roof and still be a house, but a thing that provides no protection from the elements—a thing that has altogether ceased to shelter—is no longer a house” (p.100)

In Korsgaard’s view, action, too, has constitutive standards. It also comes in degrees, in the sense that the more an action conforms to the constitutive standards, the more fully it is an action. But what are the constitutive standards of action? Korsgaard’s analysis begins by taking primitive the idea that action is essentially performed by an agent, or some person to whom we attribute the movement and its causal effects to as its author (2009, p.18). Actions are accordingly separated from other events (like mere behavior) due to the fact that it can be attributed to an agent. So, the constitutive principles of action trades on the constitutive principles of agency. The notion of an agent is then cashed out as “the autonomous and efficacious cause of her own movements” (2009, p.213). According to Korsgaard, in order to be *efficacious*, one must act in accordance with the hypothetical imperative; in order to be *autonomous*, one must act in accordance with the categorical imperative. By conforming to these principles, one *constitutes* herself as an agent. So, the principles of practical rationality—which is cashed out as Kantian categorical and hypothetical imperatives—are the constitutive principles of of agency, and so derivatively, of action.

Korsgaard’s argument in support of these claims is rich and complex, and I don’t have

the space here to do it justice. Here, I will content myself with presenting three of her key moves, relevant to the argument I will make later on: Korsgaard's reading of the categorical imperative (CI) and its relation to autonomy, her reading of the hypothetical imperative (HI) and its relation to efficacy, and her general approach to grounding normativity on human necessity.

First up is Korsgaard's reading of the categorical imperative and its relation to autonomy. Her understanding of CI relies heavily on the first formulation: "act only in accordance with that maxim through which you can at the same time will that it become a universal law" (Kant, 1785/2003, 4:421). Korsgaard rejects the prevailing reading of the CI test, which states that maxims allow us to evaluate the moral value of an action; this view is typically juxtaposed with the Aristotelian view, which insists that general rules cannot help us with evaluating specific actions, as only virtuous (or *phronetic*) perceivers can determine the goodness of an action. By contrast, Korsgaard reads Kant as an Aristotelian, in at least two major ways: Kant also thinks that moral characteristics do not—at least primarily—attach to act-types, and that what is at stake is the *form* of the maxim, understood in Aristotle's sense (Korsgaard, 2008, p.225-229). The form, according to Aristotle's metaphysics, is the arrangement of the parts of something that enables to perform its function. Given this, Korsgaard writes, "The parts of the action as described in a Kantian maxim are, primarily, the act and the end. [...] Kant's question is whether the act and the end are so arranged, or related to one another, that the maxim can serve as a universal law" (Korsgaard, 2009, p.15).

To illustrate this point, consider the following maxims (from Plato's *Republic*):

1. I will keep my weapon, because I want it for myself.
2. I will refuse to return your weapon, because I want it for myself.
3. I will refuse to return your weapon, because you have run mad and may hurt someone.

Maxims 1 and 3 describe good actions, while maxim 2 describes a bad one. Maxim 1 and 2 share the same kind of motivations (ends); maxim 2 and 3 share the same kind of

actions (means). The asymmetry of the moral valuations arise precisely from the fact that moral value does not rest in the individual parts (the content), but the way it is related (the form). Thus, “if the action and the purpose are related to one another so that the maxim can be willed as a universal law, then the maxim is good” (Korsgaard, 1996, p.108). As such, what distinguishes the agent, “is not first and foremost what he thinks about when he decides, but rather the way he deliberates when he makes his decisions” (ibid., p.48).

Second, according Korsgaard, hypothetical imperative “says that if you will an end, you must will the means to that end” (2009, p.68). In this sense, hypothetical imperatives can be understood as compliance with instrumental reason. But of course, Korsgaard rejects the Humean notion of instrumental reason (also known as *internalism* about practical reason), which, in its crudest form, says that we have normative reason to do whatever it takes (means) to satisfy our desires (end) given some background beliefs about how that may be achieved. Instead, Korsgaard follows Kant (1795/2003, 4:412) in understanding rationality as self-determination: unlike non-rational beings that behave/move in accordance with laws of nature, rational beings act in accordance with in accordance with its own representation of a law. That is, it is not enough that *my body* causes the movements for it to count as an action; *I*, the rational person, must *will* it:

“Suppose I decide to get some work done on my book today. At this moment, now, I decide, I will, to work; at the next moment, at any moment (importantly, maybe even at this moment), I will certainly want to stop. If I am to work I must will it—and that means I must determine myself to stay on its track. Timidity, idleness, and depression will exert their claims in turn, will attempt to control or overrule my will, to divert me from my work. Am I to let these forces determine my movements? At each moment I must say to them: “I am not you; my will is this work.” Desire and temptation will also take their turns. “I am not a shameful thing like timidity,” desire will say, “follow me and your life will be sweet.” But if I give in to each claim as it appears I will do nothing and I will not have a life. For to will an end is not just to cause it, not even if the cause is one of own my desires and impulses, but to consciously pick up the reins, and make myself the cause of the end. The reason that I

must conform to the hypothetical imperative is that if I don't conform to it, if I always allow myself to be derailed by timidity, idleness, or depression, then I never really will an end. The desire to pursue the end and the desires that draw me away from it each hold sway in their turn, but my will is never active. The distinction between my will and the operation of the desires and impulses in me does not exist, and that means that I, considered as an agent, do not exist" (Korsgaard, 2009, p.69-70)

Crucially, Korsgaard's conception of HI is, ultimately, deflationary, in the sense that it is superfluous to CI. It is not the case that we first will some end as a universal law (according to CI), then try to find ways to fulfill it (according to HI). Every HI, which consists of a practical judgment that leads to an intention/action, has to be underwritten by the corresponding CI, a theoretical judgment whose form can be universalized. To understand this, we need to hark back to Kant's definition of rationality as self-determination: we are only free inasmuch as we set our own laws and follow them, therefore, "every rational being must so act as if he were through his maxim always a lawmaking member in the universal kingdom of ends." Given this, if the movements we make are not underwritten by the CI, then they are not really free, which implies that the efficacy of those movements cannot be attributed to the agent. Thus, "the hypothetical imperative is not really a separate principle at all; rather, it captures an aspect of the categorical imperative: the fact that the laws of our will must be practical laws" (Korsgaard, 2009, p.70).

The third and final key point is Korsgaard's master argument for 'grounding' practical normativity. For our purposes, we need not get into the weeds of the metaphysical discussion concerning what the relevant grounding relation consists in. As Chang (2013) suggests, we can work with "the basic idea that fact *y* grounds fact *x* when *y* gives a metaphysically necessary explanation of *x* that is not causation, modal covariation, or subsumption" (p.165). All we need to say for now is that some things (like reason or rules governing action) count as genuinely normative *in virtue of* some other thing, and that this relation is transitive.

Korsgaard's master argument (2008, p.25-26), then, goes like this:¹

¹I have opted to simplify the argument slightly for the sake of concision. In fact, Korsgaard adds another

1. It is (rationally) necessary to conform to the principles of practical reason in virtue of our being an agent necessarily.
2. Action requires an agent, so the necessity of being an agent comes down to the necessity of acting.
3. The necessity of choosing and acting is is “our plight: the simple inexorable fact of the human condition” (ibid., p.2)

So, Korsgaard’s strategy is to ground the categorical imperative on the human condition, namely, the fact that we must choose and act. Since we must act and since universally willing whatever maxim we act in accordance with is what allows us to act in the first place, categorical imperatives ought to have a normative grip on us.

Now, the reasoning may sound circular. In the final section, I will deal with some major objections to Korsgaard’s constitutivism as it affects my variant, and offer my responses. But first, let me flesh out my account.

4.2 From Metaethical to Semantic Constitutivism

My strategy is to adopt Korsgaard’s general scheme for grounding practical normativity (as it applies to actions) to explain semantic normativity (as it applies to assertions). In particular, I will argue that the constitutive principles of specifically *discursive* agency, understood as the ability to be the autonomous and efficacious cause of her own moves in the game of giving and asking for reasons, are what Rahman et al. (2018) calls the ‘principles of immanent reasoning.’

4.2.1 Why Semantic Constitutivism?

Let us take a step back. What has been at stake for the preceding chapters is the following question: What is the normative principle that distinguishes good conceptual norms

inferential step between 1. and 2., namely: “The necessity of conforming to the principles of practical reason comes down to the necessity of being a unified agent. And if it is correct that agency requires unity, then the necessity of being a unified agent comes down to the necessity of being an agent.”

over others? In particular, what normative principles can we appeal to in order to avoid relativism about conceptual norms, such that concepts like ‘tonk’ and ‘Boche’ can be singled out as defective ones? Furthermore, in virtue of what are those principles genuinely normative, rather than merely conventional? This is to ask for, as Korsgaard (1996) puts it rather poetically, the “source of [semantic] normativity.”

On one hand, Brandom’s answer is to appeal to the objectivity of conceptual content. In Chapter 1, I argued that his account is at best questionable, if not outright incoherent. Brandom’s explanation for objectivity takes us far outside of pragmatism, understood as taking *use*, or actual features of discursive practices, as theoretical primitives for building a meaning theory. The question at this point was to seek a way out of relativism within a broadly pragmatist theoretical orientation.

On the other hand, the principle of harmony is just that kind of normative principle that can do the job of discriminating between good conceptual norms and bad ones. However, as we have discussed in Chapter 2, there isn’t an obvious way to ground it. The choice offered was between the Principle of Innocence, which results in drawing a sharp boundary between the semantics of logical and non-logical vocabulary and Dummett’s negative reasoning, i.e. given 1) the inferentialist approach to semantics is correct given the defects of the representationalist approach and 2) any adequate meaning theory must account for compositionality, the principle of harmony must hold.² I argued that neither of these choices are acceptable grounds for the principle of harmony. That is, even if we accept that inference rules are genuinely normative in virtue of the constitutive principle of logic (viz. harmony), there is not a viable answer to the question, in virtue of what is the principle of harmony genuinely normative?

This is where the constitutivist strategy comes in. My suggestion is that the principle of harmony can be grounded on our discursive agency. Indeed, it should be clear from the discussion in the previous chapters that many logical inferentialists are, though not explicitly proclaimed, constitutivists about logical normativity. Given the understand-

²There is another approach to grounding the principle of harmony, one different from the approaches considered, namely by appealing to evolution. This is one of the theses of Tennant (1987), and more recently, Peregrin (2024). There are reasons to doubt that normativity can in principle be grounded on non-normative facts (i.e. naturalized), but there is no space to discuss these at length.

ing of harmony (minimally) as the criterion of logicality, any rules that fail to satisfy the principle of harmony should not count as conferring meaning to a expression *qua* logical constant. The most straightforward interpretation of this idea is that the principle of harmony is a constitutive principle of any deductive system, and thus logic at large. Accordingly, any purportedly meaningful pair of rules can be evaluated against the normative standard set by the principle of harmony. Indeed, like metaethical constitutivists, logical inferentialists typically argue for a substantive normative claim—namely, that intuitionistic inference rules are better, or more justified, than its classical counterparts—based on this constitutive principle.

The relation between the rule-based conception of ethics and logic go beyond these superficial similarities. As I will show in the following sections, there is a certain kind of *identity* between the subject matter of ethics and logic.

4.2.2 Dialogical Constitutivism

My account, which I call *Dialogical Constitutivism*, has the same general shape as Korsgaard's constitutivism about practical normativity:

1. The normative necessity of conforming to the principles of discursive rationality (i.e. the principles of immanent reasoning) comes down to the necessity of being a discursive agent.
2. Discursive moves require an discursive agent; so the necessity of being a discursive agent comes down to the necessity of making discursive moves.
3. The necessity of choosing and speaking is "our plight: the simple inexorable fact of the human condition."

My claim is that semantic normativity is grounded on constitutive principles of discursive agency, which are the principles of immanent reasoning. To be a *discursive agent* is to be the autonomous and efficacious author of her moves in the game of giving and asking for reasons. In particular, we constitute ourselves as autonomous by making dis-

cursive moves in accordance with the principles of immanent reasoning, which, as it will be shown, really just can be identified with the categorical imperative.

Why should we think that the principles of immanent reasoning are constitutive principles of discursive agency? Again, according to Korsgaard, an agent is characterized by two features: *autonomy* and *efficacy* of her movements. As we saw, autonomy (i.e. to act in accordance with the categorical imperative) is what allows us to be efficient (i.e. to act in accordance with the hypothetical imperative). Accordingly, a specifically discursive agent as one that makes discursive moves autonomously and efficiently. Discursive moves, in turn, involve either endorsing a claim (via action, most obviously asserting) or challenging a claim (by asking). So, the question is: what principles allow us to assert and ask in an autonomous way? In other words, what principles allow us to ensure that the rules according to which we assert and ask are universalizable?

Now, recall that the principles of immanent reasoning are *symmetry*, which says that particle rules are player-independent, or fixed for both parties in the dialogue, and *harmony*, which says that the proponent can only state elementary propositions already conceded by the opponent. The categorical imperative, when transposed to the semantic key, yields the following prescription: When I apply a certain conceptual norm (in stating or challenging), I must will it *universally*, i.e. that the same can be done by any arbitrary agent. It is my contention that the dialogical constraint of symmetry and harmony guarantee this for any meaningful dialogue. Symmetry guarantees this at the play-level, in the sense that for any choice one player (X) makes within the dialogue in accordance with the particle rules, the same choice must be available for the other player (Y) if Y were to be in X's position. Symmetry therefore guarantees that particle rules for any expression is universalizable, at least according to both parties engaged in the dialogue.

The dialogical rendering of intrinsic harmony, as secured by the Socratic rule, is player-dependent. On first glance, this seem to run counter to the universalizability of rules demanded by the categorical imperative. But this is too quick. While the Socratic rule prescribes that the two parties have different roles within a given dialogue—namely, the proponent has the duty to defend the thesis and the opponent has the right to challenge it—the role within a dialogue (of being the proponent or the opponent) is not indexed

to any particular person. For example, Jill might be the proponent for some thesis and Jones the opponent in one dialogue, but the roles must be, in principle, reversible. In other words, structural rules like the Socratic rule are player-dependent but nonetheless *anonymous*, in the sense that it does not privilege any particular individual.

There is yet a deeper reason to think that the Socratic rule is a constitutive principle of discursive agency. What the Socratic rule ensures it that the proponent must ground the reasons for endorsing the thesis on what the opponent concedes during the dialogue, that is, in their interaction structured by the particle rules. In other words, the proponent is not allowed to bring additional information to the dialogue by stating an elementary proposition. This secures a kind of analyticity immanent to the dialogue itself: no dialogue could ever vindicate a thesis that is stronger—i.e. outstrips the information already contained in—the concessions (or the premises). Without the Socratic rule, the proponent would be able to win any play by simply stating the conclusion they are looking for!

This brings to the fore another important point about the nature of dialogues: the possibility of a dialogue hinges on one party defending or endorsing a thesis by giving reasons, and the other party challenging it by asking for reasons. Without the parties playing their respective roles, dialogues could not possibly get off the ground. This is why symmetry on its own is insufficient as the sole constitutive principle of discursive agency. Since the Socratic rule is a constitutive principle of any nontrivial dialogue and discursive moves would not be possible to make in the first place without the possibility of a dialogue, harmony is a constitutive principle of discursive agency.

I have thus far argued that the principles of immanent reasoning are the constitutive principles of discursive agency. Therefore, making discursive moves that violate these principles undermines one's discursive agency, and thus such moves cannot be said to be properly discursive. This is why, for example, drawing inferences using concepts like *tonk* and 'Boche' are all-things-considered irrational. In other words, these conceptual norms are defective because we can abide by them only by violating principles of immanent reasoning. But violating those principles diminish our autonomy, and by extension, our efficacy as discursive agents.

Conceptual norms are genuinely normative if and only if we can abide by the princi-

ples of immanent reasoning when we make discursive moves in accordance with them. So, the normativity of conceptual norms (i.e. semantic normativity) is grounded on constitutive principles of discursive agency. What, then, grounds these constitutive principles? The answer is that these constitutive principles are genuinely normative for us because, whether we like it or not, we must make discursive moves and an event would not count as a discursive move unless it is made autonomously and efficiently by the discursive agent, i.e. by acting in accordance with the constitutive principles of discursive agency.

4.2.3 Semantic and Practical Normativity are One

Since discursive moves (of stating and asking) are kinds of actions, it is now apparent that conceptual norms are really just special kinds of practical norms; it is rather misleading to speak of semantic normativity and practical normativity as separate categories. In fact, Kantian maxims—which are typically evaluated against practical or ethical norms—can themselves be interpreted as inference rules—which are subject to semantic or conceptual norms. Recall the examples I gave in the preceding section:

1. I want my weapon for myself $\vdash$ I will keep my weapon
2. I want your weapon for myself $\vdash$ I will refuse to return it
3. You have run mad and may hurt someone $\vdash$ I will refuse to return your weapon

1. and 3. are valid inferences in general, while 2. is not. The rough reason is that, when read as inference rules, 1. and 3. produce corresponding particle rules that respect symmetry and harmony, while 2. cannot be rendered as particle rules while respecting symmetry and harmony. I will not go into the formal explanation for this since these examples cannot be adequately rendered in the standard dialogical framework.³ Nonetheless, an informal reflection would help us understand why this is so. 1. (and, more contentiously,

³It is possible to do so within the Immanent Reasoning approach developed by Rahman et al. (2018), as it allows us greater flexibility when dealing with material dialogues such as these due to the expressive power endowed by Martin-Löf's Constructive Type Theory. But this step is beyond the scope of the present work.

3.) can be rendered as particle rules that respect symmetry: it does not matter which party in the dialogue states and which party challenges the statement for the validity features to be reflected in the play. But, for 2., we could not render the inference rule into particle rules that maintain player-independence. A similar point could be made about the requirement of harmony. Whereas, in the case of 1. and 3., the conditions for applying the consequent is ‘contained’ within the antecedent, the same cannot be said with respect to 2.

These two points jointly imply that 1. and 3. count as meaning-conferring rules, while 2. is incoherent. It is not merely incidental that 1. and 3. turn out to be good maxims, whereas 2. is a bad one. “Like ethics,” Frege remarked, “logic can [...] be characterized as a normative science” (Frege, 1979, p.128). The founders of the Erlangen school of Dialogical logic took this one step further, claiming that *logic has its roots in ethics*.⁴ I would like to suggest that logic and ethics are *one*, in at least two senses: first, that semantic normativity and practical normativity have the same source, which is constitutive principle of agency, i.e. the categorical imperative. Secondly, on one hand, meaning can only be constituted in actions (i.e. in dialogues), but also on the other hand, actions are not intelligible as actions without concepts. The latter point is lesson from Anscombe (1957). This is to say that a discursive move is an action and that action itself is a discursive move.

Asserting something that is incoherent (i.e. to violate the principles of immanent reasoning) is therefore the same thing as acting in an unethical way; to act unethically (i.e. to violate the categorical imperative) is to undermine the coherence of the conceptual content of the action. In either case, the act of irrationality diminishes the agent’s own autonomy and efficacy, that is, it undermines the agent herself qua agent.

4.3 The *Shmagency* Objection

My strategy to grounding semantic normativity assumed that it is in some sense inescapable for human beings to make discursive moves. The necessity of following the principles of immanent reasoning comes down to the necessity of being a discursive

⁴cf. Lorenzen (1969) and Martin-Löf (2015).

agent, and the necessity of being a discursive agent comes down to the necessity of making discursive moves, which is inescapable for discursive beings like us. Enoch (2006)'s influential objection targets this last step of the reasoning. He writes,

“If a [constitutivist] theory is going to work for agency, then, it is not sufficient to show that some aims or motives or capacities are constitutive of agency. Rather, it is also necessary to show that the “game” of agency is one we have reason to play, that we have reasons to be agents And this, of course, is a paradigmatically normative judgment. . . . [I]f we need a normative judgment . . . in order for the constitutive-of-agency strategy to kick in, then the constitutivist strategy cannot give us the whole story of normativity.” (Enoch, 2006, p. 186)

To sharpen the point, Enoch asks us to imagine a skeptic who acknowledges that agency indeed is governed by some constitutive norm, but still asks, “Why should I be an agent? . . . Why should I act?” (Enoch 2006, p. 179). This skeptic is content being a *shmagent*—a nonagent who is very similar to agents, but who lacks those features of agency that give rise to agency’s constitutive norm. Schmagents are also perfectly content to perform *shmactions*—nonaction events that are very similar to actions” but that lack those features of action that ground practical norms and reasons (Enoch 2006, p. 179). Enoch thinks that constitutivists must show that the skeptic ought to be an agent rather than a shmagent, but that no such account could possibly be given according to the constitutivist strategy: Constitutivism derives normativity from the nature of agency itself, and any such norms will be genuinely normative only for agents (and not for shmagents), as the normativity is only intelligible from the standpoint of an agent. Thus, Enoch concludes, “normativity will not come from what is constitutive of action” (p. 171).

Korsgaard would respond to Enoch’s accusation by appealing to the fact that agency is inescapable: even choosing not to act is a form of action. Katsafanas concurs, claiming that the “core idea” of constitutivism is that “the authority of universal normative claims arises from a certain form of inescapability” (2013, p. 47). But an immediate worry comes to mind. If action is indeed inescapable, then the fact of inescapability appears to under-

mine the normative significance of actions themselves. Recall the Kantian bifurcation of things that are subject to self-determined (and thus rational) norms and others that are subject to causal laws, which are not self-determined. If the normativity of norms are ultimately grounded on the fact that we cannot choose to escape agency, then the constitutivist explanation seems to contradict the very Kantian division of the world, since it denies, at the foundation, that actions are self-determined. A similar point has been made by Enoch himself.

I think there are two plausible lines of replies. The strategy pursued by Korsgaard is to appeal to the neo-Aristotelian idea that categories of objects are individuated by their respective *telos* (or function), and that these *telos* provides a normative standard of assessments for object qua members of their category.⁵ The obvious next step is to analyze the *telos* of human beings as exercising our rational agency. This is certainly a plausible path,⁶ but it should be noted that to pursue this strategy is to move away from the Kantian explanation for normativity as autonomy and efficacy.

The other approach, which is the one I think holds more promise, is to double down on the Kantian explanation for normativity. In fact, it would be hard to see how the shmagency objection could apply to constitutivism about semantic normativity. The reason is quite simple: the only point of view from which we can ask questions about agency (i.e. make discursive moves) is that of the discursive agent. So, the shmagent's question, "Why should I be a (discursive) agent?" cannot be rendered intelligible without presupposing that the shmagent is actually not a shmagent after all! What all of this rests on is the Kantian assumption that the transcendental conditions for agency are autonomy and efficacy. And unfortunately, there is no space available to examine this fundamental assumption on which our discussion has been based.

⁵Cf., for example, Foot (2001).

⁶There are many issues plaguing these theories, starting with the oft-cited idea that neo-Aristotelian naturalists commit the naturalistic fallacy, i.e. deriving 'ought' from 'is.'

4.4 Conclusion

This investigation began with the acknowledgement of the inferentialist's need to account for objectivity in order to avoid rampant relativism according to which any conceptual norm we will automatically count as genuinely normative. Brandom's own account of objectivity was found to be unsatisfactory (Chapter 1). Then, we turned to the debates in logical inferentialism which deal with this precise issue. I argued that intrinsic harmony provides a normative standard internal to the inferentialist theory (i.e. it does not appeal to any kind of objective reality), but it was found that the principle is not grounded on substantial fact/principle (Chapter 2). Then, I have shown that a proper appreciation of the dialogical constitution of meaning shows that intrinsic harmony is part and parcel of meaningful communication between discursive subjects, for logical and non-logical expressions alike. (Chapter 3).

In this chapter, I have identified the source of semantic normativity as the constitutive principles of discursive agency. We are subject to conceptual norms that we will, but these norms are themselves constrained by those constitutive principles. The constitutive principles were analyzed as the principles of immanent reasoning: that is, *symmetry* of particle rules and *harmony* that arises as a structural feature of the dialogue as a result of the coordination between the two parties involved in it. These principles allow us to weed out certain expressions governed by defective rules, such as 'Boche' and *tonk* on count of their undermining our very own discursive agency.

By identifying the principles of immanent reasoning as the constitutive principles of discursive agency, I tried to show that it is possible to account for the irrationality of certain norms without appealing to objective facts about the world; in other words, only by using the theoretical primitives afforded to the inferentialist meaning-theoretic project. In doing so, I have tried to chart the middle path between the dogmatism of the representationalist approach to meaning and the relativism that plague so many inferentialist theories of meaning.

References

- Anscombe, G. E. M. (1957). *Intention*. Cambridge, Mass.: Harvard University Press.
- Austin, J. L. (1962). *How to do things with words* (M. Sbisá & J. O. Urmson, Eds.). Oxford, UK: Clarendon Press.
- Beall, J., & Restall, G. (2000). Logical pluralism. *Australasian Journal of Philosophy*, 78(4), 475-493.
- Belnap, N. D. (1962). Tonk, plonk, plink. *Analysis*, 22, 130–134.
- Boghossian, P. (2014). What is inference? *Philosophical Studies: An International Journal for Philosophy in the Analytic Tradition*, 169(1), 1-18.
- Boghossian, P. A. (2003). The normativity of content. *Philosophical Issues*, 13(1), 31-45. doi: 10.1111/1533-6077.00003
- Bouché, G. (2020). A pure philosophy of language with an edifying intent: Brandom's reply to rorty. In G. Bouché (Ed.), *Reading brandom: On a spirit of trust* (1st Edition ed., p. 166-183). London and New York: Routledge.
- Brandom, R. (1994). *Making it explicit: Reasoning, representing, and discursive commitment*. Cambridge, Mass.: Harvard University Press.
- Brandom, R. (2000). *Articulating reasons: An introduction to inferentialism*. Cambridge, Mass.: Harvard University Press.
- Brandom, R. (2007). Inferentialism and some of its challenges. *Philosophy and Phenomenological Research*, 74(3), 651-676.

- Brandom, R. (2008). *Between saying and doing: Towards an analytic pragmatism*. Oxford, GB: Oxford University Press UK.
- Brandom, R. (2009). *Reason in philosophy: Animating ideas*. Cambridge, MA: Harvard University Press.
- Brandom, R. (2018). From logical expressivism to expressivist logic: A sketch of a program and some implementations. *Philosophical Issues*, 28(1), 70-88.
- Brandom, R. (2019). *A spirit of trust: A reading of hegel's Phenomenology*. Cambridge, Massachusetts: Harvard University Press.
- Brandom, R. B. (2022). *Pragmatism and idealism: Rorty and hegel on representation and reality*. Oxford, UK: Oxford University Press.
- Chang, R. (2013). Grounding practical normativity: Going hybrid. *Philosophical Studies*, 164(1), 163-187.
- Clerbout, N., & McConaughy, Z. (2022). Dialogical Logic. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Fall 2022 ed.). Metaphysics Research Lab, Stanford University.
- Curry, H. B. (1951). A theory of formal deducibility. *Journal of Symbolic Logic*, 16(1), 56-58.
doi: 10.2307/2268674
- Davidson, D. (1984). Thought and talk. In *Inquiries into truth and interpretation*. Clarendon Press.
- Dennett, D. C. (1991). Real patterns. *Journal of Philosophy*, 88(1), 27-51.
- Dosen, K., & Schroeder-Heister, P. (1985). Conservativeness and uniqueness. *Theoria*, 51(3), 159-173. doi: 10.1111/j.1755-2567.1985.tb00093.x
- Dummett, M. (1973). *Frege: Philosophy of language*. London: Duckworth.
- Dummett, M. (1981). *The interpretation of frege's philosophy*. Cambridge: Harvard University Press.

- Dummett, M. (1991). *The logical basis of metaphysics*. Cambridge: Harvard University Press.
- Enoch, D. (2006). Agency, shmagency: Why normativity won't come from what is constitutive of action. *Philosophical Review*, 115(2), 169-198.
- Etchemendy, J. (1990). *The concept of logical consequence*. Cambridge: Harvard University Press.
- Fitch, F. (1963). A logical analysis of some value concepts. *Journal of Symbolic Logic*, 28(2), 135-142.
- Fitch, F. B. (1944). Review: Alfred tarski, the semantic conception of truth and the foundations of semantics. *Journal of Symbolic Logic*, 9(3), 68-68.
- Fodor, J., & Lepore, E. (2007). Brandom beleaguered. *Philosophy and Phenomenological Research*, 74(3), 677-691. doi: 10.1111/j.1933-1592.2007.00045.x
- Foot, P. (2001). *Natural goodness*. New York: Oxford University Press.
- Frege, G. (1979). *Posthumous writings*. Chicago: University of Chicago Press. (Translation of Frege 1969 by Peter Long and Roger White)
- Gentzen, G. (1969 [1935]). *Investigations into logical deduction (untersuchungen über das logische schliessen)* (M. E. Szabo, Trans.). Heidelberg: Springer.
- Hale, B., & Wright, C. (2010). Assertibilist truth and objective content: Still inexplicit? In B. Weiss & J. Wanderer (Eds.), *Reading brandom: On making it explicit* (p. 276-295). Routledge.
- Jacinto, B., & Read, S. (2017). General-elimination stability. *Studia Logica*, 105(2), 361-405.
- Jaeggi, R. (2018). *Critique of forms of life* (C. Cronin, Trans.). Cambridge, MA: Harvard University Press.
- Kant, I. (1785/2003). *Groundwork for the metaphysics of morals* (T. E. Hill & A. Zweig, Eds.). New York: Oxford University Press.

- Katsafanas, P. (2013). *Agency and the foundations of ethics: Nietzschean constitutivism*. Oxford: Oxford University Press UK.
- Katsafanas, P. (2018). Constitutivism about practical reasons. In D. Star (Ed.), *The oxford handbook of reasons and normativity* (p. 367-394). Oxford University Press.
- Korsgaard, C. M. (1996). *The sources of normativity* (O. O'Neill, Ed.). New York: Cambridge University Press.
- Korsgaard, C. M. (2008). *The constitution of agency: Essays on practical reason and moral psychology*. New York: Oxford University Press.
- Korsgaard, C. M. (2009). *Self-constitution: Agency, identity, and integrity*. New York: Oxford University Press.
- Korsgaard, C. M. (2019). Constitutivism and the virtues. *Philosophical Explorations*, 22(2), 98-116.
- Levine, S. (2010). Rehabilitating objectivity: Rorty, brandom, and the new pragmatism. *Canadian Journal of Philosophy*, 40(4), 567-589.
- Lewis, D. (1970). General semantics. *Synthese*, 22(1-2), 18–67.
- Lorenzen, P. (1955). *Einführung in die operative logik und mathematik*. Berlin,: Springer.
- Lorenzen, P. (1969). *Normative logic and ethics*. Mannheim, Germany / Zürich, Switzerland: Bibliographisches Institut.
- Lorenzen, P., & Lorenz, K. (1978). *Dialogische logik*. Darmstadt: Wissenschaftliche Buchgesellschaft.
- MacFarlane, J. (2010). Pragmatism and inferentialism. In B. Weiss & J. Wanderer (Eds.), *Reading brandom: On making it explicit* (p. 81-95). London: Routledge.
- Martin-Löf, P. (1996). On the meanings of the logical constants and the justifications of the logical laws. *Nordic Journal of Philosophical Logic*, 1(1), 11-60.

- Martin-Löf, P. (1984). *An intuitionistic theory of types: Notes by giovanni sambin of a series of lectures given in padua, june 1980*. Napoli: Bibliopolis. (Notes by Giovanni Sambin)
- Martin-Löf, P. (2015). *Is logic part of normative ethics?* (Lecture held at the research unit Sciences, Normes, Décisions (FRE 3593), Paris, May 2015. Transcription by Ansten Klev)
- Milne, P. (2002). Harmony, purity, simplicity and a “seemingly magical fact”. *The Monist*, 85(4), 498-534. doi: 10.5840/monist200285427
- Peregrin, J. (2014). *Inferentialism: Why rules matter*. London and New York: Palgrave-Macmillan.
- Peregrin, J. (2018). Is inferentialism circular? *Analysis*, 78(3), 450-454.
- Peregrin, J. (2024). *Normative species: How naturalized inferentialism explains us*. Routledge.
- Prawitz, D. (1965). *Natural deduction: A proof-theoretical study*. Mineola, N.Y.: Dover Publications.
- Prior, A. N. (1960). The runabout inference-ticket. *Analysis*, 21(2), 38-39. doi: 10.1093/analys/21.2.38
- Prior, A. N. (1967). *Past, present and future*. Oxford: Clarendon Press.
- Quine, W. V. O. (1969). Epistemology naturalized. *Ontological Relativity and Other Essays*, 69-90.
- Rahman, S., Clerbout, N., Klev, A., & McConaughy, Z. (2018). *Immanent reasoning or equality in action: A plaidoyer for the play level*. Cham, Switzerland: Springer Verlag.
- Rahman, S., & Keiff, L. (2005). On how to be a dialogician. In D. Vanderveken (Ed.), *Logic, thought and action* (Vol. 2, p. 359-408). Dordrecht: Springer.
- Read, S. (2008). Harmony and modality. In C. Dégremon, L. Kieff, & H. Rückert (Eds.), *Dialogues, logics and other strange things: Essays in honour of shahid rahman* (p. 285-303). College Publications.

- Reinikainen, J. (2021). Brandom and the pragmatist quest for semantic objectivity. *Acta Philosophica Fennica*, 97, 55-78.
- Ripley, D. (2015). Anything goes. *Topoi*, 34(1), 25-36.
- Rorty, R. (1970). Incorrigibility as the mark of the mental. *The Journal of Philosophy*, 67(12), 399-424.
- Rorty, R. (1979). *Philosophy and the mirror of nature*. Princeton, NJ: Princeton University Press.
- Rosenkranz, S. (2001). Farewell to objectivity: A critique of brandom. *Philosophical Quarterly*, 51(203), 232-237.
- Rumfitt, I. (2000). Yes and no. *Mind*, 109(436), 781-823. doi: 10.1093/mind/109.436.781
- Rumfitt, I. (2017). Against harmony. In B. Hale, C. Wright, & A. Miller (Eds.), *The blackwell companion to the philosophy of language*. Blackwell.
- Schroeder-Heister, P. (1991). Uniform proof-theoretic semantics for logical constants (abstract). *Journal of Symbolic Logic*, 56, 1142. doi: 10.2307/2275092
- Schroeder-Heister, P. (2008). Lorenzen's operative justification of intuitionistic logic. In M. van Atten, P. Boldini, M. Bourdeau, & G. Heinzmann (Eds.), *One hundred years of intuitionism (1907–2007)* (p. 214-240). Basel, Switzerland: Birkhäuser.
- Schroeder-Heister, P. (2014). The calculus of higher-level rules, propositional quantification, and the foundational approach to proof-theoretic harmony. *Studia Logica*, 102(6), 1185-1216.
- Searle, J. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3), 417-57.
- Sellars, W. (1953). Inference and meaning. *Mind*, 62(247), 313–338.
- Steinberger, F. (2009). *Harmony and logical inferentialism* (PhD diss.). University of Cambridge.

- Steinberger, F. (2011). What harmony could and could not be. *Australasian Journal of Philosophy*, 89(4), 617-639. doi: 10.1080/00048402.2010.528781
- Steinberger, F., & Murzi, J. (2017). Inferentialism. In *Blackwell companion to philosophy of language* (pp. 197–224). Wiley Blackwell.
- Stevenson, J. T. (1960). Roundabout the runabout inference-ticket. *Analysis*, 21(6), 124-128.
- Street, S. (2010). What is constructivism in ethics and metaethics? *Philosophy Compass*, 5(5), 363-384. doi: 10.1111/j.1747-9991.2009.00280.x
- Sundholm, G. (1983). Constructions, proofs and the meaning of logical constants. *Journal of Philosophical Logic*, 12(2), 151-172.
- Sundholm, G. (1997). Implicit epistemic aspects of constructive logic. *Journal of Logic, Language and Information*, 6(2), 191-212. doi: 10.1023/a:1008266418092
- Tarski, A. (1935). Der wahrheitsbegriff in den formalisierten sprachen. *Studia Philosophica*, 1, 261-405.
- Tennant, N. (1987). *Anti-realism and logic: Truth as eternal*. New York: Oxford University Press.
- Tennant, N. (1997). *The taming of the true*. New York: Oxford University Press.
- Tranchini, L. (2018). Stabilizing quantum disjunction. *Journal of Philosophical Logic*, 47(6), 1029-1047.
- Tranchini, L. (2024). Harmony via reductions and expansions. In *Harmony and paradox* (Vol. 62). Cham: Springer. doi: 10.1007/978-3-031-46921-3_1
- Turbanti, G. (2011). Modality in brandom's incompatibility semantics. In M. I. Crespo, D. Gakis, & G. Weidman-Sassoon (Eds.), *Proceedings of the amsterdam graduate conference - truth, meaning, and normativity*. ILLC Publications.

- Velleman, D. (2000). *The possibility of practical reason* (J. D. Velleman, Ed.). New York: Oxford University Press.
- Weir, A. (2013). A robust non-transitive logic. *Topoi*, 34(1), 99-107.
- Williamson, T. (2007). *The philosophy of philosophy*. Malden, MA: Blackwell Publishing.
- Williamson, T. (2010). Review of robert brandon, reason in philosophy: Animating ideas. *Times Literary Supplement*(5579), 22-23.
- Wittgenstein, L. (1953). *Philosophical investigations* (G. E. M. Anscombe, Ed.). New York, NY, USA: Wiley-Blackwell.